Chapter 5: Testing and Evaluation

Table 5.1 Summary of the automated test cases

Type	Section	Test cases	Passed	Failed	Not run
Unit	5.1	156	156	0	0
Functional	5.2	115	114	0	1
Business rules (decision tables)	5.3	124	124	0	0
Integration	5.4	41	40	0	1
Total		436	434	0	2

Counts updated for Modules 4, 5 and 6: the 123 headless Flutter cases already listed in Sections 5.1-5.4 (51 unit, 21 functional, 40 business-rule, 11 integration; 123 passed on re-run) and 24 real-device integration_test cases (21 functional, 3 integration; 22 passed, 0 failed, 2 not run).

5.1 Unit Testing

Unit Testing 1: Validators and LoginGuard (Module 1, User Authentication & Onboarding)

Testing Objective: To ensure the registration validators accept valid input, reject invalid input at the stated boundaries, and that five failed logins lock the account for 15 minutes.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M1-U01	Email validator accepts a well-formed address	"name@example.com"	Accepted (no error message)	Accepted (no error message)	**Pass**
2	ATC-M1-U02	Email validator rejects an address without a domain dot	"a@b"	Rejected with an error message	Rejected with the message "Enter an email like name@example.com."	**Pass**
3	ATC-M1-U03	Email validator rejects an empty value	"" (empty string)	Rejected with an error message	Rejected with the message "Enter your email address."	**Pass**
4	ATC-M1-U04	Password with 8+ characters, uppercase, number and symbol is accepted	"Abcdef1!"	Accepted (no error message)	Accepted (no error message)	**Pass**
5	ATC-M1-U05	Password without an uppercase letter is rejected	"abcdefg1!"	Rejected with an error message	Rejected with the message "Add an uppercase letter."	**Pass**
6	ATC-M1-U06	Password without a symbol is rejected	"Abcdefg1"	Rejected with an error message	Rejected with the message "Add a symbol such as ! or @."	**Pass**
7	ATC-M1-U07	Password shorter than 8 characters is rejected (boundary)	"Ab1!" (4 characters)	Rejected with an error message	Rejected with the message "Use at least 8 characters."	**Pass**
8	ATC-M1-U08	Confirm-password must match the password	password "Abcdef1!", confirmation "Abcdef1?"	Rejected with an error message	Rejected with the message "The two passwords do not match."	**Pass**
9	ATC-M1-U09	A user who turns 13 tomorrow is too young (boundary)	date of birth 2013-10-16, today 2026-10-15	Rejected with an error message	Rejected with the message "You must be at least 13 years old."	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
10	ATC-M1-U10	A user who turns 13 today is accepted (boundary)	date of birth 2013-10-15, today 2026-10-15	Accepted (no error message)	Accepted (no error message)	**Pass**
11	ATC-M1-U11	Height below 50 cm is rejected (boundary)	"49" cm	Rejected with an error message	Rejected with the message "Height must be between 50 and 300 cm."	**Pass**
12	ATC-M1-U12	Height of 170 cm is accepted	"170" cm	Accepted (no error message)	Accepted (no error message)	**Pass**
13	ATC-M1-U13	Weight below 20 kg is rejected (boundary)	"19" kg	Rejected with an error message	Rejected with the message "Weight must be between 20 and 500 kg."	**Pass**
14	ATC-M1-U14	The weight range applies in pounds too	"30" lbs (13.6 kg)	Rejected with an error message	Rejected with the message "Weight must be between 44 and 1,100 lbs."	**Pass**
15	ATC-M1-U15	Five failed logins lock the account	five wrong passwords for a@x.com	Account locked after the 5th failure	Not locked after 4 failures; locked for 15 minutes after the 5th	**Pass**
16	ATC-M1-U16	The lock ends after 15 minutes	locked account, 14 min 59 s then 15 min 1 s later	Still locked at 14:59, unlocked at 15:01	At 14:59 locked = true; at 15:01 unlocked = true	**Pass**
17	ATC-M1-U17	A successful login clears the failure count	4 failures, then a successful login, then 1 failure	Counter restarts (4 attempts left)	Attempts left after the next failure: 4	**Pass**

Unit Testing 2: HealthCalc and the health classifiers (Module 2, AI Health Classification Engine)

Testing Objective: To ensure BMI, BMR, age and unit conversion are calculated correctly, that the rule-based classifier follows the clinical rules, and that the Logistic Regression model never overrides a reported condition.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M2-U01	BMI is calculated from height and weight	weight 70 kg, height 175 cm	BMI 22.86	BMI = 22.86	**Pass**
2	ATC-M2-U02	BMI 27 is categorised as Overweight (WHO)	BMI 27	"Overweight"	Category = "Overweight"	**Pass**
3	ATC-M2-U03	BMR (Mifflin-St Jeor) for a man	male, 70 kg, 175 cm, age 30	1648.75 kcal/day	BMR = 1648.75 kcal/day	**Pass**
4	ATC-M2-U04	BMR (Mifflin-St Jeor) for a woman	female, 60 kg, 165 cm, age 25	1345.25 kcal/day	BMR = 1345.25 kcal/day	**Pass**
5	ATC-M2-U05	Feet and pounds are converted before BMI is calculated	height 5.75 ft, weight 154 lbs	BMI about 22.7	175.3 cm, 69.9 kg, BMI = 22.7	**Pass**
6	ATC-M2-U06	Age is calculated from the date of birth (day before the birthday)	born 2000-06-15, today 2026-06-14	25	Age = 25	**Pass**
7	ATC-M2-U07	Normal BMI and no conditions classify as Healthy	male 30, 175 cm, 70 kg, no conditions	"Healthy"	Category = "Healthy"	**Pass**
8	ATC-M2-U08	BMI 27.8 classifies as Overweight	175 cm, 85 kg, no conditions	"Overweight"	Category = "Overweight"	**Pass**
9	ATC-M2-U09	BMI 32.7 classifies as Obese	175 cm, 100 kg, no conditions	"Obese"	Category = "Obese"	**Pass**
10	ATC-M2-U10	Reported heart disease overrides BMI (Cardiac)	100 kg, condition "Heart Disease"	"Cardiac"	Category = "Cardiac"	**Pass**
11	ATC-M2-U11	Reported diabetes classifies as Diabetic	100 kg, condition "Diabetes"	"Diabetic"	Category = "Diabetic"	**Pass**
12	ATC-M2-U12	Heart disease and diabetes together give the more cautious class	conditions "Heart Disease" and "Diabetes"	"Cardiac"	Category = "Cardiac"	**Pass**
13	ATC-M2-U13	Hypertension classifies as At-Risk	condition "Hypertension"	"At-Risk"	Category = "At-Risk"	**Pass**
14	ATC-M2-U14	Underweight BMI (16.3) classifies as At-Risk	175 cm, 50 kg	"At-Risk"	Category = "At-Risk"	**Pass**
15	ATC-M2-U15	Risk level and risk factors are listed	100 kg, Hypertension, Dairy allergy	Risk level High; factors include Hypertension and Dairy allergy	Risk level High; factors: Hypertension, Elevated BMI (32.7), Dairy allergy	**Pass**
16	ATC-M2-U16	The Logistic Regression model never overrides a reported condition	conditions "Heart Disease"; then "Diabetes"	"Cardiac"; "Diabetic"	Results: "Cardiac"; "Diabetic"	**Pass**
17	ATC-M2-U17	The model flags an obese 58-year-old that the rules alone would not	age 58, 175 cm, 117 kg, lightly active, nothing reported	Rules alone: "Obese"; with the model: "At-Risk"	Rules alone: "Obese"; with the model: "At-Risk"	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
18	ATC-M2-U18	The hybrid is never less cautious than the rules alone	ages 25/45/65 x weights 55/80/110 kg (9 people)	No person gets a lower caution rank	0 of 9 people were classified less cautiously	**Pass**

Unit Testing 3: Mood suggestions and the workout card (Module 3, Home Dashboard)

Testing Objective: To ensure the Home dashboard's mood suggestions are gentle for low moods and the workout card's calorie estimate grows with body weight.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M3-U01	The five moods have the report's labels in order	Mood.values	Happy, Neutral, Sad, Angry, Stressed	Labels: Happy, Neutral, Sad, Angry, Stressed	**Pass**
2	ATC-M3-U02	Every mood has a suggestion and low moods are never pushed to train hard	The suggestion for each mood	All non-empty; Sad is gentle, Stressed is low intensity, Angry is not maximal	Sad: "Be kind to yourself. A short walk or a gentle mobility session is enough today." | Stressed: "Choose a low-intensity mobility or stretching session today and breathe slowly." | Angry: "Movement can help. Try steady, controlled exercise rather than anything maximal."	**Pass**
3	ATC-M3-U03	The calorie estimate on the workout card grows with body weight	The first workout day of a plan at 50 kg and at 100 kg	Positive, and higher at 100 kg than at 50 kg	38 min; 131 kcal at 50 kg; 263 kcal at 100 kg	**Pass**

Unit Testing 4: Exercise knowledge base, safety rules and plan generator (Module 4, Workout & Exercise)

Testing Objective: To ensure the exercise dataset is imported faithfully, the safety and screening rules decide each exercise as written, MET-based energy estimates are correct, and the weekly plan respects the training profile of each health category.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-U07	The free-exercise-db dataset is imported with every usable record	exercise_catalog.json (pinned commit f00c92c)	876 records, 871 imported, 5 without instructions dropped	Records 876; imported 871; loaded 871; dropped 5	**Pass**
2	ATC-M4-U08	Exercise IDs are unique and a dataset duplicate of a curated exercise is not offered twice	unified catalogue (curated + dataset)	No duplicate ID; "Bodyweight_Squat" hidden behind curated "squat"	887 exercises; duplicates 0; Bodyweight_Squat listed: false	**Pass**
3	ATC-M4-U09	A missing optional field is tolerated, not invented	dataset record "Bicycling" (mechanic not stated)	Loaded with an empty mechanic	Loaded: true; mechanic ""	**Pass**
4	ATC-M4-U10	Equipment exercises are offered only for equipment the user owns	owned equipment {dumbbell}; then none	Every offered exercise needs a dumbbell; none offered without equipment	121 offered with a dumbbell, all dumbbell: true; without equipment: 0	**Pass**
5	ATC-M4-U11	Expert-level dataset exercises are never scheduled (program rule)	healthy user; dataset exercise at level 3	Excluded by PRG-NO-MAXIMAL	Outcome excluded; rules [PRG-NO-MAXIMAL]	**Pass**
6	ATC-M4-U12	Each health category maps to its training profile	categories Healthy, Cardiac, Obese, an unknown category	Healthy vigorous; Cardiac and Obese moderate; unknown falls back to At-Risk	Healthy vigorous; Cardiac moderate; Obese moderate; unknown → At-Risk	**Pass**
7	ATC-M4-U13	Hypertension excludes high-intensity intervals and names the rule	condition "Hypertension"; HIIT exercise	Excluded; rule MED-HTN	Outcome excluded; rules [MED-HTN]	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
8	ATC-M4-U14	Several conditions merge to the most restrictive limits	Hypertension + Diabetes; Hypertension alone	Both conditions apply; days capped at the smallest maximum	Rules applied: MED-HTN, MED-DIABETES; max days 5 (Hypertension alone 5)	**Pass**
9	ATC-M4-U15	Owned equipment brings dataset exercises in, with the reason shown	Gain Muscle; owns dumbbells	At least one dataset exercise whose first reason starts "Uses equipment you have"	8 dataset exercises; first reason "Uses equipment you have (dumbbell)"	**Pass**
10	ATC-M4-U16	The week has the chosen number of training days; the rest are rest days	healthy user, 4 days a week	7 days: 4 workouts, 3 rest days	7 days; 4 workouts; 3 rest days	**Pass**
11	ATC-M4-U17	The same inputs always give the same plan (deterministic)	same profile and variant, generated twice	Identical exercise lists	Identical: true (36 entries)	**Pass**
12	ATC-M4-U18	MET values are matched to the 2024 Compendium with a recorded match type	all 871 dataset exercises	181 exact, 606 approximate, 84 unmatched	Exact 181; approximate 606; unmatched 84	**Pass**
13	ATC-M4-U19	An exercise without a MET value is never offered for planning	every equipment type owned	All offered exercises have a MET value	482 offered; 0 without MET	**Pass**
14	ATC-M4-U20	Calories = MET × body weight × active hours	MET 3.8, 70 kg, 30 min active	133 kcal	kcal = 133.00	**Pass**
15	ATC-M4-U21	Intensity bands follow the WHO MET bands at the boundaries	MET 2.99, 3.0, 5.99, 6.0	light, moderate, moderate, vigorous	Bands: light, moderate, moderate, vigorous	**Pass**
16	ATC-M4-U22	Diabetes: no more than two rest days in a row	condition "Diabetes", 2 days a week chosen	3 workout days; longest rest run 2 days	3 workout days; longest rest 2 days	**Pass**
17	ATC-M4-U23	Warning symptoms stop plan generation and give the reason	readiness: chest pain reported	No workout days; gate needsClearance (SCR-SYMPTOMS)	Workout days 0; gate needsClearance; rules [SCR-SYMPTOMS]	**Pass**
18	ATC-M4-U24	A planned exercise counts as completed only when its target is reached	manual session: 2 of 2 sets; then skipped	Completed true; skipped false	Completed: true; when skipped: false	**Pass**

Unit Testing 5: Posture engine, repetition counting and form checks (Module 5, Posture Detection)

Testing Objective: To ensure joint angles, landmark visibility, movement phases, repetition acceptance, hold timing, form corrections and the posture score are computed correctly from pose landmarks. Inputs are synthetic landmark frames: these cases verify the logic, not detection accuracy on real people.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-U01	Joint angle at the middle landmark is computed correctly	points (1,0), (0,0), (0,1)	90°	Angle = 90.00°	**Pass**
2	ATC-M5-U02	Landmarks below the visibility threshold are not used	squat frame with visibility 0.3 (threshold 0.5)	No measurement	Measurement: none	**Pass**
3	ATC-M5-U03	Missing landmarks are named in the guidance	squat (dataset) with knees and ankles hidden	Feedback starts "Can't see your knees and ankles"	Feedback: "Can't see your knees and ankles — stand side-on to the camera, 2–3 m away, with your whole body in the frame"	**Pass**
4	ATC-M5-U04	The movement phases follow top → descending → bottom → ascending → top	one squat to 88°	Phases seen in that order	Phases: unknown → top → descending → bottom → ascending → top	**Pass**
5	ATC-M5-U05	Three full squats count three reps	3 squats to 88°	3 reps, 0 partial	Reps 3; partial 0	**Pass**
6	ATC-M5-U06	A half repetition is not counted	1 squat that turns back at 125°	0 reps, 1 partial attempt	Reps 0; partial 1	**Pass**
7	ATC-M5-U07	A movement faster than the minimum rep time is rejected as noise	squat down and up in 300 ms (minimum 600 ms)	0 reps	Reps 0	**Pass**
8	ATC-M5-U08	The angle is smoothed (exponential average)	samples 170° then 130°	Smoothed angle 150°	Smoothed = 150.0	**Pass**
9	ATC-M5-U09	A plank hold is timed only while in position	5 s straight plank, then 3 s sagging	About 5 s held	Held 4 s	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
10	ATC-M5-U10	The back-alignment rule passes upright and fails a deep forward lean	torso lean 30° and 70° (limit 60°)	Pass at 30°, fail at 70°	30°: pass; 70°: fail	**Pass**
11	ATC-M5-U11	The hip rule gives the right correction for sagging and piking	hip offset -0.12 and +0.20	"Raise your hips…" and "Lower your hips…"	Sag: "Raise your hips slightly — don't let them sag"; pike: "Lower your hips into a straight line"	**Pass**
12	ATC-M5-U12	Posture score = acceptable reps ÷ attempted reps	2 good squats and 1 shallow attempt	Score 67%	Score 67% (2 reps, 1 partial)	**Pass**
13	ATC-M5-U13	A correction is shown only after the fault persists for 3 frames	forward lean of 75° held during the descent	"Keep your back straight" appears, never on the first faulty frame	First correction at frame 26 of 39	**Pass**
14	ATC-M5-U14	Exercises map to a movement pattern by movement, not muscle	Barbell_Squat, Leg_Extensions, Dumbbell_Bicep_Curl	squat, legExtension, curl	Patterns: squat, legExtension, curl	**Pass**
15	ATC-M5-U15	An exercise the camera cannot follow is unsupported with a reason	Standing_Calf_Raises	Not supported; reason mentions a movement too small to measure	Level notSupported; reason "The movement is too small for the camera to measure reliably (for example wrists, calves, shoulder shrugs or a crunch)."	**Pass**
16	ATC-M5-U16	Camera capability of every schedulable exercise (current code)	curated library + dataset exercises for any equipment, program rule applied	Counts reported from the registry; only the 4 tuned exercises are validated	532 schedulable: validated 4; experimental 300; basic 0; unsupported 228	**Pass**
17	ATC-M5-U17	Number of movement patterns with a tracker	MovementPattern enum	Every pattern except "none" builds a tracker	26 patterns, 26 with a tracker (2 hold, 24 repetition)	**Pass**
18	ATC-M5-U18	An opening movement (overhead press) is counted on the mirrored angle	3 presses 110° → 150°; one-arm hanging arm ignored	3 reps	Reps 3	**Pass**

Unit Testing 6: Intent recognition, coach replies and voice session control (Module 6, Voice Assistant)

Testing Objective: To ensure spoken (transcribed) requests map to the right intent, replies use the live session data instead of invented numbers, and microphone, recognition and speech-output failures are handled. A simulated recogniser and speech output are used: these cases do not validate microphone capture or speech-recognition accuracy.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-U01	A rep question is recognised	"How many reps have I done?"	Intent repCount	Intent = repCount	**Pass**
2	ATC-M6-U02	A form question is recognised	"How's my form?"	Intent postureFeedback	Intent = postureFeedback	**Pass**
3	ATC-M6-U03	A pause command is recognised	"Pause workout"	Intent pauseSession	Intent = pauseSession	**Pass**
4	ATC-M6-U04	An unrelated question falls back to unknown	"what is the capital of France"	Intent unknown	Intent = unknown	**Pass**
5	ATC-M6-U05	Empty speech input is not treated as a command	" "	Intent unknown	Intent = unknown	**Pass**
6	ATC-M6-U06	A pain report is recognised as a safety intent	"My knee hurts"	Intent painOrSymptom	Intent = painOrSymptom	**Pass**
7	ATC-M6-U07	The reply uses the live rep counter, never an invented number	camera session with 2 counted squats; "how many reps"	Reply "You've completed 2 reps…"	Reply: "You've completed 2 reps of 36, 34 to go."	**Pass**
8	ATC-M6-U08	With no session the assistant says so instead of inventing numbers	no active session; "How many reps?"	Reply mentions no active workout session	Reply: "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."	**Pass**
9	ATC-M6-U09	Quick-action chips are contextual and at most three	"what is the capital of France"	1–3 chips	2 chips: What's my workout today?, How do I do a squat?	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
10	ATC-M6-U10	Microphone denied: the assistant never listens	microphone permission denied	State unavailable; 0 listen calls	State unavailable; listen calls 0; message "Microphone access is needed for voice commands. You can still tap a suggestion below."	**Pass**
11	ATC-M6-U11	Microphone permanently denied: the settings link is offered	microphone permanently denied	settingsNeeded true; no listening	Settings needed true; listen calls 0	**Pass**
12	ATC-M6-U12	Recognition errors give a clear message and nothing is spoken	no-match error; then timeout error	"didn't catch that"; "didn't hear anything"; no speech output	No match: "I didn't catch that. Tap the mic and try again."; timeout: "I didn't hear anything. Tap the mic and try again."; spoken 0	**Pass**
13	ATC-M6-U13	A speech-synthesis failure still shows the reply	text-to-speech fails; "hello"	Reply shown; speechOutputFailed true; state idle	Reply present true; output failed true; state idle	**Pass**
14	ATC-M6-U14	Closing releases the microphone and ignores a pending command	listening; close; late "pause workout"	Recognizer cancelled once; session still active	Cancel calls 1; session active	**Pass**
15	ATC-M6-U15	Replies never overlap; replay speaks the same reply once more	"hello", then replay	2 identical utterances; at most 1 at a time	Spoken 2; same true; max concurrent 1	**Pass**

Unit Testing 7: FoodParser, FoodDatabase and diet summary (Module 8, Smart Diet & Calorie Tracking)

Testing Objective: To ensure typed meals are understood, searched and totalled correctly and that the daily verdict never praises a very low intake.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M8-U01	Text entry is parsed into foods and servings	"2 rotis and a bowl of daal"	2 x roti (220 kcal) and 1 x daal	Found 2.0 x roti and 1.0 x daal; roti = 220 kcal	**Pass**
2	ATC-M8-U02	Gram quantities scale by serving weight	"150g grilled chicken breast"	About 248 kcal	Grilled chicken breast: 248.0 kcal	**Pass**
3	ATC-M8-U03	Fractions are understood	"half a cup of rice"	0.5 servings	Servings = 0.5	**Pass**
4	ATC-M8-U04	An unknown food is reported rather than guessed	"2 eggs, toast + green tea and blorptastic"	Four parts; the unknown one has no food	Parts: boiled_egg, white_bread, green_tea, UNKNOWN	**Pass**
5	ATC-M8-U05	Food search finds a food from a partial name	"chick"	Results include grilled chicken	Found: chana_masala, chicken_soup, chicken_curry, chicken_tikka, chicken_karahi	**Pass**
6	ATC-M8-U06	Macro shares always add up to 100%	protein 33.3 g, carbs 66.6 g, fat 12.1 g	Shares sum to 100	26% + 52% + 22% = 100%	**Pass**
7	ATC-M8-U07	Calories are totalled by meal slot	Breakfast 300 + 100 kcal, Lunch 500 kcal	Breakfast 400, Lunch 500, Dinner 0	Breakfast 400.0, Lunch 500.0, Dinner 0.0	**Pass**
8	ATC-M8-U08	Net calories = consumed minus burned	consumed 1800, burned 350	1450	Net = 1450	**Pass**
9	ATC-M8-U09	The daily verdict never praises a very low intake	goal Lose Weight, consumed 600, target 1800, BMR 1500	Mentions "very low intake"	Verdict: "That is a very low intake for you. If you skipped meals or did not log them, add them; if not, eat enough to fuel your body."	**Pass**
10	ATC-M8-U10	The daily verdict recognises a day on target	goal Lose Weight, consumed 1790, target 1800, BMR 1500	Says "right on your daily target"	Verdict: "You are right on your daily target. Nicely balanced."	**Pass**

Unit Testing 8: Nutrition plan and calorie targets (Module 9, Nutrition Plan)

Testing Objective: To ensure calorie and macro targets are consistent, never fall below the safe floor, and that meal plans never contain the user's allergens.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M9-U01	Protein, carbs and fat add up to the calorie target	goal Lose Weight, 175 cm, 70 kg	Macro calories within 40 kcal of the target	Macros give 2058 kcal; target 2056 kcal	**Pass**
2	ATC-M9-U02	A meal plan never contains the user's allergens	allergies Dairy, Gluten, Eggs, Nuts; 6 plan variants	No allergen in any meal of any variant	60 meal items checked across 6 variants; 0 allergen conflicts	**Pass**
3	ATC-M9-U03	A diabetic meal plan has no high-sugar or refined foods	condition "Diabetes"; 6 plan variants	None of the foods is tagged highSugar or refined	66 items checked; 0 high-sugar or refined foods	**Pass**
4	ATC-M9-U04	Meal calories land near the daily target	175 cm, 75 kg	Total of meals between 75% and 125% of the target	Meals total 2239 kcal; target 2633 kcal	**Pass**
5	ATC-M9-U05	The calorie deficit never goes below the safe floor	female, BMR 1100, maintenance 1500, BMI 22	Deficit target is 1200 kcal and a safety note is given	Deficit 1200 kcal; note: "Deficit limited to 1200 kcal/day, the minimum safe intake for you."	**Pass**
6	ATC-M9-U06	No calorie deficit is set for an underweight user	female, BMR 1300, maintenance 1800, BMI 17	Deficit target equals maintenance	Deficit 1800, maintenance 1800	**Pass**
7	ATC-M9-U07	Diabetic macros cap carbohydrates at 40% of calories	2000 kcal, 70 kg, diabetic	Carbohydrate share at most 40%	Carbohydrates 200 g = 40.0% of calories	**Pass**
8	ATC-M9-U08	Three nutrition tips a day and none repeats within a week	pool of 28 tips, 7 consecutive days	3 tips per day, 21 different tips in 7 days	Tips per day all 3 = true; different tips in 7 days = 21	**Pass**

Unit Testing 9: Gamification (Module 12, Community Ranking & Gamification)

Testing Objective: To ensure the weekly score, levels, streaks, badges, XP and K-Means cohorts are calculated as specified.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M12-U01	A perfect week scores 100	2 planned workouts done, food logged every day on target	Score 100	Score = 100.0	**Pass**
2	ATC-M12-U02	A week with nothing done scores 0	4 planned days, nothing done or logged	Score 0	Score = 0.0	**Pass**
3	ATC-M12-U03	Score parts are weighted 50 / 25 / 25	training 50%, logging 50%, calories on target 50%	Score 50	Score = 50.0	**Pass**
4	ATC-M12-U04	Score thresholds map to Beginner / Intermediate / Advanced (boundary values)	scores 39.9, 40, 69.9, 70	Beginner, Intermediate, Intermediate, Advanced	Levels: beginner, intermediate, intermediate, advanced	**Pass**
5	ATC-M12-U05	K-Means groups a cohort of 15 into three tiers	6 low, 6 mid and 6 high scorers	Low = Beginner, mid = Intermediate, high = Advanced; method is K-Means	Method "K-Means (18 users)"; low0 beginner, mid3 intermediate, high5 advanced	**Pass**
6	ATC-M12-U06	A streak counts consecutive active days and today may still be empty	active, active, active; then active, active, not yet	3; then 2	Streaks: 3 and 2	**Pass**
7	ATC-M12-U07	7-day and 30-day streak badges are earned only at their threshold	6, 7 and 30 active days	6 days: no badge; 7 days: daily_7; 30 days: daily_7 and daily_30	6 days: false; 7 days: true/false; 30 days: true/true	**Pass**
8	ATC-M12-U08	The badge catalogue marks earned badges and keeps locked ones	a 7-day streak	Catalogue of 10 badges, daily_7 earned and others locked	10 badges in the catalogue; 4 earned	**Pass**
9	ATC-M12-U09	XP levels start at 100 x L x (L-1)	XP 0, 199, 200, 599, 600	Levels 1, 1, 2, 2, 3	Levels: 1, 1, 2, 2, 3	**Pass**

Unit Testing 10: ChatbotEngine (Module 13, AI Chatbot)

Testing Objective: To ensure intents are detected, answers come from the user's own data, and symptoms and emergencies are handled safely.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M13-U01	Intent is detected from typed text	"what is my bmi"	Intent "bmi"	Intent = "bmi"	**Pass**
2	ATC-M13-U02	Calories left today come from the real food log	"how many calories left" with 600 kcal logged	Answer contains 600	Answer: "You've eaten 600 of 2056 kcal today — 1456 kcal left."	**Pass**
3	ATC-M13-U03	Calorie lookup adds up several foods	"How many calories in 2 rotis and daal?"	Intent calorie_lookup, 450 kcal (220 + 230)	Intent "calorie_lookup"; answer mentions 450 kcal = true	**Pass**
4	ATC-M13-U04	A symptom message recommends a specialist consultation	"my knee hurts after squats"	Intent health_concern with a "Book a consultation" action	Intent "health_concern"; actions: consult	**Pass**
5	ATC-M13-U05	Chest pain is treated as an emergency	"I have chest pain"	Intent emergency and the word "emergency" in the answer	Intent "emergency"	**Pass**
6	ATC-M13-U06	The bot warns when a food is on the user's allergy list	"can I eat butter chicken?" with a Dairy allergy	Answer says it is on the allergy list	Warns about the allergy = true	**Pass**
7	ATC-M13-U07	Nonsense input gets a polite fallback	"asdfgh qwerty"	Intent "fallback"	Intent = "fallback"	**Pass**

Unit Testing 11: ConsultationRules and specialist plan layering (Module 14, Online Consultation)

Testing Objective: To ensure time slots, filters, booking checks, the room-join window, cancellation and the layering of a specialist's plan over Baymax's plan follow the rules.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-U01	Time slots follow the specialist's own clock	Monday 09:00-17:00, 30-minute slots, Pakistan time (UTC+5)	16 slots, first at 04:00 UTC, last at 11:30 UTC	16 slots; first 2026-10-19T04:00:00.000Z; last 2026-10-19T11:30:00.000Z	**Pass**
2	ATC-M14-U02	A day off has no slots	Saturday for a Monday-Friday specialist	No slots	0 slots	**Pass**
3	ATC-M14-U03	Booked times and times that are too soon are not offered	08:50 now; 09:00 too soon, 09:30 booked	First open slot is 10:00; 14 slots offered	First open slot 2026-10-19T05:00:00.000Z; 14 slots	**Pass**
4	ATC-M14-U04	A specialist who is not taking bookings offers nothing	available = false	No open slots	0 slots	**Pass**
5	ATC-M14-U05	A note of exactly 500 characters is accepted (boundary)	note of 500 characters	No booking problem	No problem found	**Pass**
6	ATC-M14-U06	A note of 501 characters is rejected (boundary)	note of 501 characters	Booking problem mentioning 500	Problem: "Notes can be at most 500 characters."	**Pass**
7	ATC-M14-U07	A slot that was just taken is refused	slot 10:00 already booked	Message says the time was just taken	Problem: "That time was just taken. Please choose another."	**Pass**
8	ATC-M14-U08	The specialist list filters by type	doctor, nutritionist and trainer; filter = trainer	Only the trainer	Result: t	**Pass**
9	ATC-M14-U09	Search matches name or specialty, ignoring case	"SPORTS"	The sports nutritionist	Result: n	**Pass**
10	ATC-M14-U10	"Top rated" needs 4.5 stars and at least one review	rating 4.9/80 reviews, rating 4.7/0 reviews, rating 4.2/10 reviews	Only the first	Result: a	**Pass**
11	ATC-M14-U11	The video room opens five minutes before the start	confirmed appointment; 6 and 5 minutes before the start	Closed at -6 min, open at -5 min	At -6 min can join = false; at -5 min can join = true	**Pass**
12	ATC-M14-U12	There is nothing to join before the specialist accepts	status "requested" (no video link yet)	Cannot join	Can join = false	**Pass**
13	ATC-M14-U13	A request can be cancelled until it starts, but not after	requested appointment; 30 minutes before and at the start	Cancellable before, not at the start	Before the start: true; at the start: false	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
14	ATC-M14-U14	A request nobody answered before its start time expires	requested appointment, now = start time	Expired and no longer upcoming	Expired = true; upcoming = false	**Pass**
15	ATC-M14-U15	A specialist's diet plan replaces the calorie target and macros and adds advice first	Baymax target 2000 kcal; specialist target 1800 kcal, macros 120/200/55	Target 1800 kcal; avoid list starts with "Fried foods"; Baymax advice kept	Target 1800 kcal; macros 120/200/55; first avoid "Fried foods"; last tip "Walk after meals"	**Pass**
16	ATC-M14-U16	"Approved as is" changes nothing in the plan	specialist status approved	The Baymax plan is returned unchanged	Plan returned unchanged = true	**Pass**
17	ATC-M14-U17	The member's choice to use Baymax's plan is respected	specialist plan present; member choice = Baymax	Calorie target stays 2000	Target = 2000 kcal	**Pass**
18	ATC-M14-U18	A specialist's weekly schedule replaces the week with warm-up and cool-down	Monday walk 900 s; Wednesday squat 3x15; other days rest	2 workout days; warm-up first and cool-down last on each	2 workout days; Monday exercises: warmup > walk > cooldown	**Pass**

Unit Testing 12: Roles, reminders and inactivity (Module 16, Secure Data Management & Smart Alerts)

Testing Objective: To ensure role permissions, inactivity detection and the next reminder time are correct.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M16-U01	An unknown or missing role is treated as a plain member	null; "superuser"; "doctor"	member; member; doctor	Roles: user; user; doctor	**Pass**
2	ATC-M16-U02	A member cannot view assigned patients' data	member role	viewOwnData allowed; viewAssignedPatientData refused	Own data: true; patients' data: false	**Pass**
3	ATC-M16-U03	Only a doctor may upload prescriptions	doctor and trainer roles	Doctor allowed; trainer refused	Doctor: true; trainer: false	**Pass**
4	ATC-M16-U04	An administrator manages accounts but cannot read health data	admin role	manageAccounts allowed; viewAssignedPatientData refused	Manage accounts: true; read patient data: false	**Pass**
5	ATC-M16-U05	Rest days alone are not inactivity	planned Mon/Wed/Fri; last workout Fri 9 Oct; today Mon 12 Oct	No inactivity alert	Alert = none	**Pass**
6	ATC-M16-U06	A missed planned day over three days is inactivity	planned Mon/Wed/Fri; last workout Fri 9 Oct; today Wed 14 Oct	Alert: 5 days since workout, 1 missed planned day	Days since workout 5; missed planned days 1	**Pass**
7	ATC-M16-U07	The next workout reminder is the next occurrence of the weekday and time	now Wed 14 Oct 10:00; reminder Wed 18:30; Wed 09:00	Today 18:30; next Wednesday 09:00	Next: 2026-10-14 18:30:00.000; 2026-10-21 09:00:00.000	**Pass**

Unit Testing 13: Moving average (Module 17, Analytics & Progress Dashboard)

Testing Objective: To ensure the chart smoothing uses only logged days.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M17-U01	The moving average uses only logged days inside a 3-day window	[2000, null, 2200, 1800, null]	[2000, 2000, 2100, 2000, 2000]	Smoothed: 2000.0, 2000.0, 2100.0, 2000.0, 2000.0	**Pass**
2	ATC-M17-U02	A window with no logged day gives no value rather than zero	[null, null, null, 2000]	[null, null, null, 2000]	Smoothed: null, null, null, 2000.0	**Pass**

5.2 Functional Testing

Functional Testing 1: Sign-up screen (Module 1, User Authentication & Onboarding)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M1-F01	Create account is disabled on an empty form	Open the sign-up screen and type nothing	The Create account button is disabled	Button enabled = false	**Pass**
2	ATC-M1-F02	A weak password is explained and blocks sign-up	Name "Aamna Khan", email "aamna@example.com", password and confirmation "weak"	Message "Use at least 8 characters." shown; button disabled	Message shown = true; button enabled = false	**Pass**
3	ATC-M1-F03	Terms must be accepted before the account can be created	All fields valid, terms not ticked; then terms ticked	Disabled before ticking, enabled after	Enabled before ticking = false; after ticking = true	**Pass**
4	ATC-M1-F04	A different confirmation password disables sign-up again	Valid form, then confirmation changed to "Different1!"	Message "The two passwords do not match." shown; button disabled	Message shown = true; button enabled = false	**Pass**

Functional Testing 2: BMI and classification screens (Module 2, AI Health Classification Engine)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M2-F01	The BMI screen shows the computed BMI, the BMR method and the calorie deficit	Profile: female, 165 cm, 72 kg, Lose Weight	BMI 26.4 and the words "Mifflin-St Jeor" and "Deficit" on screen	BMI 26.4 shown = true; method shown = true; deficit shown = true	**Pass**
2	ATC-M2-F02	The classification screen lists the reported condition as a risk factor	Profile with condition "Hypertension"	"Hypertension" and "View workout plan" on screen	"Hypertension" appears 1 time(s); "View workout plan" button shown = true	**Pass**

Functional Testing 3: Home dashboard (Module 3, Home Dashboard)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M3-F01	The greeting names the user and follows the time of day	Open Home for "Aamna Khan"; the clock decides morning, afternoon or evening	The greeting for the current hour, the first name "Aamna" and the matching motivational line	Greeting "Good Evening" shown true; first name shown true; motivational line shown true	**Pass**
2	ATC-M3-F02	With no profile Home says "Welcome back" and asks for the health profile instead of inventing a plan	Open Home with no profile loaded	"Welcome back", "No plan yet" and "Complete your health profile"	Welcome back true; "No plan yet" true; prompt true	**Pass**
3	ATC-M3-F03	The activity cards show calories burned, active minutes and an honest steps card	Open Home with no workout done today	"Calories burned" and "Active minutes" with 0; steps card says "Steps (no sensor)" with a dash, not a made-up number	Calories burned true; Active minutes true; zeros shown 2; steps card "Steps (no sensor)" with dash true	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
4	ATC-M3-F04	The workout card shows today's session from the user's plan and Open goes to the Workout tab	Open Home with a generated plan and tap Open	Today's workout name with minutes and kcal, or "Rest day"; Open selects tab 2	Card shown true; today is a rest day, shown correctly true; tapping Open selected tab 2	**Pass**
5	ATC-M3-F05	The mood selector shows the chosen mood with a gentle suggestion, and tapping it again clears it	Tap "Sad", then tap "Sad" again	Label "Sad" and the suggestion appear; both disappear after the second tap	Prompt shown true; after the first tap label 1, suggestion 1; after the second tap 0 left	**Pass**
6	ATC-M3-F06	Quick Access lists the eight areas the report names	Open Home and read the Quick access grid	Workout plan, Diet tracker, Posture check, Body transform, Leaderboard, Allergy scanner, AI chatbot, Find a specialist	All 8 tiles shown	**Pass**
7	ATC-M3-F07	Quick Access tiles that belong to a tab switch to that tab	Tap the "Workout plan" tile, then the "Diet tracker" tile	Tab 2 (Workout), then tab 1 (Diet)	Workout plan tile selected tab 2; Diet tracker tile selected tab 1	**Pass**
8	ATC-M3-F08	The profile avatar shows the user's initial and opens the Profile tab	Tap the avatar in the top bar	Initial "A"; tab 4 (Profile) selected	Initial shown true; tab selected 4	**Pass**
9	ATC-M3-F09	The notification bell shows the unread count and caps it at 9+	Set the unread count to 3, then to 12	Badge "3" with tooltip "Notifications, 3 unread"; then badge "9+"	Badge "3" true; tooltip true; badge "9+" true	**Pass**
10	ATC-M3-F10	The bottom navigation bar has the five labelled tabs and each reports its index	Tap Diet, Workout, Analytics, Profile and Home in the bar	Indexes 1, 2, 3, 4, 0	Indexes reported: 1, 2, 3, 4, 0	**Pass**

Functional Testing 4: Exercise detail and workout plan screens (Module 4, Workout & Exercise)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-F03	Exercise detail shows the prescribed dose and a start/end illustration	Planned "Jumping Jacks": 2 sets of 30 s, rest 40 s	Sets, per-set time and rest shown; illustration caption shown	Sets 1; "30s" 1; rest "40s" 1; illustration caption 1	**Pass**
2	ATC-M4-F04	A dataset exercise shows its photos with the source caption	Planned "Dumbbell Bicep Curl": 3 × 10	Caption "Photos: free-exercise-db (public domain)…"; technique steps listed	Photo caption 1; Technique heading 1	**Pass**
3	ATC-M4-F05	"Start exercise" opens mode selection linked to the planned exercise	Generated plan, Monday, 4th exercise	Mode selection opens with plan link (Monday, entry 3)	Opened Push-Ups; plan day 0; entry 3	**Pass**
4	ATC-M4-F06	The workout tab shows the training profile of the health classification	Profile with "Heart Disease", doctor's clearance given	"Your training profile: Cardiac" with "Highest intensity scheduled: Moderate"	Profile title 1; intensity line 1	**Pass**

Functional Testing 5: Posture detection home, mode selection, camera, manual guidance and summary screens (Module 5, Posture Detection)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-F01	Posture Detection offers Today's workout, My plan and the Exercise library	Signed-in user with a 6-day plan	Three tabs	3 of 3 tabs present	**Pass**
2	ATC-M5-F02	Without a plan an empty state is shown, not invented exercises	No health profile	"No personalised plan yet" and "Create my health profile"	Empty state 1; button 1	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
3	ATC-M5-F03	My plan lists the real plan day with camera capability and starts an entry with its plan link	Tab "My plan", Monday, 3rd exercise	Entry and capability shown; mode selection receives exercise, sets and plan link	Capability label 1; opened Lunges (3 sets), plan entry 2	**Pass**
4	ATC-M5-F04	The generic exercise library is kept with camera and manual sections	Tab "Exercise library"	Note that it is not the personalised plan; "Camera posture detection" and "Manual guidance" sections	Generic-library note 1; camera section 1; manual section 1	**Pass**
5	ATC-M5-F05	Camera permission is not requested before the consent toggle is on	Bodyweight Squat; Continue tapped with consent off	Both modes shown; 0 permission requests; consent prompt shown	Mode options 2; camera requests 0; camera screen opened 0	**Pass**
6	ATC-M5-F06	"Skip for now" returns to the previous screen	Mode selection opened from a screen; tap "Skip for now"	Mode selection closed	Mode selection still open: 0	**Pass**
7	ATC-M5-F07	Refused camera permission keeps the camera off and offers Manual Guidance	Consent on; camera permission denied	"Camera access needed" and "Use Manual Guidance"; camera screen not opened	Dialog 1; manual offered 1; camera screen 0	**Pass**
8	ATC-M5-F08	Camera screen counts reps from pose frames and shows status and privacy notice	2 squats fed as pose frames	Reps 2; "Landmarks detected", "Angles calculated"; local-processing notice	Reps 2; status chips 1/1; notice 1	**Pass**
9	ATC-M5-F09	"Stop Session" ends the session and shows the summary with score, stats and feedback	2 good squats, then Stop Session	Summary: "Session Complete!", posture score 100%, Total Reps, Accuracy, form feedback; session saved	Summary 1; score 100%; stats 1/1; feedback 1; saved 1	**Pass**
10	ATC-M5-F10	Manual guidance: confirmed sets add reps; the form checklist and response buttons are shown	Bodyweight Squat 3 × 12; tap "Completed ✓" once	12 reps, 1 set; checklist step and Skip / Too Difficult buttons present	Reps 12; sets 1; checklist step 1; Too Difficult 1; Skip 1	**Pass**
11	ATC-M5-F11	"Next Exercise" advances to the following exercise	Plank 1 × 20 s then Squat; hold finished	Next exercise is Bodyweight Squat; plank saved in history	Current exercise squat; last finished plank	**Pass**
12	ATC-M5-F12	A plank in camera mode shows a hold timer instead of a rep counter	Plank 2 × 20 s; 6 s in position	"HOLD" badge and "In position"; about 5–6 s held	HOLD badge 1; "In position" 1; held 5 s	**Pass**
13	ATC-M5-F13	An exercise the camera cannot follow explains why and keeps Manual Guidance	Calf Raises	Camera not available with the reason; Continue opens Manual Guidance	Reason shown 1; session mode manual; camera requests 0	**Pass**

Functional Testing 6: Voice assistant screen (Module 6, Voice Assistant)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-F01	Opening the assistant asks for the microphone and shows the listening state	Voice assistant opened	"Listening..." and "Cancel"; 1 microphone request	Listening 1; Cancel 1; microphone requests 1	**Pass**
2	ATC-M6-F02	A spoken question shows the transcript, the reply card, at most three chips and is spoken aloud	Recognised speech "how do I do a squat" (simulated recogniser)	Transcript shown; reply about Bodyweight Squat spoken; ≤ 3 chips	Transcript 1; spoken "Here is how to do Bodyweight Squat"; chips 2	**Pass**
3	ATC-M6-F03	A refused microphone is explained and a retry is offered	Microphone permission denied	"Voice input unavailable", the reason and "Try again"; not listening	Unavailable 1; reason 1; retry 1; listen calls 0	**Pass**
4	ATC-M6-F04	Leaving the assistant releases the microphone without running a command	Assistant listening; screen closed; a late "pause workout"	Recognizer cancelled once; no command executed	Cancel calls 1; session active	**Pass**

Functional Testing 7: Diet screens (Module 8, Smart Diet & Calorie Tracking)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M8-F01	The daily summary shows consumed, burned and net calories with a verdict	Logged 1000 kcal, burned 250, goal Lose Weight, BMR 1500	Consumed 1000, burned 250, net +750 and the words "very low intake"	Consumed shown true; burned shown true; net "+750" shown true; verdict shown true	**Pass**
2	ATC-M8-F02	An empty day says nothing is logged and invents no macro split	Day with no food entries	"Nothing is logged" and the prompt to log food	"Nothing is logged" shown true; prompt shown true	**Pass**
3	ATC-M8-F03	Typing a local name finds the food by its other name	Type "chawal"	Suggestions with "Also called" and kcal values	"Also called" appears 5 time(s); kcal shown 5 time(s)	**Pass**
4	ATC-M8-F04	Picking a suggestion adds it to the meal and closes the list	Type "banana", tap the add button of the first result	"Your meal" appears and the add buttons disappear	"Your meal" before: 0, after: 1; add buttons left: 0	**Pass**
5	ATC-M8-F05	A food that matches the user's allergy is flagged in the search results	User allergic to Dairy types "milk"	"on your allergy list" warning with a warning icon	Allergy warning shown 4 time(s); warning icons 4	**Pass**
6	ATC-M8-F06	A search with no result says so instead of showing an empty box	Type "qqzzxx"	"No matching foods"	Message shown = true	**Pass**

Functional Testing 8: Nutrition plan screen (Module 9, Nutrition Plan)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M9-F01	The nutrition plan shows the calorie target and lists the user's allergen foods under "avoid"	Profile with Dairy and Nuts allergies	The target number and "Dairy products" on screen	Target 1508 kcal shown = true; "Dairy products" under avoid = true	**Pass**

Functional Testing 9: My badges screen (Module 12, Community Ranking & Gamification)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M12-F01	My badges shows the level, XP and the unlock rule of a locked badge	Open My badges with 1 workout day, 2 meal days and 1 badge	Level 1, 170 XP, 30 XP to level 2, and the "30-Day Streak" badge listed	Level true; XP true; XP to next true; locked badge listed true	**Pass**
2	ATC-M12-F02	A failed load shows no invented numbers	The data loader throws "offline"	"Level 1" is not shown	"Level 1" shown 0 time(s)	**Pass**

Functional Testing 10: Find a specialist, Book appointment and My appointments screens (Module 14, Online Consultation)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-F01	The specialist list shows a count and disables Book for a specialist who is not taking bookings	Open "Find a specialist" with 3 specialists (1 closed)	"3 Specialists Found", "Not taking bookings", 3 Book buttons of which 1 is disabled	Count text true; "Not taking bookings" true; Book buttons 3, disabled 1	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
2	ATC-M14-F02	The type chip filters the list	Tap the "Nutritionist" chip	"1 Nutritionist Found"; only Sara Khan remains	Count text true; Sara Khan shown 1; Dr. Hina Malik shown 0	**Pass**
3	ATC-M14-F03	Search by specialty finds the trainer	Type "yoga" with the type filter on All	"1 Specialist Found" and Omar Closed	Count text true; Omar Closed shown true	**Pass**
4	ATC-M14-F04	A specialist with no reviews is shown as New and not hidden	One specialist with 0 reviews	"Newly Joined" listed with a "New" tag	Name shown true; "New" tag shown true	**Pass**
5	ATC-M14-F05	A failed load is explained and can be retried	The first load fails; tap "Try again"	Error text, then "3 Specialists Found"	Error shown true; after retry list shown true	**Pass**
6	ATC-M14-F06	Confirm booking stays disabled until a time is chosen, and nothing is shared by default	Open the booking screen and choose nothing	Confirm disabled; "Not chosen yet"; every sharing tick box off	Confirm disabled true; "Not chosen yet" 1; sharing boxes 3, any ticked false	**Pass**
7	ATC-M14-F07	Booking sends exactly the chosen time, type, shares and note, then opens the payment screen	Pick the first time (09:00), "Chat only", share "Diet log, last 7 days", note "Help with a meal plan", tap Confirm	Slot 04:00 UTC, chat, diet log shared only, note kept; the Payment screen opens	Slot 2026-10-19T04:00:00.000Z; type chat; diet log shared true, health summary shared false; chat fee shown true; Payment screen opened true	**Pass**
8	ATC-M14-F08	A time someone else just took is refused and the screen stays	Choose a time; the booking call answers "just taken"	The message "just taken by someone else" and the booking screen still open	Message shown true; still on the booking screen true	**Pass**
9	ATC-M14-F09	A time that is already booked is not offered	The 09:00 slot is booked by someone else	09:00 is not listed, 09:30 is	09:00 listed 0 time(s); 09:30 listed 1 time(s)	**Pass**
10	ATC-M14-F10	The notes box stops at 500 characters	Paste 600 characters into the note	Only 500 characters are kept	Characters kept: 500	**Pass**
11	ATC-M14-F11	My appointments separates upcoming from past and shows the specialist's notes only under Past	One upcoming and one completed appointment; open the Past tab	"Upcoming (1)"; notes hidden first, visible under Past with "Completed"	Upcoming tab true; notes before 0, after 1; "Completed" 1	**Pass**
12	ATC-M14-F12	A request shows as awaiting acceptance, has no video link and can be cancelled	Appointment with status "requested"	"Awaiting acceptance", "Waiting for your specialist", no meeting room, a Cancel request button	Awaiting label true; waiting text true; meeting room shown 0; cancel button true	**Pass**
13	ATC-M14-F13	A declined request shows the reason and offers no cancel	Appointment declined with reason "On leave that day"	"Not accepted" and the reason; no Cancel appointment button	"Not accepted" true; reason shown true; cancel button 0	**Pass**
14	ATC-M14-F14	A confirmed appointment shows the meeting link and the join window rule	Confirmed appointment tomorrow	Meeting room, the link and "opens 5 minutes before"	Meeting room true; link true; join rule text true	**Pass**
15	ATC-M14-F15	The room is announced as open inside the join window	Confirmed appointment starting in 3 minutes	"The room is open now"	Room-open text shown true	**Pass**
16	ATC-M14-F16	Cancelling asks for confirmation first and then shows the new status	Tap Cancel appointment, choose "Keep it", then cancel again and confirm	Nothing cancelled after "Keep it"; afterwards status "Cancelled" and no cancel button	Confirmation asked true; nothing cancelled after "Keep it" true; cancelled id u1; status shown true; cancel button left 0	**Pass**
17	ATC-M14-F17	An appointment that has already started has no cancel button	Confirmed appointment that started 5 minutes ago	No Cancel appointment button	Cancel button shown 0 time(s)	**Pass**

Functional Testing 11: Notifications screen (Module 16, Secure Data Management & Smart Alerts)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M16-F01	Notifications show unread counts per tab and filter by tab	4 notifications (3 unread); tap the Diet tab	"All (3)" and "Diet (1)"; Diet shows only the meal reminder	"All (3)" true; "Diet (1)" true; meal reminder shown 1; workout reminder shown 0	**Pass**
2	ATC-M16-F02	Mark all read reports exactly the unread items	Tap "Mark all read"	The callback receives 3 unread items	Callback received 4 item(s), 3 unread	**Pass**
3	ATC-M16-F03	A failed load shows an error, not an empty inbox	The loader throws "offline"	"Could not load your notifications." and no "No notifications yet"	Error shown true; empty-inbox text shown 0 time(s)	**Pass**

Functional Testing 12: Analytics screen (Module 17, Analytics & Progress Dashboard)

Objective: to ensure the screen shows the correct content and reacts correctly to the user's actions.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M17-F01	Analytics shows the calorie chart, the weekly score and an honest scope note	Open analytics with 6 weigh-ins	"Calories vs your target", "Weekly fitness score", and the note "not part of this version"	Calorie chart true; weekly score true; scope note true	**Pass**
2	ATC-M17-F02	A failed analytics load offers a retry	The loader throws	"Could not load your analytics." and a "Try again" button	Error shown true; retry button shown true	**Pass**

Functional Testing 13: Access control of the specialist API and admin routes (API)

Objective: To ensure only signed-in specialists reach the portal and only administrators reach the admin routes.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M16-F04	A request without a valid sign-in is turned away	GET /api/me with no token, a garbage token and a Basic scheme	401 three times; /health stays open (200)	No token 401; garbage token 401; Basic scheme 401; /health 200	**Pass**
2	ATC-M16-F05	A member or administrator cannot open the specialist screens even by typing the address	GET /api/dashboard, /api/appointments, /api/appointments/a1 as a member, an administrator and an unknown user	403 for every one of the 9 requests	Status codes: 403, 403, 403, 403, 403, 403, 403, 403, 403	**Pass**
3	ATC-M16-F06	Another specialist's appointment looks like it does not exist	Doctor opens appointment "other" that belongs to the nutritionist	404	Status 404	**Pass**
4	ATC-M16-F07	Only an administrator reaches the admin routes	GET /api/admin/specialists as doctor, nutritionist, member, unknown user and anonymous; and as admin	401 anonymous; 403 for the four others; 200 for the admin	Anonymous 401; doctor/nutritionist/member/unknown 403/403/403/403; administrator 200	**Pass**

Functional Testing 14: Requests, accept and decline, and appointment actions (API)

Objective: To ensure a booked request can be accepted (creating the video room) or declined, expired requests cannot be answered, and notes, prescriptions and cancellations behave correctly.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-F21	The dashboard lists open requests soonest first, hides expired ones and counts them	Doctor opens the dashboard with 2 live requests, 1 expired request and 1 confirmed session	Requests r1 then r2; today shows the confirmed session; 2 pending; no video link on a request	Requests: r1,r2; today: ok; pending 2; link on request ""	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
2	ATC-M14-F22	Accepting a request creates the video room on the server, confirms it and tells the patient	Doctor accepts r1, then accepts it again	200 and status confirmed; a meet.jit.si room; patient notified; slot kept; the second accept is 400	Accept 200 (confirmed); room https://meet.jit.si/baymax-6m8gbshsj784epg7ksgf; patient notice "Your appointment was accepted"; slot kept true; second accept 400	**Pass**
3	ATC-M14-F23	Declining frees the time, keeps the reason and tells the patient	Doctor declines r2 with reason " On leave that day "	Status declined, reason trimmed, slot freed, other slot kept, patient notified, no room made, accepting afterwards is 400	Decline 200; status declined; reason "On leave that day"; slot freed true; other slot kept true; accept afterwards 400	**Pass**
4	ATC-M14-F24	A request whose time has passed is shown as expired and cannot be answered	Doctor lists appointments and tries to accept "late" (start was 30 minutes ago)	phase expired, canRespond false, accept answers 400 "expired"	Phase expired; canRespond false; accept 400 "The start time has passed, so this request expired. The patient can book another time."	**Pass**
5	ATC-M14-F25	Only the specialist the request was sent to can answer it	Doctor answers the nutritionist's request; a member and an anonymous caller answer r1	404, 403 and 401; r1 stays requested	Other specialist's request 404; member 403; anonymous 401; r1 is requested	**Pass**
6	ATC-M14-F26	Notes are saved after the start, the patient is told, and a note before the start is refused	Doctor writes notes for a1 (starts in 60 min) and for a2 (started 10 min ago)	a1: 400 "started"; a2: 200, note and prescription stored, patient notified	Before start 400; after start 200; prescription "Paracetamol 500mg"; patient notified true	**Pass**
7	ATC-M14-F27	Only a doctor can write a prescription	Nutritionist saves notes with a prescription	400 mentioning "doctor"	Status 400: "Only a doctor can write a prescription."	**Pass**
8	ATC-M14-F28	Cancelling frees the slot and tells the patient; a finished appointment cannot be changed again	Doctor cancels a1, then tries to mark it completed	200, slot gone, notice "Appointment cancelled"; then 400 "already cancelled"	Cancel 200; slot freed true; second change 400 "This appointment is already cancelled."	**Pass**

Functional Testing 15: Patients, plans and food-log review (API)

Objective: To ensure specialists see only their own patients and can set, validate and remove diet and workout plans and review shared food logs.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-F29	My patients lists only people with an accepted appointment and supports search	Doctor opens /api/patients, then searches "bil"; a member tries the same	Ayesha Khan and Bilal Ahmed only; search gives Bilal; member gets 403	Patients: Ayesha Khan, Bilal Ahmed; search "bil": Bilal Ahmed; member 403	**Pass**
2	ATC-M14-F30	A nutritionist saves a diet plan that is stored with its author and announced to the patient	PUT /api/patients/pat1/plans/diet with 1900 kcal, 120/210/60 g	200; stored target 1900 with author Sara Khan (nutritionist); patient notified; visible on the record	Status 200; stored target 1900; author Sara Khan (nutritionist); notice "Your diet plan was updated"; record shows 1900	**Pass**
3	ATC-M14-F31	Diet plan numbers are checked and each problem is explained	Targets 300 and 6000 kcal; macros that do not add up to 1900; only some macros; an unknown status; an empty change	Each refused with 400 and a clear message	400 "The calorie target must be a whole number from 800 to 5000." | 400 "The calorie target must be a whole number from 800 to 5000." | 400 "Protein, carbohydrate and fat add up to about 4100 kcal, which differs from the 1900 kcal target by more than 15%. Check the numbers." | 400 "Give protein, carbohydrate and fat together, or none of them." | 400 "Choose "Approve as is" or "Save changes"." | 400 "Change at least one thing, or choose "Approve as is"."	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
4	ATC-M14-F32	A trainer builds a workout plan from the exercise library only and the schedule is validated	Valid 7-day plan; then a plan with an invented exercise, 6 days, no sets, and a workout on a rest day	Valid plan 200 and stored; each bad plan 400	Valid plan 200, 7 days stored; bad plans: 400 "The schedule needs all seven days." | 400 ""made_up_move" is not an exercise in the Baymax library." | 400 "Sets for Sit-to-Stand (Chair Squat) must be a whole number from 1 to 8." | 400 "A rest day cannot have exercises."	**Pass**
5	ATC-M14-F33	A specialist can remove their plan and the patient is told Baymax's plan applies again	Nutritionist sets and then deletes the diet plan; a trainer tries to delete; delete again	Trainer 403; nutritionist 200 and plan gone; patient notified "back to Baymax"; second delete 404	Trainer 403; nutritionist 200; plan gone true; patient told true; second delete 404	**Pass**
6	ATC-M14-F34	A nutritionist approves or flags a day of the shared food log, and a flag needs a reason	PUT diet-reviews: approve 2026-10-19; flag 2026-10-18 with a note; flag without a note	200, 200 with the patient told "flagged", then 400 "why you are flagging"	Approve 200; flag 200, patient told true; flag without reason 400 "Say why you are flagging this day, so the patient knows what to change."	**Pass**

Functional Testing 16: Registration, approval, administration and account closing (API)

Objective: To ensure specialists register as pending, an administrator approves or rejects, administrators add specialists without setting passwords, and closing an account cleans up correctly.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-F35	Registering as a specialist stores a pending application and grants nothing	Signed-in member "newbie" posts the form (and tries to set role=admin and another email)	201 pending; email taken from the account; no role; not listed; licence number not echoed back	Status 201 (pending); stored email newbie@example.com; role undefined; listed false; role seen by the app user	**Pass**
2	ATC-M14-F36	The licence number is required and the form is checked	Registration without a licence number; with a bad type; with a negative fee	Each 400	400 "Enter your professional licence or registration number (3 to 40 letters, digits, spaces, - . /)." | 400 "Type must be one of: doctor, nutritionist, trainer." | 400 "Fees must be whole numbers of PKR between 0 and 1,000,000."	**Pass**
3	ATC-M14-F37	An administrator approves an application: role, verified public profile, and the specialist can sign in	Admin approves "newbie"; the licence number is checked; the new specialist opens the dashboard; approve again	Role doctor; profile verified, no licence published; dashboard 200; second approval 409	Admin sees licence "PMDC-12345"; approve 200; role doctor; verified true; licence published false; new specialist dashboard 200; second approval 409	**Pass**
4	ATC-M14-F38	Rejecting an application records the reason and never creates a role	Admin rejects "newbie" with a 301-character reason, then without a reason	400, then 200; no role	Long reason 400; reject 200; status rejected; role undefined	**Pass**
5	ATC-M14-F39	An administrator adds a specialist without ever setting a password	POST /api/admin/specialists with a new email	201; account created without a password; role and profile set; a link to choose a password is returned	Status 201; created true; role doctor; link returned true	**Pass**
6	ATC-M14-F40	Closing an account needs the word DELETE and a recent sign-in	DELETE /api/me with an empty body; with DELETE but an hour-old sign-in	400; then 401 "reauth-required"; nothing deleted	Empty 400; stale sign-in 401 (reauth-required); role kept doctor	**Pass**
7	ATC-M14-F41	Closing a specialist account cancels future bookings, frees the times, tells patients and keeps past records	Doctor closes the account (DELETE, recent sign-in)	2 bookings cancelled; past record kept; slots gone; patient told; role, profile and sign-in removed; others untouched	Status 200; cancelled 2; future cancelled/cancelled; past completed; slots left false; sign-in deleted doc1; nutritionist kept nutritionist	**Pass**

Functional Testing 17: Real-device UI automation of Modules 4, 5 and 6 (Flutter integration_test, Redmi Note 11 (2201117TG))

Objective: to run the real Baymax screens on the physical phone and operate them automatically (taps, scrolling and navigation through Flutter integration_test), with a controlled in-memory test user. Camera cases use the phone's camera and on-device MediaPipe; voice cases use the phone's speech and text-to-speech engines. These cases do not measure posture-detection or speech-recognition accuracy.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-DEV-S01	The app shell starts on the phone and the bottom navigation opens the Workout tab and returns Home	Controlled test user "Device Tester" (in memory); tap "Workout", then "Home"	Home shows "TODAY'S WORKOUT"; Workout tab shows "Your workout plan"; Home shown again	Home shown (greeting "Device" 1); Workout tab "Your workout plan" 1; returned Home	**Pass**
2	ATC-DEV-M4-01	The Workout tab shows the weekly plan generated for the profile	Test user: healthy, 6 training days a week; open the Workout tab	"Your workout plan", "<intensity> intensity · 6 days a week", 7 day buttons of which 1 is a rest day	Summary "Moderate intensity · 6 days a week" shown 1; day buttons 7; rest-day buttons 1	**Pass**
3	ATC-DEV-M4-02	Selecting a workout day shows that day's session and its exercise count	Workout tab; tap the first training day that is not already selected	Header "<day name> · <focus>" and an "Exercises" count equal to the day's main exercises	Monday: header "Full-Body Fitness · Total body" shown 1; 9 planned entries (5 main); exercise cards on screen 9	**Pass**
4	ATC-DEV-M4-03	An exercise opens its detail screen with the planned sets, reps or time and rest, and Back returns to the plan	Workout tab; tap the first exercise card; then Back	Exercise Detail shows the exercise name and the planned dose; the plan is shown again after Back	"Dynamic Warm-Up" 1 sets × 3m, rest 0 s: name 1, sets 1, reps/time 2, rest 1; back on the plan	**Pass**
5	ATC-DEV-M4-04	Regenerate plan produces a different exercise selection (cloud save unavailable in the test session)	Workout tab; "Regenerate plan" → confirm "Regenerate"	The weekly plan changes; the app reports that the new plan could not be saved to the server (no account in the test)	Plan changed: true; message "New plan generated, but it could not be saved to the server."	**Pass**
6	ATC-DEV-M4-05	The training profile card names the profile of the health classification	Test user with "Hypertension"; Workout tab	"Your training profile: <category>" for the classifier's category	Category At-Risk; card "Your training profile: At-Risk" shown 1	**Pass**
7	ATC-DEV-M4-06	A reported warning symptom blocks plan generation and the reason is shown (negative case)	Test user; readiness answer: chest pain; Workout tab	"A doctor's clearance comes first" and rule SCR-SYMPTOMS; no exercise cards	Gate shown; rule SCR-SYMPTOMS shown 1; exercise cards 0	**Pass**
8	ATC-DEV-M4-07	Without a health profile (no classification) no plan is invented (negative case)	Test session with no profile; Workout tab	"Complete your health profile to get a personalised workout plan."	Empty-state message shown 1; plan null	**Pass**
9	ATC-DEV-M5-01	Posture check opens Posture Detection with Today's workout, My plan and the Exercise library	Home → menu → "Posture check"	Tabs "Today's workout", "My plan", "Exercise library"	Tabs shown: Today's workout, My plan, Exercise library	**Pass**
10	ATC-DEV-M5-02	Today's workout lists today's plan entries (or the rest day) from the personalised plan	Posture Detection → "Today's workout"	One entry per exercise of today's plan day, or the rest-day message when today is a rest day	Today (weekday 6) is a rest day in the plan; rest-day message shown 1	**Pass**
11	ATC-DEV-M5-03	My plan shows every day of the complete plan with each entry's camera capability	Posture Detection → "My plan" → each training day	For every training day: its entries with a capability label	6 training days checked. Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1; Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1	**Pass**
12	ATC-DEV-M5-04	The generic Exercise library keeps its camera and manual sections	Posture Detection → "Exercise library"	"Camera posture detection" and "Manual guidance" sections	Camera section shown; manual section shown 1	**Pass**
13	ATC-DEV-M5-05	A planned camera exercise opens mode selection; Continue without consent does not start the camera	My plan → a camera-tracked planned exercise → Continue with the consent toggle off	Both modes and the consent switch; the consent message; no camera screen	Entry 0-1: modes 2, consent switch 1, consent message shown, camera screen 0	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
14	ATC-DEV-M5-06	With consent the real camera session starts on the phone, and Stop Session ends it with a summary	Planned camera exercise → consent on → Continue → about 8 s of live camera → "Stop Session"	Live camera overlay running, exercise name or setup hint, privacy notice; after Stop: "Session Summary" whose reps equal the session counter; saved in memory only	Blocked: camera permission for the installed test build is "denied" on the phone (a reinstall resets it); the tests do not grant permissions. Allow Camera for Baymax in Android Settings and rerun	**Blocked**
15	ATC-DEV-M5-07	Manual Guidance shows the set counter, checklist and controls, and Pause stops the clock	Planned exercise → Manual Guidance → Continue → Pause → Resume	"Manual Guidance", "Proper Form", "Completed ✓", "Skip", "Too Difficult"; Pause switches to "Resume" and back	Shown: Proper Form, Completed ✓, Skip, Too Difficult; Resume after Pause 1; Pause after Resume 1	**Pass**
16	ATC-DEV-M5-08	A refused camera permission keeps the camera off and offers Manual Guidance (denial simulated in the app, no system dialog)	Camera permission answer "denied" (AppPermissions override); consent on → Continue → "Use Manual Guidance" → Continue	"Camera access needed" dialog; no camera screen; the mode switches to Manual Guidance and Continue opens it	Dialog shown; camera screen 0; mode switched to Manual Guidance; Continue opened Manual Guidance	**Pass**
17	ATC-DEV-M5-09	An exercise the camera cannot follow says why and still offers Manual Guidance	Exercise library → first exercise in "Manual guidance" section	Mode selection explains that camera detection is not available, with the reason; Manual Guidance available	Ankle Circles: reason "Stretches and mobility moves are held in many different positions, so there is no single movement the camera could follow reliably" shown true; Manual Guidance option 1	**Pass**
18	ATC-DEV-M6-01	The chatbot's mic opens the Module 6 voice assistant, which shows the state matching the phone's microphone permission	Home → AI chatbot (Module 13) → "Voice assistant"; microphone permission as set on the phone	Voice assistant screen (not the chatbot); granted → "Listening..."; not granted → "Voice input unavailable" with the reason and "Try again"	Phone microphone permission: denied; assistant state: unavailable; reason shown 1; "Try again" 1; chatbot still on top: 0	**Pass**
19	ATC-DEV-M6-02	A quick-action chip is answered from the real session state and spoken with the phone's text-to-speech	No active workout; tap the chip "How many reps?"	Reply card says there is no active workout session (no invented numbers); text-to-speech reports no failure; at most 3 chips	Reply: "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."; engine On-device rules (Dialogflow unavailable); speech output failed false; chips 2	**Pass**
20	ATC-DEV-M6-03	"What's my workout today?" is answered from the personalised plan	Test user's plan; tap the chip "What's my workout today?"	Reply names today's plan day, or says today is a rest day, matching the plan	Plan today: rest day; reply "Today is a rest day in your plan. Light walking or stretching is fine if you feel like moving."	**Pass**
21	ATC-DEV-M6-04	The close button leaves the voice assistant and the app stays usable	Home → chatbot → voice assistant → "Close voice assistant" → Back	Voice assistant closed; chatbot shown; Back returns to Home	Voice assistant closed; chatbot shown 1; Home shown after Back	**Pass**

5.3 Business Rules Testing

Business Rule Testing 1: DT-M2-1 Health category (reported conditions always override BMI)

Table 5.2 Decision table: DT-M2-1 Health category (reported conditions always override BMI)

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8	Rule 9
Reported condition	None	None	None	None	Hypertension	Diabetes	Diabetes	Heart disease	Heart disease and diabetes
BMI band	Normal	Overweight	Obese	Underweight	Normal	Normal	Obese	Normal	Obese
Action / expected result	Healthy	Overweight	Obese	At-Risk	At-Risk	Diabetic	Diabetic	Cardiac	Cardiac

Table 5.3 Test cases for DT-M2-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M2-B01	Rule 1: Health category: None, BMI Normal	Reported "None"; BMI band Normal (175 cm, 70 kg)	Category "Healthy"	Category = "Healthy"	**Pass**
2	ATC-M2-B02	Rule 2: Health category: None, BMI Overweight	Reported "None"; BMI band Overweight (175 cm, 85 kg)	Category "Overweight"	Category = "Overweight"	**Pass**
3	ATC-M2-B03	Rule 3: Health category: None, BMI Obese	Reported "None"; BMI band Obese (175 cm, 100 kg)	Category "Obese"	Category = "Obese"	**Pass**
4	ATC-M2-B04	Rule 4: Health category: None, BMI Underweight	Reported "None"; BMI band Underweight (175 cm, 50 kg)	Category "At-Risk"	Category = "At-Risk"	**Pass**
5	ATC-M2-B05	Rule 5: Health category: Hypertension, BMI Normal	Reported "Hypertension"; BMI band Normal (175 cm, 70 kg)	Category "At-Risk"	Category = "At-Risk"	**Pass**
6	ATC-M2-B06	Rule 6: Health category: Diabetes, BMI Normal	Reported "Diabetes"; BMI band Normal (175 cm, 70 kg)	Category "Diabetic"	Category = "Diabetic"	**Pass**
7	ATC-M2-B07	Rule 7: Health category: Diabetes, BMI Obese	Reported "Diabetes"; BMI band Obese (175 cm, 100 kg)	Category "Diabetic"	Category = "Diabetic"	**Pass**
8	ATC-M2-B08	Rule 8: Health category: Heart disease, BMI Normal	Reported "Heart disease"; BMI band Normal (175 cm, 70 kg)	Category "Cardiac"	Category = "Cardiac"	**Pass**
9	ATC-M2-B09	Rule 9: Health category: Heart disease and diabetes, BMI Obese	Reported "Heart disease and diabetes"; BMI band Obese (175 cm, 100 kg)	Category "Cardiac"	Category = "Cardiac"	**Pass**

Business Rule Testing 2: DT-M13-1 Which answer comes first

Table 5.4 Decision table: DT-M13-1 Which answer comes first

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Message is about	a possible emergency	an emergency and an exercise	a symptom	BMI	no known topic
Action / expected result	Intent emergency	Intent emergency	Intent health_concern	Intent bmi	Intent fallback

Table 5.5 Test cases for DT-M13-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M13-B01	Rule 1: Chatbot: message about a possible emergency	User types "I have chest pain"	Intent "emergency"	Intent = "emergency"	**Pass**
2	ATC-M13-B02	Rule 2: Chatbot: message about an emergency and an exercise	User types "I have chest pain after squats"	Intent "emergency"	Intent = "emergency"	**Pass**
3	ATC-M13-B03	Rule 3: Chatbot: message about a symptom	User types "my knee hurts after squats"	Intent "health_concern"	Intent = "health_concern"	**Pass**
4	ATC-M13-B04	Rule 4: Chatbot: message about BMI	User types "what is my bmi"	Intent "bmi"	Intent = "bmi"	**Pass**
5	ATC-M13-B05	Rule 5: Chatbot: message about no known topic	User types "asdfgh qwerty"	Intent "fallback"	Intent = "fallback"	**Pass**

Business Rule Testing 3: DT-M14-1 Can this booking go ahead?

Table 5.6 Decision table: DT-M14-1 Can this booking go ahead?

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7
Taking bookings	Yes	No	Yes	Yes	Yes	Yes	Yes
Time chosen	Yes	Yes	No	Yes	Yes	Yes	Yes
Time free	Yes	Yes	-	No	Yes	Yes	Yes
15+ min ahead	Yes	Yes	-	Yes	No	Yes	Yes
Within working hours	Yes	Yes	-	Yes	Yes	No	Yes
Note 500 or less	Yes	Yes	Yes	Yes	Yes	Yes	No
Action / expected result	Booking allowed	Refused: This specialist is not taking bookings right now.	Refused: Choose a date and a time.	Refused: That time was just taken. Please choose another.	Refused: That time is too soon. Please choose a later one.	Refused: That time is outside the specialist's working hours.	Refused: Notes can be at most 500 characters.

Table 5.7 Test cases for DT-M14-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B01	Rule 1: Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours Yes, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: Yes	Booking allowed	Booking allowed	**Pass**
2	ATC-M14-B02	Rule 2: Booking check: Taking bookings No, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours Yes, Note 500 or less Yes	Taking bookings: No; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: Yes	Refused: This specialist is not taking bookings right now.	Refused: This specialist is not taking bookings right now.	**Pass**
3	ATC-M14-B03	Rule 3: Booking check: Taking bookings Yes, Time chosen No, Time free -, 15+ min ahead -, Within working hours -, Note 500 or less Yes	Taking bookings: Yes; Time chosen: No; Time free: -; 15+ min ahead: -; Within working hours: -; Note 500 or less: Yes	Refused: Choose a date and a time.	Refused: Choose a date and a time.	**Pass**
4	ATC-M14-B04	Rule 4: Booking check: Taking bookings Yes, Time chosen Yes, Time free No, 15+ min ahead Yes, Within working hours Yes, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: No; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: Yes	Refused: That time was just taken. Please choose another.	Refused: That time was just taken. Please choose another.	**Pass**
5	ATC-M14-B05	Rule 5: Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead No, Within working hours Yes, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: No; Within working hours: Yes; Note 500 or less: Yes	Refused: That time is too soon. Please choose a later one.	Refused: That time is too soon. Please choose a later one.	**Pass**
6	ATC-M14-B06	Rule 6: Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours No, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: No; Note 500 or less: Yes	Refused: That time is outside the specialist's working hours.	Refused: That time is outside the specialist's working hours.	**Pass**
7	ATC-M14-B07	Rule 7: Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours Yes, Note 500 or less No	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: No	Refused: Notes can be at most 500 characters.	Refused: Notes can be at most 500 characters.	**Pass**

Business Rule Testing 4: DT-M14-2 Join and cancel by status and time

Table 5.8 Decision table: DT-M14-2 Join and cancel by status and time

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Status	Confirmed	Confirmed	Confirmed	Confirmed	Requested	Requested	Declined	Completed
Video link	Yes	Yes	Yes	Yes	No	No	No	Yes
Time	6 minutes before start	5 minutes before start	10 minutes after start	after the slot ended	30 minutes before start	at the start time	30 minutes before start	1 hour after start
Action / expected result	Join no; cancel yes	Join yes; cancel yes	Join yes; cancel no	Join no; cancel no	Join no; cancel yes	Join no; cancel no	Join no; cancel no	Join no; cancel no

Table 5.9 Test cases for DT-M14-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B08	Rule 1: Appointment Confirmed, 6 minutes before start	Status "Confirmed"; video link: Yes; time: 6 minutes before start	Join not allowed; cancel allowed	Join not allowed; cancel allowed	Pass
2	ATC-M14-B09	Rule 2: Appointment Confirmed, 5 minutes before start	Status "Confirmed"; video link: Yes; time: 5 minutes before start	Join allowed; cancel allowed	Join allowed; cancel allowed	Pass
3	ATC-M14-B10	Rule 3: Appointment Confirmed, 10 minutes after start	Status "Confirmed"; video link: Yes; time: 10 minutes after start	Join allowed; cancel not allowed	Join allowed; cancel not allowed	Pass
4	ATC-M14-B11	Rule 4: Appointment Confirmed, after the slot ended	Status "Confirmed"; video link: Yes; time: after the slot ended	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass
5	ATC-M14-B12	Rule 5: Appointment Requested, 30 minutes before start	Status "Requested"; video link: No; time: 30 minutes before start	Join not allowed; cancel allowed	Join not allowed; cancel allowed	Pass
6	ATC-M14-B13	Rule 6: Appointment Requested, at the start time	Status "Requested"; video link: No; time: at the start time	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass
7	ATC-M14-B14	Rule 7: Appointment Declined, 30 minutes before start	Status "Declined"; video link: No; time: 30 minutes before start	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass
8	ATC-M14-B15	Rule 8: Appointment Completed, 1 hour after start	Status "Completed"; video link: Yes; time: 1 hour after start	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass

Business Rule Testing 5: DT-M14-3 Whose diet plan applies

Table 5.10 Decision table: DT-M14-3 Whose diet plan applies

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Specialist plan	None	None	Approved as is	Changed (1800 kcal)	Changed (1800 kcal)
Member's choice	Specialist	Baymax	Specialist	Specialist	Baymax
Action / expected result	Target 2000 kcal	Target 2000 kcal	Target 2000 kcal	Target 1800 kcal	Target 2000 kcal

Table 5.11 Test cases for DT-M14-3

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B16	Rule 1: Diet plan: specialist plan None, member chose Specialist	Baymax target 2000 kcal; specialist plan: None; member chose: Specialist	Daily target 2000 kcal	Daily target 2000 kcal	Pass
2	ATC-M14-B17	Rule 2: Diet plan: specialist plan None, member chose Baymax	Baymax target 2000 kcal; specialist plan: None; member chose: Baymax	Daily target 2000 kcal	Daily target 2000 kcal	Pass

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
3	ATC-M14-B18	Rule 3: Diet plan: specialist plan Approved as is, member chose Specialist	Baymax target 2000 kcal; specialist plan: Approved as is; member chose: Specialist	Daily target 2000 kcal	Daily target 2000 kcal	**Pass**
4	ATC-M14-B19	Rule 4: Diet plan: specialist plan Changed (1800 kcal), member chose Specialist	Baymax target 2000 kcal; specialist plan: Changed (1800 kcal); member chose: Specialist	Daily target 1800 kcal	Daily target 1800 kcal	**Pass**
5	ATC-M14-B20	Rule 5: Diet plan: specialist plan Changed (1800 kcal), member chose Baymax	Baymax target 2000 kcal; specialist plan: Changed (1800 kcal); member chose: Baymax	Daily target 2000 kcal	Daily target 2000 kcal	**Pass**

Business Rule Testing 6: DT-M14-4 May this account set a plan?

Table 5.12 Decision table: DT-M14-4 May this account set a plan?

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8	Rule 9	Rule 10
Account	Nutritionist	Doctor	Trainer	Trainer	Doctor	Nutritionist	Doctor	Doctor	Nutritionist	Member
Plan type	diet	diet	diet	workout	workout	workout	diet	diet	diet	diet
Patient	own patient, recent appointment	own patient, recent appointment	own patient, recent appointment	own patient, recent appointment	own patient, recent appointment	own patient, recent appointment	own patient, appointment 40 days ago	only a pending request	another specialist's patient	own patient, recent appointment
Action / expected result	Status 200	Status 200	Status 403	Status 200	Status 200	Status 403	Status 400	Status 404	Status 404	Status 403

Table 5.13 Test cases for DT-M14-4

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B21	Rule 1: Plan change: Nutritionist, diet plan, own patient, recent appointment	Nutritionist sends a diet plan for own patient, recent appointment	Status 200	Status 200	**Pass**
2	ATC-M14-B22	Rule 2: Plan change: Doctor, diet plan, own patient, recent appointment	Doctor sends a diet plan for own patient, recent appointment	Status 200	Status 200	**Pass**
3	ATC-M14-B23	Rule 3: Plan change: Trainer, diet plan, own patient, recent appointment	Trainer sends a diet plan for own patient, recent appointment	Status 403	Status 403 ("Only doctors and nutritionists can change a diet plan.")	**Pass**
4	ATC-M14-B24	Rule 4: Plan change: Trainer, workout plan, own patient, recent appointment	Trainer sends a workout plan for own patient, recent appointment	Status 200	Status 200	**Pass**
5	ATC-M14-B25	Rule 5: Plan change: Doctor, workout plan, own patient, recent appointment	Doctor sends a workout plan for own patient, recent appointment	Status 200	Status 200	**Pass**
6	ATC-M14-B26	Rule 6: Plan change: Nutritionist, workout plan, own patient, recent appointment	Nutritionist sends a workout plan for own patient, recent appointment	Status 403	Status 403 ("Only doctors and trainers can change a workout plan.")	**Pass**
7	ATC-M14-B27	Rule 7: Plan change: Doctor, diet plan, own patient, appointment 40 days ago	Doctor sends a diet plan for own patient, appointment 40 days ago	Status 400	Status 400 ("You can change a patient's plan from when you accept their appointment until 30 days after it.")	**Pass**
8	ATC-M14-B28	Rule 8: Plan change: Doctor, diet plan, only a pending request	Doctor sends a diet plan for only a pending request	Status 404	Status 404 ("Patient not found.")	**Pass**
9	ATC-M14-B29	Rule 9: Plan change: Nutritionist, diet plan, another specialist's patient	Nutritionist sends a diet plan for another specialist's patient	Status 404	Status 404 ("Patient not found.")	**Pass**
10	ATC-M14-B30	Rule 10: Plan change: Member, diet plan, own patient, recent appointment	Member sends a diet plan for own patient, recent appointment	Status 403	Status 403 ("This area is only for Baymax specialists.")	**Pass**

Business Rule Testing 7: DT-M14-6 Status change by time

Table 5.14 Decision table: DT-M14-6 Status change by time

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
Change to	completed	completed	missed	missed	cancelled	cancelled
Time	1 minute before the start	at the start	9 minutes after the start	10 minutes after the start	10 minutes after the start	30 minutes after the start (slot ended)
Action / expected result	Status 400	Status 200	Status 400	Status 200	Status 200	Status 400

Table 5.15 Test cases for DT-M14-6

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B31	Rule 1: Mark completed, 1 minute before the start	Doctor marks a confirmed appointment as completed 1 minute before the start	Status 400	Status 400 ("The session has not started yet.")	**Pass**
2	ATC-M14-B32	Rule 2: Mark completed, at the start	Doctor marks a confirmed appointment as completed at the start	Status 200	Status 200	**Pass**
3	ATC-M14-B33	Rule 3: Mark missed, 9 minutes after the start	Doctor marks a confirmed appointment as missed 9 minutes after the start	Status 400	Status 400 ("The patient can only be marked as missed 10 minutes after the start time.")	**Pass**
4	ATC-M14-B34	Rule 4: Mark missed, 10 minutes after the start	Doctor marks a confirmed appointment as missed 10 minutes after the start	Status 200	Status 200	**Pass**
5	ATC-M14-B35	Rule 5: Mark cancelled, 10 minutes after the start	Doctor marks a confirmed appointment as cancelled 10 minutes after the start	Status 200	Status 200	**Pass**
6	ATC-M14-B36	Rule 6: Mark cancelled, 30 minutes after the start (slot ended)	Doctor marks a confirmed appointment as cancelled 30 minutes after the start (slot ended)	Status 400	Status 400 ("This appointment has already ended.")	**Pass**

Business Rule Testing 8: DT-M14-7 Answering a request

Table 5.16 Decision table: DT-M14-7 Answering a request

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7
Action	accept	decline	accept	decline	accept	accept	accept
Request is	waiting and in the future	waiting and in the future	past its start time	past its start time	already declined	already confirmed	sent to another specialist
Action / expected result	Status 200	Status 200	Status 400	Status 400	Status 400	Status 400	Status 404

Table 5.17 Test cases for DT-M14-7

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B37	Rule 1: accept a request that is waiting and in the future	Doctor tries to accept a request that is waiting and in the future	Status 200	Status 200	**Pass**
2	ATC-M14-B38	Rule 2: decline a request that is waiting and in the future	Doctor tries to decline a request that is waiting and in the future	Status 200	Status 200	**Pass**
3	ATC-M14-B39	Rule 3: accept a request that is past its start time	Doctor tries to accept a request that is past its start time	Status 400	Status 400 ("The start time has passed, so this request expired. The patient can book another time.")	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
4	ATC-M14-B40	Rule 4: decline a request that is past its start time	Doctor tries to decline a request that is past its start time	Status 400	Status 400 ("The start time has passed, so this request expired. The patient can book another time.")	Pass
5	ATC-M14-B41	Rule 5: accept a request that is already declined	Doctor tries to accept a request that is already declined	Status 400	Status 400 ("This request is already declined.")	Pass
6	ATC-M14-B42	Rule 6: accept a request that is already confirmed	Doctor tries to accept a request that is already confirmed	Status 400	Status 400 ("This request was already accepted.")	Pass
7	ATC-M14-B43	Rule 7: accept a request that is sent to another specialist	Doctor tries to accept a request that is sent to another specialist	Status 404	Status 404 ("Appointment not found.")	Pass

Business Rule Testing 9: DT-M14-8 Registration decisions

Table 5.18 Decision table: DT-M14-8 Registration decisions

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7
Application	pending	pending	approved	approved	holds another role	pending	pending
Who acts	an administrator	an administrator	an administrator	an administrator	an administrator	a doctor	the applicant
Action	approve	reject	approve	reject	approve	approve	approve
Action / expected result	Status 200	Status 200	Status 409	Status 409	Status 409	Status 403	Status 403

Table 5.19 Test cases for DT-M14-8

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B44	Rule 1: Application pending: an administrator chooses approve	Application is pending; an administrator sends "approve"	Status 200	Status 200	Pass
2	ATC-M14-B45	Rule 2: Application pending: an administrator chooses reject	Application is pending; an administrator sends "reject"	Status 200	Status 200	Pass
3	ATC-M14-B46	Rule 3: Application approved: an administrator chooses approve	Application is approved; an administrator sends "approve"	Status 409	Status 409 ("This application was already approved.")	Pass
4	ATC-M14-B47	Rule 4: Application approved: an administrator chooses reject	Application is approved; an administrator sends "reject"	Status 409	Status 409 ("This application was already approved.")	Pass
5	ATC-M14-B48	Rule 5: Application holds another role: an administrator chooses approve	Application is holds another role; an administrator sends "approve"	Status 409	Status 409 ("This account already has the role "trainer".")	Pass
6	ATC-M14-B49	Rule 6: Application pending: a doctor chooses approve	Application is pending; a doctor sends "approve"	Status 403	Status 403 ("This area is only for Baymax specialists.")	Pass
7	ATC-M14-B50	Rule 7: Application pending: the applicant chooses approve	Application is pending; the applicant sends "approve"	Status 403	Status 403 ("This area is only for Baymax specialists.")	Pass

Business Rule Testing 10: DT-M14-9 Closing an account

Table 5.20 Decision table: DT-M14-9 Closing an account

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Typed DELETE	Yes	No	Wrong word	Yes
Signed in	recently	recently	recently	an hour ago

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Action / expected result	Status 200	Status 400	Status 400	Status 401

Table 5.21 Test cases for DT-M14-9

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M14-B51	Rule 1: Close account: confirmation Yes, sign-in recently	DELETE /api/me with Yes; sign-in recently	Status 200	Status 200; role kept false	**Pass**
2	ATC-M14-B52	Rule 2: Close account: confirmation No, sign-in recently	DELETE /api/me with No; sign-in recently	Status 400	Status 400; role kept true	**Pass**
3	ATC-M14-B53	Rule 3: Close account: confirmation Wrong word, sign-in recently	DELETE /api/me with Wrong word; sign-in recently	Status 400	Status 400; role kept true	**Pass**
4	ATC-M14-B54	Rule 4: Close account: confirmation Yes, sign-in an hour ago	DELETE /api/me with Yes; sign-in an hour ago	Status 401	Status 401; role kept true	**Pass**

Business Rule Testing 11: DT-M16-1 Workout reminder due

Table 5.22 Decision table: DT-M16-1 Workout reminder due

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Reminder switched on	Yes	No	Yes	Yes	Yes
Planned today	Yes	Yes	No	Yes	Yes
Workout done	No	No	No	Yes	No
Reminder time passed	Yes	Yes	Yes	Yes	No
Action / expected result	Raise reminder	Nothing	Nothing	Nothing	Nothing

Table 5.23 Test cases for DT-M16-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M16-B01	Rule 1: Workout reminder: switch Yes, planned day Yes, workout done No, time passed Yes	Reminder switch Yes; planned today: Yes; workout done: No; 18:00 reached (now 19:00 or 17:00): Yes	Reminder raised	Reminder raised	**Pass**
2	ATC-M16-B02	Rule 2: Workout reminder: switch No, planned day Yes, workout done No, time passed Yes	Reminder switch No; planned today: Yes; workout done: No; 18:00 reached (now 19:00 or 17:00): Yes	No reminder	No reminder	**Pass**
3	ATC-M16-B03	Rule 3: Workout reminder: switch Yes, planned day No, workout done No, time passed Yes	Reminder switch Yes; planned today: No; workout done: No; 18:00 reached (now 19:00 or 17:00): Yes	No reminder	No reminder	**Pass**
4	ATC-M16-B04	Rule 4: Workout reminder: switch Yes, planned day Yes, workout done Yes, time passed Yes	Reminder switch Yes; planned today: Yes; workout done: Yes; 18:00 reached (now 19:00 or 17:00): Yes	No reminder	No reminder	**Pass**
5	ATC-M16-B05	Rule 5: Workout reminder: switch Yes, planned day Yes, workout done No, time passed No	Reminder switch Yes; planned today: Yes; workout done: No; 18:00 reached (now 19:00 or 17:00): No	No reminder	No reminder	**Pass**

Business Rule Testing 12: DT-M16-2 Meal log reminder due

Table 5.24 Decision table: DT-M16-2 Meal log reminder due

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Reminder switched on	Yes	No	Yes	Yes
Food logged today	No	No	Yes	No
Reminder time passed	Yes	Yes	Yes	No
Action / expected result	Raise reminder	Nothing	Nothing	Nothing

Table 5.25 Test cases for DT-M16-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M16-B06	Rule 1: Meal reminder: switch Yes, food logged No, time passed Yes	Meal reminder switch Yes; food logged today: No; 20:00 reached (now 21:00 or 19:00): Yes	Reminder raised	Reminder raised	**Pass**
2	ATC-M16-B07	Rule 2: Meal reminder: switch No, food logged No, time passed Yes	Meal reminder switch No; food logged today: No; 20:00 reached (now 21:00 or 19:00): Yes	No reminder	No reminder	**Pass**
3	ATC-M16-B08	Rule 3: Meal reminder: switch Yes, food logged Yes, time passed Yes	Meal reminder switch Yes; food logged today: Yes; 20:00 reached (now 21:00 or 19:00): Yes	No reminder	No reminder	**Pass**
4	ATC-M16-B09	Rule 4: Meal reminder: switch Yes, food logged No, time passed No	Meal reminder switch Yes; food logged today: No; 20:00 reached (now 21:00 or 19:00): No	No reminder	No reminder	**Pass**

Business Rule Testing 13: DT-M14-5 Is the patient's shared data shown?

Table 5.26 Decision table: DT-M14-5 Is the patient's shared data shown?

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7
Patient ticked sharing	Yes	No	Yes	Yes	Yes	Yes	Yes
Appointment status	confirmed	confirmed	confirmed	confirmed	confirmed	requested	cancelled
Doctor opens it	20 minutes before the start	20 minutes before the start	40 minutes before the start	10 minutes after the start	95 minutes after the start (slot ended + 65 min)	20 minutes before the start	20 minutes before the start
Action / expected result	Show and log	Hide	Hide	Show and log	Hide	Hide	Hide

Table 5.27 Test cases for DT-M14-5

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M16-B10	Rule 1: Shared data: patient ticked Yes, confirmed, 20 minutes before the start	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 20 minutes before the start	Shared data shown and the access logged	Shared data shown, access logged true	**Pass**
2	ATC-M16-B11	Rule 2: Shared data: patient ticked No, confirmed, 20 minutes before the start	Patient ticked sharing: No; appointment confirmed; doctor opens it 20 minutes before the start	Shared data not shown	Shared data hidden	**Pass**
3	ATC-M16-B12	Rule 3: Shared data: patient ticked Yes, confirmed, 40 minutes before the start	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 40 minutes before the start	Shared data not shown	Shared data hidden ("Shared data opens 30 minutes before the appointment.")	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
4	ATC-M16-B13	Rule 4: Shared data: patient ticked Yes, confirmed, 10 minutes after the start	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 10 minutes after the start	Shared data shown and the access logged	Shared data shown, access logged true	Pass
5	ATC-M16-B14	Rule 5: Shared data: patient ticked Yes, confirmed, 95 minutes after the start (slot ended + 65 min)	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 95 minutes after the start (slot ended + 65 min)	Shared data not shown	Shared data hidden ("The viewing period for this appointment has ended.")	Pass
6	ATC-M16-B15	Rule 6: Shared data: patient ticked Yes, requested, 20 minutes before the start	Patient ticked sharing: Yes; appointment requested; doctor opens it 20 minutes before the start	Shared data not shown	Shared data hidden ("Accept the request to see what the patient chose to share. It opens 30 minutes before the appointment.")	Pass
7	ATC-M16-B16	Rule 7: Shared data: patient ticked Yes, cancelled, 20 minutes before the start	Patient ticked sharing: Yes; appointment cancelled; doctor opens it 20 minutes before the start	Shared data not shown	Shared data hidden ("This appointment was cancelled.")	Pass

Business Rule Testing 14: DT-M4-1 Screening gate (may a plan be generated?)

Table 5.33 Decision table: DT-M4-1 Screening gate (may a plan be generated?)

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Warning symptoms	No	Yes	No	No	No
Heart condition	No	No	Yes	Yes	No
Doctor's clearance	Not needed	Not needed	No	Yes	Not needed
Weight recorded	Yes	Yes	Yes	Yes	No
Action / expected result	Generate plan	Block plan: needsClearance (SCR-SYMPTOMS)	Block plan: needsClearance (SCR-CARDIAC-CLEARANCE)	Generate plan	Block plan: needsInfo (SCR-MISSING-WEIGHT)

Table 5.34 Test cases for DT-M4-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-B01	Rule 1: Screening gate: symptoms No, heart condition No, clearance Not needed, profile data Yes	Conditions none; symptoms none; clearance null; weight 70 kg	Plan generated	Gate eligible; workout days 4	Pass
2	ATC-M4-B02	Rule 2: Screening gate: symptoms Yes, heart condition No, clearance Not needed, profile data Yes	Conditions none; symptoms chest_pain; clearance null; weight 70 kg	No plan; gate needsClearance (SCR-SYMPTOMS)	Gate needsClearance (SCR-SYMPTOMS); workout days 0	Pass
3	ATC-M4-B03	Rule 3: Screening gate: symptoms No, heart condition Yes, clearance No, profile data Yes	Conditions Heart Disease; symptoms none; clearance false; weight 70 kg	No plan; gate needsClearance (SCR-CARDIAC-CLEARANCE)	Gate needsClearance (SCR-CARDIAC-CLEARANCE); workout days 0	Pass
4	ATC-M4-B04	Rule 4: Screening gate: symptoms No, heart condition Yes, clearance Yes, profile data Yes	Conditions Heart Disease; symptoms none; clearance true; weight 70 kg	Plan generated	Gate eligible; workout days 4	Pass
5	ATC-M4-B05	Rule 5: Screening gate: symptoms No, heart condition No, clearance Not needed, profile data No	Conditions none; symptoms none; clearance null; weight 0 kg	No plan; gate needsInfo (SCR-MISSING-WEIGHT)	Gate needsInfo (SCR-MISSING-WEIGHT); workout days 0	Pass

Business Rule Testing 15: DT-M4-2 Exercise eligibility (rules checked in order: impact, tags, intensity, injured joint)

Table 5.35 Decision table: DT-M4-2 Exercise eligibility (rules checked in order: impact, tags, intensity, injured joint)

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
Health profile	Healthy	Hypertension	Obese	Healthy	Healthy	Hypertension
Exercise impact	Low	Low	High	High	Low	Low
Restricted tag	None	HIIT	None	None	None	None
Injured joint loaded	No	No	No	Knee	Knee	No
MET recorded	Yes	Yes	Yes	Yes	Yes	No
Action / expected result	eligible	excluded	excluded	excluded	eligibleWithModifications	requiresInformation

Table 5.36 Test cases for DT-M4-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-B06	Rule 1: Exercise eligibility: Healthy, impact Low, tag None, injury No, MET Yes	Profile Healthy; exercise impact low, tags none, MET 4.0; injuries none	Outcome eligible	Outcome eligible	**Pass**
2	ATC-M4-B07	Rule 2: Exercise eligibility: Hypertension, impact Low, tag HIIT, injury No, MET Yes	Profile At-Risk; exercise impact low, tags hiit, MET 4.0; injuries none	Outcome excluded	Outcome excluded; rules MED-HTN	**Pass**
3	ATC-M4-B08	Rule 3: Exercise eligibility: Obese, impact High, tag None, injury No, MET Yes	Profile Obese; exercise impact high, tags none, MET 4.0; injuries none	Outcome excluded	Outcome excluded; rules BODY-OBESE	**Pass**
4	ATC-M4-B09	Rule 4: Exercise eligibility: Healthy, impact High, tag None, injury Knee, MET Yes	Profile Healthy; exercise impact high, tags none, MET 4.0; injuries knee	Outcome excluded	Outcome excluded; rules INJ-KNEE	**Pass**
5	ATC-M4-B10	Rule 5: Exercise eligibility: Healthy, impact Low, tag None, injury Knee, MET Yes	Profile Healthy; exercise impact low, tags none, MET 4.0; injuries knee	Outcome eligibleWithModifications	Outcome eligibleWithModifications; rules INJ-KNEE	**Pass**
6	ATC-M4-B11	Rule 6: Exercise eligibility: Hypertension, impact Low, tag None, injury No, MET No	Profile At-Risk; exercise impact low, tags none, MET 0.0; injuries none	Outcome requiresInformation	Outcome requiresInformation; rules PRF-AT-RISK	**Pass**

Business Rule Testing 16: DT-M4-3 Equipment availability (dataset exercises)

Table 5.37 Decision table: DT-M4-3 Equipment availability (dataset exercises)

Conditions	Rule 1	Rule 2	Rule 3
Equipment owned	Dumbbell	None	Dumbbell
Exercise needs	Dumbbell	Dumbbell	Barbell
Action / expected result	Offer	Do not offer	Do not offer

Table 5.38 Test cases for DT-M4-3

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-B12	Rule 1: Equipment rule: owns Dumbbell, exercise needs Dumbbell	Owned equipment dumbbell; exercise Dumbbell_Bicep_Curl	Offered for planning	Offered: true	**Pass**
2	ATC-M4-B13	Rule 2: Equipment rule: owns None, exercise needs Dumbbell	Owned equipment none; exercise Dumbbell_Bicep_Curl	Not offered	Offered: false	**Pass**
3	ATC-M4-B14	Rule 3: Equipment rule: owns Dumbbell, exercise needs Barbell	Owned equipment dumbbell; exercise Barbell_Squat	Not offered	Offered: false	**Pass**

Business Rule Testing 17: DT-M5-1 Camera capability level → features offered

Table 5.39 Decision table: DT-M5-1 Camera capability level → features offered

Conditions	Rule 1	Rule 2	Rule 3
Capability level	validatedForm	experimentalForm	notSupported
Action / expected result	Camera; validated posture score and corrections	Camera; corrections, score not labelled validated	Manual Guidance only, reason shown

Table 5.40 Test cases for DT-M5-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-B01	Rule 1: Camera capability of squat: validatedForm	Exercise squat	Camera available; form checks yes; score labelled validated yes	Level validatedForm; camera true; form true; validated score true	**Pass**
2	ATC-M5-B02	Rule 2: Camera capability of Barbell_Squat: experimentalForm	Exercise Barbell_Squat	Camera available; form checks yes; score labelled validated no	Level experimentalForm; camera true; form true; validated score false	**Pass**
3	ATC-M5-B03	Rule 3: Camera capability of Standing_Calf_Raises: notSupported	Exercise Standing_Calf_Raises	Camera not available; form checks no; score labelled validated no	Level notSupported; camera false; form false; validated score false; reason "The movement is too small for the camera to measure reliably (for example wrists, calves, shoulder shrugs or a crunch)."	**Pass**

Business Rule Testing 18: DT-M5-2 Repetition acceptance

Table 5.41 Decision table: DT-M5-2 Repetition acceptance

Conditions	Rule 1	Rule 2	Rule 3
Turn-around depth reached	Yes	No	Yes
Duration ≥ 600 ms	Yes	Yes	No
Action / expected result	Count rep	Partial attempt (lowers score)	Ignore as noise

Table 5.42 Test cases for DT-M5-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-B04	Rule 1: Rep acceptance: depth reached Yes, duration ≥ minimum Yes	One squat (to 88°, 1.8 s)	1 rep, 0 partial	Reps 1; partial 0	**Pass**
2	ATC-M5-B05	Rule 2: Rep acceptance: depth reached No, duration ≥ minimum Yes	One squat (turning at 125°, 1.8 s)	0 rep, 1 partial	Reps 0; partial 1	**Pass**
3	ATC-M5-B06	Rule 3: Rep acceptance: depth reached Yes, duration ≥ minimum No	One squat (to 88°, 0.3 s)	0 rep, 0 partial	Reps 0; partial 0	**Pass**

Business Rule Testing 19: DT-M5-3 Landmark visibility

Table 5.43 Decision table: DT-M5-3 Landmark visibility

Conditions	Rule 1	Rule 2
Required landmarks visible (≥ 0.5)	Yes	No
Action / expected result	Measure joint angles	Skip frame, ask user to move into view

Table 5.44 Test cases for DT-M5-3

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-B07	Rule 1: Landmark visibility 0.9 (threshold 0.5)	Squat frame at 120°, landmark visibility 0.9	Angle measured	Measured 120°	**Pass**
2	ATC-M5-B08	Rule 2: Landmark visibility 0.3 (threshold 0.5)	Squat frame at 120°, landmark visibility 0.3	No measurement; nothing counted	Measurement: none	**Pass**

Business Rule Testing 20: DT-M5-4 Correction feedback trigger

Table 5.45 Decision table: DT-M5-4 Correction feedback trigger

Conditions	Rule 1	Rule 2	Rule 3
Form fault present	No	Yes	Yes
Persists ≥ 3 frames	n/a	No	Yes
Action / expected result	No correction	No correction	Show and speak correction

Table 5.46 Test cases for DT-M5-4

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-B09	Rule 1: Correction trigger: fault No, persists ≥ 3 frames n/a	Descent with 10° forward lean for 600 ms (limit 60°)	No correction	Frames with the correction: 0 of 45	**Pass**
2	ATC-M5-B10	Rule 2: Correction trigger: fault Yes, persists ≥ 3 frames No	Descent with 75° forward lean for 66 ms (limit 60°)	No correction	Frames with the correction: 0 of 29	**Pass**
3	ATC-M5-B11	Rule 3: Correction trigger: fault Yes, persists ≥ 3 frames Yes	Descent with 75° forward lean for 600 ms (limit 60°)	"Keep your back straight" shown	Frames with the correction: 16 of 45	**Pass**

Business Rule Testing 21: DT-M5-5 Planned exercise completion

Table 5.47 Decision table: DT-M5-5 Planned exercise completion

Conditions	Rule 1	Rule 2	Rule 3
All planned sets done	Yes	No	Yes
Skipped	No	No	Yes
Action / expected result	Mark done today	Not marked done	Not marked done

Table 5.48 Test cases for DT-M5-5

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-B12	Rule 1: Plan completion: all sets Yes, skipped No	Manual squat session, 2 sets planned; 2 set(s) recorded; Finish	completedTarget true	completedTarget true	Pass
2	ATC-M5-B13	Rule 2: Plan completion: all sets No, skipped No	Manual squat session, 2 sets planned; 1 set(s) recorded; Finish	completedTarget false	completedTarget false	Pass
3	ATC-M5-B14	Rule 3: Plan completion: all sets Yes, skipped Yes	Manual squat session, 2 sets planned; 2 set(s) recorded; Skip	completedTarget false	completedTarget false	Pass

Business Rule Testing 22: DT-M6-1 Voice session control

Table 5.49 Decision table: DT-M6-1 Voice session control

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Command	pause workout	resume workout	pause workout	my knee hurts
Session state	active	paused	none	active
Action / expected result	Session paused; reply "Paused…"	Session active; reply "Resuming…"	Session idle; reply "no active workout session…"	Session paused; reply "Please stop the exercise now…"

Table 5.50 Test cases for DT-M6-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-B01	Rule 1: Voice command "pause workout" with session active	Recognised text "pause workout"; session active	Session paused; reply contains "Paused"	Session paused; reply "Paused. Say resume when you are ready."	Pass
2	ATC-M6-B02	Rule 2: Voice command "resume workout" with session paused	Recognised text "resume workout"; session paused	Session active; reply contains "Resuming"	Session active; reply "Resuming. Let's go."	Pass
3	ATC-M6-B03	Rule 3: Voice command "pause workout" with session none	Recognised text "pause workout"; session none	Session idle; reply contains "no active workout session"	Session idle; reply "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."	Pass
4	ATC-M6-B04	Rule 4: Voice command "my knee hurts" with session active	Recognised text "my knee hurts"; session active	Session paused; reply contains "Please stop the exercise now"	Session paused; reply "Please stop the exercise now and rest. Pain, dizziness or chest discomfort are signals to stop, not to push through. I can't diagnose injuries — if it is severe or doesn't settle, contact a doctor or emergency services. I've paused your session."	Pass

Business Rule Testing 23: DT-M6-2 Microphone permission

Table 5.51 Decision table: DT-M6-2 Microphone permission

Conditions	Rule 1	Rule 2	Rule 3
Microphone permission	granted	denied	permanently denied
Action / expected result	Listen	Explain, offer Try again	Explain, offer Open Settings

Table 5.52 Test cases for DT-M6-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-B05	Rule 1: Microphone permission granted	Assistant opened; microphone granted	State listening; settings link false	State listening; settings link false; listen calls 1	Pass
2	ATC-M6-B06	Rule 2: Microphone permission denied	Assistant opened; microphone denied	State unavailable; settings link false	State unavailable; settings link false; listen calls 0	Pass
3	ATC-M6-B07	Rule 3: Microphone permission permanently denied	Assistant opened; microphone permanently denied	State unavailable; settings link true	State unavailable; settings link true; listen calls 0	Pass

Business Rule Testing 24: DT-M6-3 Speech recognition result (simulated recogniser)

Table 5.53 Decision table: DT-M6-3 Speech recognition result (simulated recogniser)

Conditions	Rule 1	Rule 2	Rule 3
Recognition result	final text "how many reps"	no match	timeout (silence)
Action / expected result	Answer and speak	Show retry message	Show retry message

Table 5.54 Test cases for DT-M6-3

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-B08	Rule 1: Recognition result: final text "how many reps"	Simulated recogniser returns final text "how many reps"	Reply shown and spoken	Reply "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."; spoken 1	Pass
2	ATC-M6-B09	Rule 2: Recognition result: no match	Simulated recogniser returns no match	Message "didn't catch that"; nothing spoken	Message "I didn't catch that. Tap the mic and try again."; spoken 0	Pass
3	ATC-M6-B10	Rule 3: Recognition result: timeout (silence)	Simulated recogniser returns timeout (silence)	Message "didn't hear anything"; nothing spoken	Message "I didn't hear anything. Tap the mic and try again."; spoken 0	Pass

Business Rule Testing 25: DT-M6-4 Spoken reply

Table 5.55 Decision table: DT-M6-4 Spoken reply

Conditions	Rule 1	Rule 2
Speech output works	Yes	No
Action / expected result	Show and speak	Show reply, flag that it was not spoken

Table 5.56 Test cases for DT-M6-4

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-B11	Rule 1: Text-to-speech works	"hello"; speech output available	Reply shown and spoken	Reply shown true; output failed false	Pass
2	ATC-M6-B12	Rule 2: Text-to-speech fails	"hello"; speech output fails	Reply still shown; failure flagged	Reply shown true; output failed true	Pass

5.4 Integration Testing

Integration Testing 1: Cross-module scenarios inside the app(Modules 2, 4, 8, 9, 12, 13, 14, 16, 17)

Testing Objective: to ensure data flows correctly between the classification, plan, diet, gamification, chatbot, consultation and alert modules.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-01	A diabetic user with a dairy allergy gets a safe plan from profile to meals (M2-FR10, M4-FR1, M9-FR6)	Female, 165 cm, 72 kg, Diabetes, Dairy allergy, goal Lose Weight	Category Diabetic; meals have no dairy, no high-sugar or refined food; macros agree with the calorie target	Category Diabetic; 11 meal items; dairy items 0; high-sugar or refined items 0; macros give 1495 kcal against a target of 1494	**Pass**
2	ATC-INT-02	A cardiac user gets a low-intensity plan and a calorie target above the safe floor (M2-FR10, M4-FR3, M9-FR1)	Male, 165 cm, 100 kg, Heart Disease, goal Lose Weight, asks for 6 days	Category Cardiac; intensity Low; at most 4 workout days; no HIIT; target not below 1500 kcal for a man	Category Cardiac; intensity Low; 4 workout days; 0 HIIT exercises; target 2107 kcal	**Pass**
3	ATC-INT-03	Typed meals become a daily summary with the correct totals and a fair verdict (M8-FR8, M8-FR13)	Breakfast "2 rotis", Lunch "a bowl of daal"; burned 150 kcal; goal Lose Weight; target 1800; BMR 1400	Consumed 450, net 300, verdict not claiming a surplus	Consumed 450 kcal (Breakfast 220, Lunch 230); net 300; verdict: "That is a very low intake for you. If you skipped meals or did not log them, add them; if not, eat enough to fuel your body."	**Pass**
4	ATC-INT-04	A week of activity earns a badge and raises no inactivity alert; a long gap raises one (M12-FR6, M12-FR8, M16-FR5)	7 consecutive active days; then no activity for 6 days with a plan for Mon/Wed/Fri	7-day streak badge earned and no "We miss you" notice; with the gap, a "We miss you" notice	Busy week: streak 7, badge daily_7 true, inactivity alert raised false; idle week: alert raised true	**Pass**
5	ATC-INT-05	A booking is checked, paid through the test gateway and becomes a cancellable request without a video link (M14-FR2)	Valid slot on Monday 10:00, video, PKR 2,000 + PKR 150 platform fee, card 4242 4242 4242 4242	No booking problem; total PKR 2,150; payment succeeds with a reference; request is upcoming, cancellable, and cannot be joined yet	Booking problem: none; Pay enabled true; total PKR 2150; payment succeeded (reference sbx_c_3js4swwr2zra3f); upcoming true; cancellable true; joinable at the start false	**Pass**
6	ATC-INT-06	A specialist's diet plan changes the calorie target but never puts an allergen back into the meals (M14-FR5, M9-FR6)	Baymax plan for a user allergic to Dairy and Nuts; nutritionist sets 1800 kcal	Target 1800 kcal; meals identical to Baymax's; still no allergen	Target 1800 kcal; meals unchanged true; allergen items 0 of 11	**Pass**
7	ATC-INT-07	The chatbot answers calories left from the user's own plan and food log (M13-FR3)	Plan target for a Lose Weight user; 600 kcal logged; "how many calories left"	The answer contains target minus 600	Target 1494 kcal; expected "894" in the answer; answer starts: "You've eaten 600 of 1494 kcal today — 894 kcal left."	**Pass**

Integration Testing 2: Module 2 health classification to Module 4 workout plan

Testing Objective: to ensure the health category and conditions from Module 2 set the restrictions and intensity of the generated plan.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-34	The Module 2 health category sets the intensity ceiling of the generated plan	Profile with "Heart Disease" (classified Cardiac), clearance given	Every scheduled exercise with a MET value is light or moderate	Category Cardiac; 36 exercises checked; vigorous 0	**Pass**
2	ATC-INT-35	A condition recorded in Module 2 removes restricted exercises from the Module 4 plan	Profile with "Hypertension"	No HIIT- or plyometric-tagged exercise in the week	44 planned entries; restricted entries: none	**Pass**

Integration Testing 3: Module 4 workout plan and Module 5 posture sessions (both directions)

Testing Objective: to ensure a planned exercise starts a posture session with its planned dose, and that only a session that reached its target completes the planned exercise and the plan day.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-36	A planned exercise starts a posture session with the planned dose and a camera configuration	Generated plan, Monday; every entry converted to a session	Sets, reps and seconds carried over; every entry resolves a camera capability	9 entries; dose mismatches 0; 4 with camera tracking, 5 Manual Guidance only	**Pass**
2	ATC-INT-37	Completing every main exercise of the day marks the plan day done once	Monday plan; each main entry finished in Manual Guidance with all sets	Day marked once in the Module 4 activity record	Marked 1× (2026-10-19 Full-Body Fitness); done today 5 entries	**Pass**
3	ATC-INT-38	A skipped or incomplete session does not complete the planned exercise	Monday entry 1: one set of several, then skipped	Entry not done; day not marked	Entry done false; day marks 0	**Pass**

Integration Testing 4: Module 5 posture detection and Module 6 voice assistant (both directions)

Testing Objective: to ensure live posture results reach the voice coach (spoken corrections, form and rep answers) and voice commands control the posture session. Pose frames, microphone and speaker are simulated.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-39	A posture correction from the camera engine is spoken once by the cue announcer	Camera squat session; about 3 s at the bottom with 75° forward lean (limit 60°)	"Keep your back straight" spoken exactly once (rate-limited)	Spoken 1: Keep your back straight — lift your chest	**Pass**
2	ATC-INT-40	Asking "how's my form" answers from the live posture analysis	Same faulty descent; then "How's my form?"	Reply "Here is what to fix: Keep your back straight…"	Reply "Here is what to fix: Keep your back straight — lift your chest."	**Pass**
3	ATC-INT-41	The rep count spoken by the assistant equals the camera counter	3 squats counted by the camera engine; "how many reps"	Reply states 3 reps	Camera counter 3; reply "You've completed 3 reps of 36, 33 to go."	**Pass**
4	ATC-INT-42	A spoken "pause" stops rep counting until "resume"	Camera session; "pause workout"; 2 squats while paused; "resume workout"; 1 squat	Paused then active; frames while paused ignored	After pause: paused; reps shown while paused 0; after resume: active	**Pass**
5	ATC-INT-43	Ending a session by voice needs confirmation before the session is asked to finish	"stop workout", then "yes"	Confirmation asked first, session still running; after "yes" a finish is requested from the session screen	After "stop": still active true, asked "Do you want to end this Bodyweight Squat session?"; after "yes": finish requested true, active true	**Pass**

Integration Testing 5: End-to-end Modules 2, 4, 5 and 6

Testing Objective: to ensure a profile produces a plan whose exercises can be done as sessions, queried by voice and recorded as a completed plan day.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-44	Profile → plan → posture sessions with voice queries → plan day completed	Healthy profile; Monday plan; each main entry done in Manual Guidance; "how many reps" asked in each	Voice replies match the logged reps; Monday marked done once; every session saved	5 sessions; voice reply mismatches 0; saved 5; day marked 1×	**Pass**

Integration Testing 6: App, API and admin scripts: the whole consultation flow

Testing Objective: to ensure the administrator set-up, specialist registration and approval, booking by the phone, accepting a request, plans, food-log review, account closing and deletion work together on real emulated Auth and Database.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-10	Nobody is an administrator until the one-time command is run on the server (M16-FR2)	Call the admin API with an ordinary account, run the one-time make-admin command on the server, then call again	Refused before the command, accepted after it	All checks in the scenario held (0.2 s)	**Pass**
2	ATC-INT-11	A doctor registers; they get nothing until the administrator approves (M14-FR11)	A doctor registers through the API, tries the specialist screens, the administrator approves, the doctor tries again	No access while pending; access to the specialist API after approval	All checks in the scenario held (0.0 s)	**Pass**
3	ATC-INT-12	The administrator adds a nutritionist directly (no password is ever set by the administrator) (M14-FR15)	The administrator adds a nutritionist by email	Account created without any password; role and public profile created	All checks in the scenario held (0.0 s)	**Pass**
4	ATC-INT-13	The member (the phone) sees the specialists, holds a time, pays, and it becomes a request (M14-FR2)	A signed-in member (the phone) lists specialists, holds a time, pays	The booking becomes a request with the payment recorded and the time kept	All checks in the scenario held (0.0 s)	**Pass**
5	ATC-INT-14	The doctor sees the request, not the patient's data, and accepting creates the video room (M14-FR3, M14-FR12)	The doctor opens the request, then accepts it	Request visible without the patient's data; after accepting a video room exists and the member is told	All checks in the scenario held (0.0 s)	**Pass**
6	ATC-INT-15	My patients: the doctor now has a patient, and a record to work in (M14-FR13)	The doctor opens My patients and the patient's record	The patient is listed and the record opens	All checks in the scenario held (0.0 s)	**Pass**
7	ATC-INT-16	The doctor sets a diet and a workout plan; the member can read them but never write them (M14-FR5)	The doctor sets a diet plan and a workout plan; the member then reads and tries to write them	Member can read both but the database refuses any write	All checks in the scenario held (0.0 s)	**Pass**
8	ATC-INT-17	The nutritionist reviews a day of the food log (inside the sharing window); the member sees the verdict (M14-FR9)	The nutritionist reviews a day of the shared food log inside the sharing window	The verdict is stored and the member sees it	All checks in the scenario held (0.0 s)	**Pass**
9	ATC-INT-18	A specialist closes their account: future bookings are cancelled and freed, the patient is told, the sign-in is deleted (M14-FR14)	A specialist closes their account	Future bookings cancelled and freed, patient told, sign-in deleted	All checks in the scenario held (0.0 s)	**Pass**
10	ATC-INT-19	The administrator can delete another specialist, but not an administrator or themselves (M14-FR15)	The administrator deletes another specialist, then tries an administrator and themselves	The specialist is deleted; the other two are refused	All checks in the scenario held (0.0 s)	**Pass**

Integration Testing 7: App booking against the real database rules

Testing Objective: to ensure the deployed Realtime Database rules allow exactly what the app and the portal need and refuse everything else, for real signed-in users.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-20	Nothing is readable without signing in, and members cannot read or write the protected nodes (M16-FR2)	Read the database without signing in, then as a member read and write protected nodes	Every read and write refused by the real database rules	All checks in the scenario held (0.3 s)	**Pass**
2	ATC-INT-21	A member can hold a free time, and nobody else can take it while it is held (M14-FR2)	A member holds a free time; a second member tries the same time	First hold succeeds; the second is refused	All checks in the scenario held (0.1 s)	**Pass**
3	ATC-INT-22	A hold must look right: own patient id, real fee, a hold of at most ten minutes, a real specialist, a future time (M14-FR2)	Holds with a wrong patient id, a wrong fee, a hold over ten minutes, a made-up specialist and a past time	Every malformed hold refused	All checks in the scenario held (0.0 s)	**Pass**
4	ATC-INT-25	Only the two people involved can read an appointment, and a member cannot touch the specialist's notes or anyone else's booking (M16-FR2)	A third party and the two involved people read an appointment; a member edits notes and another member's booking	Only the two involved can read; edits refused	All checks in the scenario held (0.0 s)	**Pass**
5	ATC-INT-26	Cancelling a held or requested booking frees the time; an expired hold can be taken over (M14-FR2)	Cancel a held and a requested booking; take over an expired hold	Times freed; the expired hold can be taken over	All checks in the scenario held (0.0 s)	**Pass**
6	ATC-INT-27	Once the specialist accepts (portal server), the member keeps reading it but cannot cancel after the start or rewrite it (M14-FR3)	After the specialist accepts, the member reads, tries to cancel after the start and to rewrite the booking	Reading allowed; cancelling after the start and rewriting refused	All checks in the scenario held (0.0 s)	**Pass**
7	ATC-INT-30	Specialist plans and food log reviews: a member reads only their own and can never write them; the plan choice is theirs (M14-FR5, M14-FR9)	A member reads and tries to write specialist plans and reviews; sets the plan choice	Reads only their own; writes refused; the plan choice is theirs	All checks in the scenario held (0.0 s)	**Pass**

Integration Testing 8: Admin account command

Testing Objective: to ensure the command that creates the administrator login saves a working private login, never prints the password and never takes over an existing account.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-31	The saved login signs in and is accepted by the admin API; the file is private and the password is not printed (M16-FR2)	Run create-admin; sign in with the saved login; call the admin API	The login works, the file is private and the password is never printed	All checks in the scenario held (0.2 s)	**Pass**
2	ATC-INT-32	Running it again does not change the password or the file, unless a reset is requested (M16-FR2)	Run create-admin a second time for the same email, then with a reset	Password and file unchanged unless a reset is requested	All checks in the scenario held (0.3 s)	**Pass**
3	ATC-INT-33	An ordinary member who is not an admin cannot use the admin API, and an existing member account is not hijacked (M16-FR2)	An ordinary member calls the admin API; create-admin on an existing member account	Refused; the existing member account is not hijacked	All checks in the scenario held (0.1 s)	**Pass**

Integration Testing 9: Cross-module flows on the real phone (Modules 4, 5 and 6, Flutter integration_test)

Testing Objective: to follow the user journey across modules on the physical phone: plan to posture session to recorded completion (4 → 5 → 4), the voice assistant answering from a live camera session (5 ↔ 6), and a voice command pausing a session (6 → 5).

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-DEV-INT-01	Home → plan → exercise → Manual Guidance session → summary → the planned exercise is recorded as completed (Modules 4 → 5 → 4)	Test user (6 days a week); Workout tab → training day (today, or the first training day when today is a rest day) → its quickest repetition exercise → Start exercise → Manual Guidance → "Completed ✓" for every set (rest timers run) → Finish → Back to Workout Plan	Summary "Confirmed N reps" with N = sets × reps; target reached; session saved with its plan link; plan progress records the entry as done	Monday (today is a rest day): Bodyweight Squat 3×12 (entry 1, plan link 0/1); summary opened by itself after the last set, label "Confirmed 36 reps" 1; target reached true; saved with plan link true (in memory); plan progress done true	**Pass**
2	ATC-DEV-INT-02	During a live camera session the voice assistant answers from that session and the session continues (Modules 5 ↔ 6)	Planned camera exercise → consent → Continue → menu "Ask Baymax (voice)" → chip "How many reps?" → close	Reply states the camera counter's reps for that exercise; after closing, the camera session is still active	Blocked: camera permission for the installed test build is "denied" on the phone (a reinstall resets it); the tests do not grant permissions. Allow Camera for Baymax in Android Settings and rerun	**Blocked**
3	ATC-DEV-INT-03	The voice chip "Pause workout" pauses the running posture session (Module 6 → 5)	Planned exercise → Manual Guidance → "Ask Baymax" → chip "How many reps?" → chip "Pause workout" → close	Session paused; the guidance screen shows "Resume"	Session paused; "Resume" shown 1	**Pass**

5.5 Automated Testing

5.5.1 Automated Testing Strategy

Unit, functional, business-rule (decision-table) and integration tests are automated with Flutter test (app), the Node.js test runner (API) and the Firebase emulators (end-to-end).

For Modules 4 (Workout Plan Generation), 5 (Posture Detection) and 6 (Voice Assistant) the same layers are used: Flutter test for unit, widget (functional), decision-table and integration cases, run headless on the development PC and recorded through ATC_LOG. On top of these, Flutter's official integration_test package runs the real application on a physically connected Android phone: the app is built and installed, and the test code drives the real screens through Flutter's gesture system. Unlike the headless tests, this exercises the real device rendering, the bundled datasets, the camera plugin with on-device MediaPipe, and the speech and text-to-speech engines. A fictional test user is held in memory (nobody is signed in), finished sessions are kept in an in-memory log, and the phone's permission state is read but never changed, so no real account or health record is touched. Real-person posture accuracy is a separate, manual validation (Section 5.5.6). Module 6 (voice assistant) is distinct from the Module 13 AI chatbot, which is only used as an entry point.

5.5.2 Automation Tools and Environment

Table 5.28 Automated Testing Tools and Environment

Tool / Framework	Purpose	Version	Test Type	Platform/Environment
flutter_test (Flutter test framework)	Unit, widget (screen) and cross-module tests of the mobile app	Flutter 3.41.9 • channel stable • https://github.com/flutter/flutter.git	Unit / Functional / Business rules / Integration	Dart, headless on macOS (no phone or emulator)
Node.js built-in test runner (node:test)	API tests of the Express server (functional and business rules) on an in-memory database; runs the end-to-end tests	Node v26.11.0	API / Functional / Business rules	Node.js, macOS
Firebase Emulator Suite (Auth + Realtime Database)	Runs the real database security rules and sign-in for the end-to-end tests	firebase-tools 15.33.0	Integration / End-to-end	Local emulators on 127.0.0.1, project demo-baymax
Firebase JS SDK and Admin SDK	Sign in as real emulated users; seed and inspect data	firebase 13.0.0, firebase-admin 14.5.0	Integration / End-to-end	Node.js
Java runtime	Needed by the database emulator	openjdk version "26" 2026-03-17	-	macOS

Table 5.57 Automation tools and environment added for Modules 4, 5 and 6

Tool / Framework	Purpose	Version	Test Type	Platform/Environment
flutter_test (Flutter test framework)	Unit, widget, decision-table and integration cases of Modules 4, 5 and 6 (headless)	Flutter 3.47.7, Dart 3.13.5 (stable)	Unit / Functional / Business rules / Integration	Windows 11 Home, headless
integration_test (Flutter SDK package) with IntegrationTestWidgetsFlutterBinding	Automated UI tests of the installed app on the physical phone	integration_test from the Flutter 3.47.7 SDK	Functional / Integration (real device)	Redmi Note 11 (2201117TG), Android 13 (API 33), MIUI V816
flutter drive + test_driver/integration_test.dart (integration_test_driver_extended)	Builds, installs and launches the test build; collects results and screenshots on the PC	Flutter 3.47.7 tools	Test runner (real device)	Windows 11, USB
Android Debug Bridge (adb)	Device connection, install, device properties	adb 1.0.41 (platform-tools 37.0.1)	-	Windows 11, USB to the phone
Android SDK / Gradle (via flutter build)	Debug build of the app and test APK	Android SDK 36, build-tools 36.0.0, AGP 8.11.1	-	Windows 11, JDK 21.0.10
On-device components under test	Camera plugin, MediaPipe pose landmarker, speech_to_text, flutter_tts	as locked in pubspec.lock	Functional / Integration (real device)	Redmi Note 11 (2201117TG), Android 13 (API 33), MIUI V816

5.5.3 Automated Test Cases

Table 5.29 Automated Test Cases

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M1-U01	M1-FR5	Email validator accepts a well-formed address	"name@example.com"	Accepted (no error message)	Accepted (no error message)	**Pass**
ATC-M1-U02	M1-FR5	Email validator rejects an address without a domain dot	"a@b"	Rejected with an error message	Rejected with the message "Enter an email like name@example.com."	**Pass**
ATC-M1-U03	M1-FR5	Email validator rejects an empty value	"" (empty string)	Rejected with an error message	Rejected with the message "Enter your email address."	**Pass**

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M1-U04	M1-FR5	Password with 8+ characters, uppercase, number and symbol is accepted	"Abcdef1!"	Accepted (no error message)	Accepted (no error message)	Pass
ATC-M1-U05	M1-FR5	Password without an uppercase letter is rejected	"abcdefg1!"	Rejected with an error message	Rejected with the message "Add an uppercase letter."	Pass
ATC-M1-U06	M1-FR5	Password without a symbol is rejected	"Abcdefg1"	Rejected with an error message	Rejected with the message "Add a symbol such as ! or @."	Pass
ATC-M1-U07	M1-FR5	Password shorter than 8 characters is rejected (boundary)	"Ab1!" (4 characters)	Rejected with an error message	Rejected with the message "Use at least 8 characters."	Pass
ATC-M1-U08	M1-FR5	Confirm-password must match the password	password "Abcdef1!", confirmation "Abcdef1?"	Rejected with an error message	Rejected with the message "The two passwords do not match."	Pass
ATC-M1-U09	M1-FR5	A user who turns 13 tomorrow is too young (boundary)	date of birth 2013-10-16, today 2026-10-15	Rejected with an error message	Rejected with the message "You must be at least 13 years old."	Pass
ATC-M1-U10	M1-FR5	A user who turns 13 today is accepted (boundary)	date of birth 2013-10-15, today 2026-10-15	Accepted (no error message)	Accepted (no error message)	Pass
ATC-M1-U11	M1-FR5	Height below 50 cm is rejected (boundary)	"49" cm	Rejected with an error message	Rejected with the message "Height must be between 50 and 300 cm."	Pass
ATC-M1-U12	M1-FR5	Height of 170 cm is accepted	"170" cm	Accepted (no error message)	Accepted (no error message)	Pass
ATC-M1-U13	M1-FR5	Weight below 20 kg is rejected (boundary)	"19" kg	Rejected with an error message	Rejected with the message "Weight must be between 20 and 500 kg."	Pass
ATC-M1-U14	M1-FR5	The weight range applies in pounds too	"30" lbs (13.6 kg)	Rejected with an error message	Rejected with the message "Weight must be between 44 and 1,100 lbs."	Pass
ATC-M1-U15	M1-FR2	Five failed logins lock the account	five wrong passwords for a@x.com	Account locked after the 5th failure	Not locked after 4 failures; locked for 15 minutes after the 5th	Pass
ATC-M1-U16	M1-FR2	The lock ends after 15 minutes	locked account, 14 min 59 s then 15 min 1 s later	Still locked at 14:59, unlocked at 15:01	At 14:59 locked = true; at 15:01 unlocked = true	Pass
ATC-M1-U17	M1-FR2	A successful login clears the failure count	4 failures, then a successful login, then 1 failure	Counter restarts (4 attempts left)	Attempts left after the next failure: 4	Pass
ATC-M2-U01	M2-FR2	BMI is calculated from height and weight	weight 70 kg, height 175 cm	BMI 22.86	BMI = 22.86	Pass
ATC-M2-U02	M2-FR2	BMI 27 is categorised as Overweight (WHO)	BMI 27	"Overweight"	Category = "Overweight"	Pass
ATC-M2-U03	M2-FR5	BMR (Mifflin-St Jeor) for a man	male, 70 kg, 175 cm, age 30	1648.75 kcal/day	BMR = 1648.75 kcal/day	Pass
ATC-M2-U04	M2-FR5	BMR (Mifflin-St Jeor) for a woman	female, 60 kg, 165 cm, age 25	1345.25 kcal/day	BMR = 1345.25 kcal/day	Pass
ATC-M2-U05	M2-FR2	Feet and pounds are converted before BMI is calculated	height 5.75 ft, weight 154 lbs	BMI about 22.7	175.3 cm, 69.9 kg, BMI = 22.7	Pass
ATC-M2-U06	M2-FR1	Age is calculated from the date of birth (day before the birthday)	born 2000-06-15, today 2026-06-14	25	Age = 25	Pass
ATC-M2-U07	M2-FR10	Normal BMI and no conditions classify as Healthy	male 30, 175 cm, 70 kg, no conditions	"Healthy"	Category = "Healthy"	Pass
ATC-M2-U08	M2-FR10	BMI 27.8 classifies as Overweight	175 cm, 85 kg, no conditions	"Overweight"	Category = "Overweight"	Pass
ATC-M2-U09	M2-FR10	BMI 32.7 classifies as Obese	175 cm, 100 kg, no conditions	"Obese"	Category = "Obese"	Pass
ATC-M2-U10	M2-FR10	Reported heart disease overrides BMI (Cardiac)	100 kg, condition "Heart Disease"	"Cardiac"	Category = "Cardiac"	Pass
ATC-M2-U11	M2-FR10	Reported diabetes classifies as Diabetic	100 kg, condition "Diabetes"	"Diabetic"	Category = "Diabetic"	Pass
ATC-M2-U12	M2-FR10	Heart disease and diabetes together give the more cautious class	conditions "Heart Disease" and "Diabetes"	"Cardiac"	Category = "Cardiac"	Pass
ATC-M2-U13	M2-FR10	Hypertension classifies as At-Risk	condition "Hypertension"	"At-Risk"	Category = "At-Risk"	Pass
ATC-M2-U14	M2-FR10	Underweight BMI (16.3) classifies as At-Risk	175 cm, 50 kg	"At-Risk"	Category = "At-Risk"	Pass
ATC-M2-U15	M2-FR11	Risk level and risk factors are listed	100 kg, Hypertension, Dairy allergy	Risk level High; factors include Hypertension and Dairy allergy	Risk level High; factors: Hypertension, Elevated BMI (32.7), Dairy allergy	Pass
ATC-M2-U16	M2-FR9	The Logistic Regression model never overrides a reported condition	conditions "Heart Disease"; then "Diabetes"	"Cardiac"; "Diabetic"	Results: "Cardiac"; "Diabetic"	Pass
ATC-M2-U17	M2-FR9	The model flags an obese 58-year-old that the rules alone would not	age 58, 175 cm, 117 kg, lightly active, nothing reported	Rules alone: "Obese"; with the model: "At-Risk"	Rules alone: "Obese"; with the model: "At-Risk"	Pass
ATC-M2-U18	M2-FR9	The hybrid is never less cautious than the rules alone	ages 25/45/65 x weights 55/80/110 kg (9 people)	No person gets a lower caution rank	0 of 9 people were classified less cautiously	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M3-U01	M3-FR4	The five moods have the report's labels in order	Mood.values	Happy, Neutral, Sad, Angry, Stressed	Labels: Happy, Neutral, Sad, Angry, Stressed	Pass
ATC-M3-U02	M3-FR4	Every mood has a suggestion and low moods are never pushed to train hard	The suggestion for each mood	All non-empty; Sad is gentle, Stressed is low intensity, Angry is not maximal	Sad: "Be kind to yourself. A short walk or a gentle mobility session is enough today." | Stressed: "Choose a low-intensity mobility or stretching session today and breathe slowly." | Angry: "Movement can help. Try steady, controlled exercise rather than anything maximal."	Pass
ATC-M3-U03	M3-FR3	The calorie estimate on the workout card grows with body weight	The first workout day of a plan at 50 kg and at 100 kg	Positive, and higher at 100 kg than at 50 kg	38 min; 131 kcal at 50 kg; 263 kcal at 100 kg	Pass
ATC-M4-U01	M4-FR1	Cardiac plan has only low-impact exercises and no HIIT or isometric holds	condition "Heart Disease", goal Lose Weight, 6 days requested	Low intensity, at most 4 days, no high-impact / HIIT / isometric exercise	Intensity Low; 4 workout days; 0 restricted exercises	Pass
ATC-M4-U02	M4-FR1	Obese plan contains no high-impact exercise	175 cm, 105 kg, goal Lose Weight, 5 days	Zero high-impact exercises	Category Obese; 0 high-impact exercises	Pass
ATC-M4-U03	M4-FR3	Hypertension keeps isometric holds to 20 seconds or less	condition "Hypertension"	Every timed hold is at most 20 s; no HIIT	Longest isometric hold 20 s; 0 HIIT exercises	Pass
ATC-M4-U04	M4-FR5	Every workout day starts with a warm-up and ends with a cool-down	default healthy profile, 4 days	Warm-up first and cool-down last on every day	4 workout days checked; warm-up first and cool-down last on all = true	Pass
ATC-M4-U05	M4-FR2	Balance training is added from age 65, not before	age 64 and age 66	No balance exercise at 64; at least one at 66	Balance exercises at 64: 0; at 66: 4	Pass
ATC-M4-U06	M4-FR5	Regenerating the plan changes the exercises	plan variant 0 versus variant 1	Different exercise selection	Variant 0 and variant 1 give different exercises	Pass
ATC-M8-U01	M8-FR8	Text entry is parsed into foods and servings	"2 rotis and a bowl of daal"	2 x roti (220 kcal) and 1 x daal	Found 2.0 x roti and 1.0 x daal; roti = 220 kcal	Pass
ATC-M8-U02	M8-FR8	Gram quantities scale by serving weight	"150g grilled chicken breast"	About 248 kcal	Grilled chicken breast: 248.0 kcal	Pass
ATC-M8-U03	M8-FR8	Fractions are understood	"half a cup of rice"	0.5 servings	Servings = 0.5	Pass
ATC-M8-U04	M8-FR8	An unknown food is reported rather than guessed	"2 eggs, toast + green tea and blorptastic"	Four parts; the unknown one has no food	Parts: boiled_egg, white_bread, green_tea, UNKNOWN	Pass
ATC-M8-U05	M8-FR7	Food search finds a food from a partial name	"chick"	Results include grilled chicken	Found: chana_masala, chicken_soup, chicken_curry, chicken_tikka, chicken_karahi	Pass
ATC-M8-U06	M8-FR2	Macro shares always add up to 100%	protein 33.3 g, carbs 66.6 g, fat 12.1 g	Shares sum to 100	26% + 52% + 22% = 100%	Pass
ATC-M8-U07	M8-FR4	Calories are totalled by meal slot	Breakfast 300 + 100 kcal, Lunch 500 kcal	Breakfast 400, Lunch 500, Dinner 0	Breakfast 400.0, Lunch 500.0, Dinner 0.0	Pass
ATC-M8-U08	M8-FR13	Net calories = consumed minus burned	consumed 1800, burned 350	1450	Net = 1450	Pass
ATC-M8-U09	M8-FR13	The daily verdict never praises a very low intake	goal Lose Weight, consumed 600, target 1800, BMR 1500	Mentions "very low intake"	Verdict: "That is a very low intake for you. If you skipped meals or did not log them, add them; if not, eat enough to fuel your body."	Pass
ATC-M8-U10	M8-FR13	The daily verdict recognises a day on target	goal Lose Weight, consumed 1790, target 1800, BMR 1500	Says "right on your daily target"	Verdict: "You are right on your daily target. Nicely balanced."	Pass
ATC-M9-U01	M9-FR2	Protein, carbs and fat add up to the calorie target	goal Lose Weight, 175 cm, 70 kg	Macro calories within 40 kcal of the target	Macros give 2058 kcal; target 2056 kcal	Pass
ATC-M9-U02	M9-FR6	A meal plan never contains the user's allergens	allergies Dairy, Gluten, Eggs, Nuts; 6 plan variants	No allergen in any meal of any variant	60 meal items checked across 6 variants; 0 allergen conflicts	Pass
ATC-M9-U03	M9-FR6	A diabetic meal plan has no high-sugar or refined foods	condition "Diabetes"; 6 plan variants	None of the foods is tagged highSugar or refined	66 items checked; 0 high-sugar or refined foods	Pass
ATC-M9-U04	M9-FR5	Meal calories land near the daily target	175 cm, 75 kg	Total of meals between 75% and 125% of the target	Meals total 2239 kcal; target 2633 kcal	Pass

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ATC-M9-U05	M9-FR1	The calorie deficit never goes below the safe floor	female, BMR 1100, maintenance 1500, BMI 22	Deficit target is 1200 kcal and a safety note is given	Deficit 1200 kcal; note: "Deficit limited to 1200 kcal/day, the minimum safe intake for you."	Pass
ATC-M9-U06	M9-FR1	No calorie deficit is set for an underweight user	female, BMR 1300, maintenance 1800, BMI 17	Deficit target equals maintenance	Deficit 1800, maintenance 1800	Pass
ATC-M9-U07	M9-FR1	Diabetic macros cap carbohydrates at 40% of calories	2000 kcal, 70 kg, diabetic	Carbohydrate share at most 40%	Carbohydrates 200 g = 40.0% of calories	Pass
ATC-M9-U08	M9-FR7	Three nutrition tips a day and none repeats within a week	pool of 28 tips, 7 consecutive days	3 tips per day, 21 different tips in 7 days	Tips per day all 3 = true; different tips in 7 days = 21	Pass
ATC-M12-U01	M12-FR1	A perfect week scores 100	2 planned workouts done, food logged every day on target	Score 100	Score = 100.0	Pass
ATC-M12-U02	M12-FR1	A week with nothing done scores 0	4 planned days, nothing done or logged	Score 0	Score = 0.0	Pass
ATC-M12-U03	M12-FR1	Score parts are weighted 50 / 25 / 25	training 50%, logging 50%, calories on target 50%	Score 50	Score = 50.0	Pass
ATC-M12-U04	M12-FR4	Score thresholds map to Beginner / Intermediate / Advanced (boundary values)	scores 39.9, 40, 69.9, 70	Beginner, Intermediate, Intermediate, Advanced	Levels: beginner, intermediate, intermediate, advanced	Pass
ATC-M12-U05	M12-FR4	K-Means groups a cohort of 15 into three tiers	6 low, 6 mid and 6 high scorers	Low = Beginner, mid = Intermediate, high = Advanced; method is K-Means	Method "K-Means (18 users)"; low0 beginner, mid3 intermediate, high5 advanced	Pass
ATC-M12-U06	M12-FR8	A streak counts consecutive active days and today may still be empty	active, active, active; then active, active, not yet	3; then 2	Streaks: 3 and 2	Pass
ATC-M12-U07	M12-FR6	7-day and 30-day streak badges are earned only at their threshold	6, 7 and 30 active days	6 days: no badge; 7 days: daily_7; 30 days: daily_7 and daily_30	6 days: false; 7 days: true/false; 30 days: true/true	Pass
ATC-M12-U08	M12-FR7	The badge catalogue marks earned badges and keeps locked ones	a 7-day streak	Catalogue of 10 badges, daily_7 earned and others locked	10 badges in the catalogue; 4 earned	Pass
ATC-M12-U09	M12-FR10	XP levels start at 100 x L x (L-1)	XP 0, 199, 200, 599, 600	Levels 1, 1, 2, 2, 3	Levels: 1, 1, 2, 2, 3	Pass
ATC-M13-U01	M13-FR1	Intent is detected from typed text	"what is my bmi"	Intent "bmi"	Intent = "bmi"	Pass
ATC-M13-U02	M13-FR3	Calories left today come from the real food log	"how many calories left" with 600 kcal logged	Answer contains 600	Answer: "You've eaten 600 of 2056 kcal today — 1456 kcal left."	Pass
ATC-M13-U03	M13-FR3	Calorie lookup adds up several foods	"How many calories in 2 rotis and daal?"	Intent calorie_lookup, 450 kcal (220 + 230)	Intent "calorie_lookup"; answer mentions 450 kcal = true	Pass
ATC-M13-U04	M13-FR4	A symptom message recommends a specialist consultation	"my knee hurts after squats"	Intent health_concern with a "Book a consultation" action	Intent "health_concern"; actions: consult	Pass
ATC-M13-U05	M13-FR4	Chest pain is treated as an emergency	"I have chest pain"	Intent emergency and the word "emergency" in the answer	Intent "emergency"	Pass
ATC-M13-U06	M13-FR3	The bot warns when a food is on the user's allergy list	"can I eat butter chicken?" with a Dairy allergy	Answer says it is on the allergy list	Warns about the allergy = true	Pass
ATC-M13-U07	M13-FR2	Nonsense input gets a polite fallback	"asdfgh qwerty"	Intent "fallback"	Intent = "fallback"	Pass
ATC-M14-U01	M14-FR2	Time slots follow the specialist's own clock	Monday 09:00-17:00, 30-minute slots, Pakistan time (UTC+5)	16 slots, first at 04:00 UTC, last at 11:30 UTC	16 slots; first 2026-10-19T04:00:00.000Z; last 2026-10-19T11:30:00.000Z	Pass
ATC-M14-U02	M14-FR2	A day off has no slots	Saturday for a Monday-Friday specialist	No slots	0 slots	Pass
ATC-M14-U03	M14-FR2	Booked times and times that are too soon are not offered	08:50 now; 09:00 too soon, 09:30 booked	First open slot is 10:00; 14 slots offered	First open slot 2026-10-19T05:00:00.000Z; 14 slots	Pass
ATC-M14-U04	M14-FR2	A specialist who is not taking bookings offers nothing	available = false	No open slots	0 slots	Pass
ATC-M14-U05	M14-FR2	A note of exactly 500 characters is accepted (boundary)	note of 500 characters	No booking problem	No problem found	Pass
ATC-M14-U06	M14-FR2	A note of 501 characters is rejected (boundary)	note of 501 characters	Booking problem mentioning 500	Problem: "Notes can be at most 500 characters."	Pass
ATC-M14-U07	M14-FR2	A slot that was just taken is refused	slot 10:00 already booked	Message says the time was just taken	Problem: "That time was just taken. Please choose another."	Pass
ATC-M14-U08	M14-FR1	The specialist list filters by type	doctor, nutritionist and trainer; filter = trainer	Only the trainer	Result: t	Pass
ATC-M14-U09	M14-FR1	Search matches name or specialty, ignoring case	"SPORTS"	The sports nutritionist	Result: n	Pass
ATC-M14-U10	M14-FR1	"Top rated" needs 4.5 stars and at least one review	rating 4.9/80 reviews, rating 4.7/0 reviews, rating 4.2/10 reviews	Only the first	Result: a	Pass

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ATC-M14-U11	M14-FR3	The video room opens five minutes before the start	confirmed appointment; 6 and 5 minutes before the start	Closed at -6 min, open at -5 min	At -6 min can join = false; at -5 min can join = true	Pass
ATC-M14-U12	M14-FR3	There is nothing to join before the specialist accepts	status "requested" (no video link yet)	Cannot join	Can join = false	Pass
ATC-M14-U13	M14-FR2	A request can be cancelled until it starts, but not after	requested appointment; 30 minutes before and at the start	Cancellable before, not at the start	Before the start: true; at the start: false	Pass
ATC-M14-U14	M14-FR2	A request nobody answered before its start time expires	requested appointment, now = start time	Expired and no longer upcoming	Expired = true; upcoming = false	Pass
ATC-M14-U15	M14-FR5	A specialist's diet plan replaces the calorie target and macros and adds advice first	Baymax target 2000 kcal; specialist target 1800 kcal, macros 120/200/55	Target 1800 kcal; avoid list starts with "Fried foods"; Baymax advice kept	Target 1800 kcal; macros 120/200/55; first avoid "Fried foods"; last tip "Walk after meals"	Pass
ATC-M14-U16	M14-FR5	"Approved as is" changes nothing in the plan	specialist status approved	The Baymax plan is returned unchanged	Plan returned unchanged = true	Pass
ATC-M14-U17	M14-FR5	The member's choice to use Baymax's plan is respected	specialist plan present; member choice = Baymax	Calorie target stays 2000	Target = 2000 kcal	Pass
ATC-M14-U18	M14-FR5	A specialist's weekly schedule replaces the week with warm-up and cool-down	Monday walk 900 s; Wednesday squat 3x15; other days rest	2 workout days; warm-up first and cool-down last on each	2 workout days; Monday exercises: warmup > walk > cooldown	Pass
ATC-M16-U01	M16-FR2	An unknown or missing role is treated as a plain member	null; "superuser"; "doctor"	member; member; doctor	Roles: user; user; doctor	Pass
ATC-M16-U02	M16-FR2	A member cannot view assigned patients' data	member role	viewOwnData allowed; viewAssignedPatientData refused	Own data: true; patients' data: false	Pass
ATC-M16-U03	M16-FR2	Only a doctor may upload prescriptions	doctor and trainer roles	Doctor allowed; trainer refused	Doctor: true; trainer: false	Pass
ATC-M16-U04	M16-FR2	An administrator manages accounts but cannot read health data	admin role	manageAccounts allowed; viewAssignedPatientData refused	Manage accounts: true; read patient data: false	Pass
ATC-M16-U05	M16-FR5	Rest days alone are not inactivity	planned Mon/Wed/Fri; last workout Fri 9 Oct; today Mon 12 Oct	No inactivity alert	Alert = none	Pass
ATC-M16-U06	M16-FR5	A missed planned day over three days is inactivity	planned Mon/Wed/Fri; last workout Fri 9 Oct; today Wed 14 Oct	Alert: 5 days since workout, 1 missed planned day	Days since workout 5; missed planned days 1	Pass
ATC-M16-U07	M16-FR3	The next workout reminder is the next occurrence of the weekday and time	now Wed 14 Oct 10:00; reminder Wed 18:30; Wed 09:00	Today 18:30; next Wednesday 09:00	Next: 2026-10-14 18:30:00.000; 2026-10-21 09:00:00.000	Pass
ATC-M17-U01	M17-FR5	The moving average uses only logged days inside a 3-day window	[2000, null, 2200, 1800, null]	[2000, 2000, 2100, 2000, 2000]	Smoothed: 2000.0, 2000.0, 2100.0, 2000.0, 2000.0	Pass
ATC-M17-U02	M17-FR5	A window with no logged day gives no value rather than zero	[null, null, null, 2000]	[null, null, null, 2000]	Smoothed: null, null, null, 2000.0	Pass
ATC-M1-F01	M1-FR4	Create account is disabled on an empty form	Open the sign-up screen and type nothing	The Create account button is disabled	Button enabled = false	Pass
ATC-M1-F02	M1-FR5	A weak password is explained and blocks sign-up	Name "Aamna Khan", email "aamna@example.com", password and confirmation "weak"	Message "Use at least 8 characters." shown; button disabled	Message shown = true; button enabled = false	Pass
ATC-M1-F03	M1-FR4	Terms must be accepted before the account can be created	All fields valid, terms not ticked; then terms ticked	Disabled before ticking, enabled after	Enabled before ticking = false; after ticking = true	Pass
ATC-M1-F04	M1-FR5	A different confirmation password disables sign-up again	Valid form, then confirmation changed to "Different1!"	Message "The two passwords do not match." shown; button disabled	Message shown = true; button enabled = false	Pass
ATC-M2-F01	M2-FR2	The BMI screen shows the computed BMI, the BMR method and the calorie deficit	Profile: female, 165 cm, 72 kg, Lose Weight	BMI 26.4 and the words "Mifflin-St Jeor" and "Deficit" on screen	BMI 26.4 shown = true; method shown = true; deficit shown = true	Pass
ATC-M2-F02	M2-FR10	The classification screen lists the reported condition as a risk factor	Profile with condition "Hypertension"	"Hypertension" and "View workout plan" on screen	"Hypertension" appears 1 time(s); "View workout plan" button shown = true	Pass
ATC-M3-F01	M3-FR1	The greeting names the user and follows the time of day	Open Home for "Aamna Khan"; the clock decides morning, afternoon or evening	The greeting for the current hour, the first name "Aamna" and the matching motivational line	Greeting "Good Evening" shown true; first name shown true; motivational line shown true	Pass
ATC-M3-F02	M3-FR1	With no profile Home says "Welcome back" and asks for the health profile instead of inventing a plan	Open Home with no profile loaded	"Welcome back", "No plan yet" and "Complete your health profile"	Welcome back true; "No plan yet" true; prompt true	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M3-F03	M3-FR2	The activity cards show calories burned, active minutes and an honest steps card	Open Home with no workout done today	"Calories burned" and "Active minutes" with 0; steps card says "Steps (no sensor)" with a dash, not a made-up number	Calories burned true; Active minutes true; zeros shown 2; steps card "Steps (no sensor)" with dash true	Pass
ATC-M3-F04	M3-FR3	The workout card shows today's session from the user's plan and Open goes to the Workout tab	Open Home with a generated plan and tap Open	Today's workout name with minutes and kcal, or "Rest day"; Open selects tab 2	Card shown true; today is a rest day, shown correctly true; tapping Open selected tab 2	Pass
ATC-M3-F05	M3-FR4	The mood selector shows the chosen mood with a gentle suggestion, and tapping it again clears it	Tap "Sad", then tap "Sad" again	Label "Sad" and the suggestion appear; both disappear after the second tap	Prompt shown true; after the first tap label 1, suggestion 1; after the second tap 0 left	Pass
ATC-M3-F06	M3-FR5	Quick Access lists the eight areas the report names	Open Home and read the Quick access grid	Workout plan, Diet tracker, Posture check, Body transform, Leaderboard, Allergy scanner, AI chatbot, Find a specialist	All 8 tiles shown	Pass
ATC-M3-F07	M3-FR5	Quick Access tiles that belong to a tab switch to that tab	Tap the "Workout plan" tile, then the "Diet tracker" tile	Tab 2 (Workout), then tab 1 (Diet)	Workout plan tile selected tab 2; Diet tracker tile selected tab 1	Pass
ATC-M3-F08	M3-FR1	The profile avatar shows the user's initial and opens the Profile tab	Tap the avatar in the top bar	Initial "A"; tab 4 (Profile) selected	Initial shown true; tab selected 4	Pass
ATC-M3-F09	M3-FR8	The notification bell shows the unread count and caps it at 9+	Set the unread count to 3, then to 12	Badge "3" with tooltip "Notifications, 3 unread"; then badge "9+"	Badge "3" true; tooltip true; badge "9+" true	Pass
ATC-M3-F10	M3-FR7	The bottom navigation bar has the five labelled tabs and each reports its index	Tap Diet, Workout, Analytics, Profile and Home in the bar	Indexes 1, 2, 3, 4, 0	Indexes reported: 1, 2, 3, 4, 0	Pass
ATC-M4-F01	M4-FR5	A cardiac workout plan shows the category, warm-up, and a rest day when a day is tapped	Profile with "Heart Disease"; tap Sunday	Title, "Cardiac", "Dynamic Warm-Up"; Sunday says rest day	Title 1, category label 1, warm-up 1, rest-day text 1	Pass
ATC-M8-F01	M8-FR13	The daily summary shows consumed, burned and net calories with a verdict	Logged 1000 kcal, burned 250, goal Lose Weight, BMR 1500	Consumed 1000, burned 250, net +750 and the words "very low intake"	Consumed shown true; burned shown true; net "+750" shown true; verdict shown true	Pass
ATC-M8-F02	M8-FR13	An empty day says nothing is logged and invents no macro split	Day with no food entries	"Nothing is logged" and the prompt to log food	"Nothing is logged" shown true; prompt shown true	Pass
ATC-M8-F03	M8-FR7	Typing a local name finds the food by its other name	Type "chawal"	Suggestions with "Also called" and kcal values	"Also called" appears 5 time(s); kcal shown 5 time(s)	Pass
ATC-M8-F04	M8-FR8	Picking a suggestion adds it to the meal and closes the list	Type "banana", tap the add button of the first result	"Your meal" appears and the add buttons disappear	"Your meal" before: 0, after: 1; add buttons left: 0	Pass
ATC-M8-F05	M8-FR7	A food that matches the user's allergy is flagged in the search results	User allergic to Dairy types "milk"	"on your allergy list" warning with a warning icon	Allergy warning shown 4 time(s); warning icons 4	Pass
ATC-M8-F06	M8-FR7	A search with no result says so instead of showing an empty box	Type "qqzzxx"	"No matching foods"	Message shown = true	Pass
ATC-M9-F01	M9-FR6	The nutrition plan shows the calorie target and lists the user's allergen foods under "avoid"	Profile with Dairy and Nuts allergies	The target number and "Dairy products" on screen	Target 1508 kcal shown = true; "Dairy products" under avoid = true	Pass
ATC-M12-F01	M12-FR9	My badges shows the level, XP and the unlock rule of a locked badge	Open My badges with 1 workout day, 2 meal days and 1 badge	Level 1, 170 XP, 30 XP to level 2, and the "30-Day Streak" badge listed	Level true; XP true; XP to next true; locked badge listed true	Pass
ATC-M12-F02	M12-FR7	A failed load shows no invented numbers	The data loader throws "offline"	"Level 1" is not shown	"Level 1" shown 0 time(s)	Pass
ATC-M14-F01	M14-FR1	The specialist list shows a count and disables Book for a specialist who is not taking bookings	Open "Find a specialist" with 3 specialists (1 closed)	"3 Specialists Found", "Not taking bookings", 3 Book buttons of which 1 is disabled	Count text true; "Not taking bookings" true; Book buttons 3, disabled 1	Pass
ATC-M14-F02	M14-FR1	The type chip filters the list	Tap the "Nutritionist" chip	"1 Nutritionist Found"; only Sara Khan remains	Count text true; Sara Khan shown 1; Dr. Hina Malik shown 0	Pass
ATC-M14-F03	M14-FR1	Search by specialty finds the trainer	Type "yoga" with the type filter on All	"1 Specialist Found" and Omar Closed	Count text true; Omar Closed shown true	Pass
ATC-M14-F04	M14-FR1	A specialist with no reviews is shown as New and not hidden	One specialist with 0 reviews	"Newly Joined" listed with a "New" tag	Name shown true; "New" tag shown true	Pass
ATC-M14-F05	M14-FR1	A failed load is explained and can be retried	The first load fails; tap "Try again"	Error text, then "3 Specialists Found"	Error shown true; after retry list shown true	Pass
ATC-M14-F06	M14-FR2	Confirm booking stays disabled until a time is chosen, and nothing is shared by default	Open the booking screen and choose nothing	Confirm disabled; "Not chosen yet"; every sharing tick box off	Confirm disabled true; "Not chosen yet" 1; sharing boxes 3, any ticked false	Pass

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ATC-M14-F07	M14-FR2	Booking sends exactly the chosen time, type, shares and note, then opens the payment screen	Pick the first time (09:00), "Chat only", share "Diet log, last 7 days", note "Help with a meal plan", tap Confirm	Slot 04:00 UTC, chat, diet log shared only, note kept; the Payment screen opens	Slot 2026-10-19T04:00:00.000Z; type chat; diet log shared true, health summary shared false; chat fee shown true; Payment screen opened true	Pass
ATC-M14-F08	M14-FR2	A time someone else just took is refused and the screen stays	Choose a time; the booking call answers "just taken"	The message "just taken by someone else" and the booking screen still open	Message shown true; still on the booking screen true	Pass
ATC-M14-F09	M14-FR2	A time that is already booked is not offered	The 09:00 slot is booked by someone else	09:00 is not listed, 09:30 is	09:00 listed 0 time(s); 09:30 listed 1 time(s)	Pass
ATC-M14-F10	M14-FR2	The notes box stops at 500 characters	Paste 600 characters into the note	Only 500 characters are kept	Characters kept: 500	Pass
ATC-M14-F11	M14-FR3	My appointments separates upcoming from past and shows the specialist's notes only under Past	One upcoming and one completed appointment; open the Past tab	"Upcoming (1)"; notes hidden first, visible under Past with "Completed"	Upcoming tab true; notes before 0, after 1; "Completed" 1	Pass
ATC-M14-F12	M14-FR3	A request shows as awaiting acceptance, has no video link and can be cancelled	Appointment with status "requested"	"Awaiting acceptance", "Waiting for your specialist", no meeting room, a Cancel request button	Awaiting label true; waiting text true; meeting room shown 0; cancel button true	Pass
ATC-M14-F13	M14-FR3	A declined request shows the reason and offers no cancel	Appointment declined with reason "On leave that day"	"Not accepted" and the reason; no Cancel appointment button	"Not accepted" true; reason shown true; cancel button 0	Pass
ATC-M14-F14	M14-FR3	A confirmed appointment shows the meeting link and the join window rule	Confirmed appointment tomorrow	Meeting room, the link and "opens 5 minutes before"	Meeting room true; link true; join rule text true	Pass
ATC-M14-F15	M14-FR3	The room is announced as open inside the join window	Confirmed appointment starting in 3 minutes	"The room is open now"	Room-open text shown true	Pass
ATC-M14-F16	M14-FR2	Cancelling asks for confirmation first and then shows the new status	Tap Cancel appointment, choose "Keep it", then cancel again and confirm	Nothing cancelled after "Keep it"; afterwards status "Cancelled" and no cancel button	Confirmation asked true; nothing cancelled after "Keep it" true; cancelled id u1; status shown true; cancel button left 0	Pass
ATC-M14-F17	M14-FR3	An appointment that has already started has no cancel button	Confirmed appointment that started 5 minutes ago	No Cancel appointment button	Cancel button shown 0 time(s)	Pass
ATC-M14-F21	M14-FR12	The dashboard lists open requests soonest first, hides expired ones and counts them	Doctor opens the dashboard with 2 live requests, 1 expired request and 1 confirmed session	Requests r1 then r2; today shows the confirmed session; 2 pending; no video link on a request	Requests: r1,r2; today: ok; pending 2; link on request ""	Pass
ATC-M14-F22	M14-FR3, M14-FR12	Accepting a request creates the video room on the server, confirms it and tells the patient	Doctor accepts r1, then accepts it again	200 and status confirmed; a meet.jit.si room; patient notified; slot kept; the second accept is 400	Accept 200 (confirmed); room https://meet.jit.si/baymax-6m8gbshsj784epg7ksgf; patient notice "Your appointment was accepted"; slot kept true; second accept 400	Pass
ATC-M14-F23	M14-FR12	Declining frees the time, keeps the reason and tells the patient	Doctor declines r2 with reason " On leave that day "	Status declined, reason trimmed, slot freed, other slot kept, patient notified, no room made, accepting afterwards is 400	Decline 200; status declined; reason "On leave that day"; slot freed true; other slot kept true; accept afterwards 400	Pass
ATC-M14-F24	M14-FR12	A request whose time has passed is shown as expired and cannot be answered	Doctor lists appointments and tries to accept "late" (start was 30 minutes ago)	phase expired, canRespond false, accept answers 400 "expired"	Phase expired; canRespond false; accept 400 "The start time has passed, so this request expired. The patient can book another time."	Pass
ATC-M14-F25	M14-FR12	Only the specialist the request was sent to can answer it	Doctor answers the nutritionist's request; a member and an anonymous caller answer r1	404, 403 and 401; r1 stays requested	Other specialist's request 404; member 403; anonymous 401; r1 is requested	Pass
ATC-M14-F26	M14-FR8	Notes are saved after the start, the patient is told, and a note before the start is refused	Doctor writes notes for a1 (starts in 60 min) and for a2 (started 10 min ago)	a1: 400 "started"; a2: 200, note and prescription stored, patient notified	Before start 400; after start 200; prescription "Paracetamol 500mg"; patient notified true	Pass
ATC-M14-F27	M14-FR6	Only a doctor can write a prescription	Nutritionist saves notes with a prescription	400 mentioning "doctor"	Status 400: "Only a doctor can write a prescription."	Pass
ATC-M14-F28	M14-FR7	Cancelling frees the slot and tells the patient; a finished appointment cannot be changed again	Doctor cancels a1, then tries to mark it completed	200, slot gone, notice "Appointment cancelled"; then 400 "already cancelled"	Cancel 200; slot freed true; second change 400 "This appointment is already cancelled."	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M14-F29	M14-FR13	My patients lists only people with an accepted appointment and supports search	Doctor opens /api/patients, then searches "bil"; a member tries the same	Ayesha Khan and Bilal Ahmed only; search gives Bilal; member gets 403	Patients: Ayesha Khan, Bilal Ahmed; search "bil": Bilal Ahmed; member 403	Pass
ATC-M14-F30	M14-FR5	A nutritionist saves a diet plan that is stored with its author and announced to the patient	PUT /api/patients/pat1/plans/diet with 1900 kcal, 120/210/60 g	200; stored target 1900 with author Sara Khan (nutritionist); patient notified; visible on the record	Status 200; stored target 1900; author Sara Khan (nutritionist); notice "Your diet plan was updated"; record shows 1900	Pass
ATC-M14-F31	M14-FR5	Diet plan numbers are checked and each problem is explained	Targets 300 and 6000 kcal; macros that do not add up to 1900; only some macros; an unknown status; an empty change	Each refused with 400 and a clear message	400 "The calorie target must be a whole number from 800 to 5000." | 400 "The calorie target must be a whole number from 800 to 5000." | 400 "Protein, carbohydrate and fat add up to about 4100 kcal, which differs from the 1900 kcal target by more than 15%. Check the numbers." | 400 "Give protein, carbohydrate and fat together, or none of them." | 400 "Choose "Approve as is" or "Save changes"." | 400 "Change at least one thing, or choose "Approve as is"."	Pass
ATC-M14-F32	M14-FR10	A trainer builds a workout plan from the exercise library only and the schedule is validated	Valid 7-day plan; then a plan with an invented exercise, 6 days, no sets, and a workout on a rest day	Valid plan 200 and stored; each bad plan 400	Valid plan 200, 7 days stored; bad plans: 400 "The schedule needs all seven days." | 400 ""made_up_move" is not an exercise in the Baymax library." | 400 "Sets for Sit-to-Stand (Chair Squat) must be a whole number from 1 to 8." | 400 "A rest day cannot have exercises."	Pass
ATC-M14-F33	M14-FR5	A specialist can remove their plan and the patient is told Baymax's plan applies again	Nutritionist sets and then deletes the diet plan; a trainer tries to delete; delete again	Trainer 403; nutritionist 200 and plan gone; patient notified "back to Baymax"; second delete 404	Trainer 403; nutritionist 200; plan gone true; patient told true; second delete 404	Pass
ATC-M14-F34	M14-FR9	A nutritionist approves or flags a day of the shared food log, and a flag needs a reason	PUT diet-reviews: approve 2026-10-19; flag 2026-10-18 with a note; flag without a note	200, 200 with the patient told "flagged", then 400 "why you are flagging"	Approve 200; flag 200, patient told true; flag without reason 400 "Say why you are flagging this day, so the patient knows what to change."	Pass
ATC-M14-F35	M14-FR11	Registering as a specialist stores a pending application and grants nothing	Signed-in member "newbie" posts the form (and tries to set role=admin and another email)	201 pending; email taken from the account; no role; not listed; licence number not echoed back	Status 201 (pending); stored email newbie@example.com; role undefined; listed false; role seen by the app user	Pass
ATC-M14-F36	M14-FR11	The licence number is required and the form is checked	Registration without a licence number; with a bad type; with a negative fee	Each 400	400 "Enter your professional licence or registration number (3 to 40 letters, digits, spaces, - . /)." | 400 "Type must be one of: doctor, nutritionist, trainer." | 400 "Fees must be whole numbers of PKR between 0 and 1,000,000."	Pass
ATC-M14-F37	M14-FR11	An administrator approves an application: role, verified public profile, and the specialist can sign in	Admin approves "newbie"; the licence number is checked; the new specialist opens the dashboard; approve again	Role doctor; profile verified, no licence published; dashboard 200; second approval 409	Admin sees licence "PMDC-12345"; approve 200; role doctor; verified true; licence published false; new specialist dashboard 200; second approval 409	Pass
ATC-M14-F38	M14-FR11	Rejecting an application records the reason and never creates a role	Admin rejects "newbie" with a 301-character reason, then without a reason	400, then 200; no role	Long reason 400; reject 200; status rejected; role undefined	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M14-F39	M14-FR15	An administrator adds a specialist without ever setting a password	POST /api/admin/specialists with a new email	201; account created without a password; role and profile set; a link to choose a password is returned	Status 201; created true; role doctor; link returned true	Pass
ATC-M14-F40	M14-FR14	Closing an account needs the word DELETE and a recent sign-in	DELETE /api/me with an empty body; with DELETE but an hour-old sign-in	400; then 401 "reauth-required"; nothing deleted	Empty 400; stale sign-in 401 (reauth-required); role kept doctor	Pass
ATC-M14-F41	M14-FR14	Closing a specialist account cancels future bookings, frees the times, tells patients and keeps past records	Doctor closes the account (DELETE, recent sign-in)	2 bookings cancelled; past record kept; slots gone; patient told; role, profile and sign-in removed; others untouched	Status 200; cancelled 2; future cancelled/cancelled; past completed; slots left false; sign-in deleted doc1; nutritionist kept nutritionist	Pass
ATC-M16-F01	M16-FR8	Notifications show unread counts per tab and filter by tab	4 notifications (3 unread); tap the Diet tab	"All (3)" and "Diet (1)"; Diet shows only the meal reminder	"All (3)" true; "Diet (1)" true; meal reminder shown 1; workout reminder shown 0	Pass
ATC-M16-F02	M16-FR9	Mark all read reports exactly the unread items	Tap "Mark all read"	The callback receives 3 unread items	Callback received 4 item(s), 3 unread	Pass
ATC-M16-F03	M16-FR8	A failed load shows an error, not an empty inbox	The loader throws "offline"	"Could not load your notifications." and no "No notifications yet"	Error shown true; empty-inbox text shown 0 time(s)	Pass
ATC-M16-F04	M16-FR2	A request without a valid sign-in is turned away	GET /api/me with no token, a garbage token and a Basic scheme	401 three times; /health stays open (200)	No token 401; garbage token 401; Basic scheme 401; /health 200	Pass
ATC-M16-F05	M16-FR2	A member or administrator cannot open the specialist screens even by typing the address	GET /api/dashboard, /api/appointments, /api/appointments/a1 as a member, an administrator and an unknown user	403 for every one of the 9 requests	Status codes: 403, 403, 403, 403, 403, 403, 403, 403, 403	Pass
ATC-M16-F06	M16-FR2	Another specialist's appointment looks like it does not exist	Doctor opens appointment "other" that belongs to the nutritionist	404	Status 404	Pass
ATC-M16-F07	M16-FR2	Only an administrator reaches the admin routes	GET /api/admin/specialists as doctor, nutritionist, member, unknown user and anonymous; and as admin	401 anonymous; 403 for the four others; 200 for the admin	Anonymous 401; doctor/nutritionist/member/unknown 403/403/403/403; administrator 200	Pass
ATC-M17-F01	M17-FR2	Analytics shows the calorie chart, the weekly score and an honest scope note	Open analytics with 6 weigh-ins	"Calories vs your target", "Weekly fitness score", and the note "not part of this version"	Calorie chart true; weekly score true; scope note true	Pass
ATC-M17-F02	M17-FR2	A failed analytics load offers a retry	The loader throws	"Could not load your analytics." and a "Try again" button	Error shown true; retry button shown true	Pass
ATC-M2-B01	M2-FR10	Health category: None, BMI Normal	Reported "None"; BMI band Normal (175 cm, 70 kg)	Category "Healthy"	Category = "Healthy"	Pass
ATC-M2-B02	M2-FR10	Health category: None, BMI Overweight	Reported "None"; BMI band Overweight (175 cm, 85 kg)	Category "Overweight"	Category = "Overweight"	Pass
ATC-M2-B03	M2-FR10	Health category: None, BMI Obese	Reported "None"; BMI band Obese (175 cm, 100 kg)	Category "Obese"	Category = "Obese"	Pass
ATC-M2-B04	M2-FR10	Health category: None, BMI Underweight	Reported "None"; BMI band Underweight (175 cm, 50 kg)	Category "At-Risk"	Category = "At-Risk"	Pass
ATC-M2-B05	M2-FR10	Health category: Hypertension, BMI Normal	Reported "Hypertension"; BMI band Normal (175 cm, 70 kg)	Category "At-Risk"	Category = "At-Risk"	Pass
ATC-M2-B06	M2-FR10	Health category: Diabetes, BMI Normal	Reported "Diabetes"; BMI band Normal (175 cm, 70 kg)	Category "Diabetic"	Category = "Diabetic"	Pass
ATC-M2-B07	M2-FR10	Health category: Diabetes, BMI Obese	Reported "Diabetes"; BMI band Obese (175 cm, 100 kg)	Category "Diabetic"	Category = "Diabetic"	Pass
ATC-M2-B08	M2-FR10	Health category: Heart disease, BMI Normal	Reported "Heart disease"; BMI band Normal (175 cm, 70 kg)	Category "Cardiac"	Category = "Cardiac"	Pass
ATC-M2-B09	M2-FR10	Health category: Heart disease and diabetes, BMI Obese	Reported "Heart disease and diabetes"; BMI band Obese (175 cm, 100 kg)	Category "Cardiac"	Category = "Cardiac"	Pass
ATC-M13-B01	M13-FR4	Chatbot: message about a possible emergency	User types "I have chest pain"	Intent "emergency"	Intent = "emergency"	Pass
ATC-M13-B02	M13-FR4	Chatbot: message about an emergency and an exercise	User types "I have chest pain after squats"	Intent "emergency"	Intent = "emergency"	Pass
ATC-M13-B03	M13-FR4	Chatbot: message about a symptom	User types "my knee hurts after squats"	Intent "health_concern"	Intent = "health_concern"	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M13-B04	M13-FR4	Chatbot: message about BMI	User types "what is my bmi"	Intent "bmi"	Intent = "bmi"	Pass
ATC-M13-B05	M13-FR4	Chatbot: message about no known topic	User types "asdfgh qwerty"	Intent "fallback"	Intent = "fallback"	Pass
ATC-M14-B01	M14-FR2	Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours Yes, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: Yes	Booking allowed	Booking allowed	Pass
ATC-M14-B02	M14-FR2	Booking check: Taking bookings No, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours Yes, Note 500 or less Yes	Taking bookings: No; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: Yes	Refused: This specialist is not taking bookings right now.	Refused: This specialist is not taking bookings right now.	Pass
ATC-M14-B03	M14-FR2	Booking check: Taking bookings Yes, Time chosen No, Time free -, 15+ min ahead -, Within working hours -, Note 500 or less Yes	Taking bookings: Yes; Time chosen: No; Time free: -; 15+ min ahead: -; Within working hours: -; Note 500 or less: Yes	Refused: Choose a date and a time.	Refused: Choose a date and a time.	Pass
ATC-M14-B04	M14-FR2	Booking check: Taking bookings Yes, Time chosen Yes, Time free No, 15+ min ahead Yes, Within working hours Yes, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: No; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: Yes	Refused: That time was just taken. Please choose another.	Refused: That time was just taken. Please choose another.	Pass
ATC-M14-B05	M14-FR2	Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead No, Within working hours Yes, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: No; Within working hours: Yes; Note 500 or less: Yes	Refused: That time is too soon. Please choose a later one.	Refused: That time is too soon. Please choose a later one.	Pass
ATC-M14-B06	M14-FR2	Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours No, Note 500 or less Yes	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: No; Note 500 or less: Yes	Refused: That time is outside the specialist's working hours.	Refused: That time is outside the specialist's working hours.	Pass
ATC-M14-B07	M14-FR2	Booking check: Taking bookings Yes, Time chosen Yes, Time free Yes, 15+ min ahead Yes, Within working hours Yes, Note 500 or less No	Taking bookings: Yes; Time chosen: Yes; Time free: Yes; 15+ min ahead: Yes; Within working hours: Yes; Note 500 or less: No	Refused: Notes can be at most 500 characters.	Refused: Notes can be at most 500 characters.	Pass
ATC-M14-B08	M14-FR3	Appointment Confirmed, 6 minutes before start	Status "Confirmed"; video link: Yes; time: 6 minutes before start	Join not allowed; cancel allowed	Join not allowed; cancel allowed	Pass
ATC-M14-B09	M14-FR3	Appointment Confirmed, 5 minutes before start	Status "Confirmed"; video link: Yes; time: 5 minutes before start	Join allowed; cancel allowed	Join allowed; cancel allowed	Pass
ATC-M14-B10	M14-FR3	Appointment Confirmed, 10 minutes after start	Status "Confirmed"; video link: Yes; time: 10 minutes after start	Join allowed; cancel not allowed	Join allowed; cancel not allowed	Pass
ATC-M14-B11	M14-FR3	Appointment Confirmed, after the slot ended	Status "Confirmed"; video link: Yes; time: after the slot ended	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass
ATC-M14-B12	M14-FR3	Appointment Requested, 30 minutes before start	Status "Requested"; video link: No; time: 30 minutes before start	Join not allowed; cancel allowed	Join not allowed; cancel allowed	Pass
ATC-M14-B13	M14-FR3	Appointment Requested, at the start time	Status "Requested"; video link: No; time: at the start time	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass
ATC-M14-B14	M14-FR3	Appointment Declined, 30 minutes before start	Status "Declined"; video link: No; time: 30 minutes before start	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass
ATC-M14-B15	M14-FR3	Appointment Completed, 1 hour after start	Status "Completed"; video link: Yes; time: 1 hour after start	Join not allowed; cancel not allowed	Join not allowed; cancel not allowed	Pass
ATC-M14-B16	M14-FR5	Diet plan: specialist plan None, member chose Specialist	Baymax target 2000 kcal; specialist plan: None; member chose: Specialist	Daily target 2000 kcal	Daily target 2000 kcal	Pass
ATC-M14-B17	M14-FR5	Diet plan: specialist plan None, member chose Baymax	Baymax target 2000 kcal; specialist plan: None; member chose: Baymax	Daily target 2000 kcal	Daily target 2000 kcal	Pass
ATC-M14-B18	M14-FR5	Diet plan: specialist plan Approved as is, member chose Specialist	Baymax target 2000 kcal; specialist plan: Approved as is; member chose: Specialist	Daily target 2000 kcal	Daily target 2000 kcal	Pass
ATC-M14-B19	M14-FR5	Diet plan: specialist plan Changed (1800 kcal), member chose Specialist	Baymax target 2000 kcal; specialist plan: Changed (1800 kcal); member chose: Specialist	Daily target 1800 kcal	Daily target 1800 kcal	Pass
ATC-M14-B20	M14-FR5	Diet plan: specialist plan Changed (1800 kcal), member chose Baymax	Baymax target 2000 kcal; specialist plan: Changed (1800 kcal); member chose: Baymax	Daily target 2000 kcal	Daily target 2000 kcal	Pass
ATC-M14-B21	M14-FR5	Plan change: Nutritionist, diet plan, own patient, recent appointment	Nutritionist sends a diet plan for own patient, recent appointment	Status 200	Status 200	Pass
ATC-M14-B22	M14-FR5	Plan change: Doctor, diet plan, own patient, recent appointment	Doctor sends a diet plan for own patient, recent appointment	Status 200	Status 200	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M14-B23	M14-FR5	Plan change: Trainer, diet plan, own patient, recent appointment	Trainer sends a diet plan for own patient, recent appointment	Status 403	Status 403 ("Only doctors and nutritionists can change a diet plan.")	Pass
ATC-M14-B24	M14-FR5	Plan change: Trainer, workout plan, own patient, recent appointment	Trainer sends a workout plan for own patient, recent appointment	Status 200	Status 200	Pass
ATC-M14-B25	M14-FR5	Plan change: Doctor, workout plan, own patient, recent appointment	Doctor sends a workout plan for own patient, recent appointment	Status 200	Status 200	Pass
ATC-M14-B26	M14-FR5	Plan change: Nutritionist, workout plan, own patient, recent appointment	Nutritionist sends a workout plan for own patient, recent appointment	Status 403	Status 403 ("Only doctors and trainers can change a workout plan.")	Pass
ATC-M14-B27	M14-FR5	Plan change: Doctor, diet plan, own patient, appointment 40 days ago	Doctor sends a diet plan for own patient, appointment 40 days ago	Status 400	Status 400 ("You can change a patient's plan from when you accept their appointment until 30 days after it.")	Pass
ATC-M14-B28	M14-FR5	Plan change: Doctor, diet plan, only a pending request	Doctor sends a diet plan for only a pending request	Status 404	Status 404 ("Patient not found.")	Pass
ATC-M14-B29	M14-FR5	Plan change: Nutritionist, diet plan, another specialist's patient	Nutritionist sends a diet plan for another specialist's patient	Status 404	Status 404 ("Patient not found.")	Pass
ATC-M14-B30	M14-FR5	Plan change: Member, diet plan, own patient, recent appointment	Member sends a diet plan for own patient, recent appointment	Status 403	Status 403 ("This area is only for Baymax specialists.")	Pass
ATC-M14-B31	M14-FR7	Mark completed, 1 minute before the start	Doctor marks a confirmed appointment as completed 1 minute before the start	Status 400	Status 400 ("The session has not started yet.")	Pass
ATC-M14-B32	M14-FR7	Mark completed, at the start	Doctor marks a confirmed appointment as completed at the start	Status 200	Status 200	Pass
ATC-M14-B33	M14-FR7	Mark missed, 9 minutes after the start	Doctor marks a confirmed appointment as missed 9 minutes after the start	Status 400	Status 400 ("The patient can only be marked as missed 10 minutes after the start time.")	Pass
ATC-M14-B34	M14-FR7	Mark missed, 10 minutes after the start	Doctor marks a confirmed appointment as missed 10 minutes after the start	Status 200	Status 200	Pass
ATC-M14-B35	M14-FR7	Mark cancelled, 10 minutes after the start	Doctor marks a confirmed appointment as cancelled 10 minutes after the start	Status 200	Status 200	Pass
ATC-M14-B36	M14-FR7	Mark cancelled, 30 minutes after the start (slot ended)	Doctor marks a confirmed appointment as cancelled 30 minutes after the start (slot ended)	Status 400	Status 400 ("This appointment has already ended.")	Pass
ATC-M14-B37	M14-FR12	accept a request that is waiting and in the future	Doctor tries to accept a request that is waiting and in the future	Status 200	Status 200	Pass
ATC-M14-B38	M14-FR12	decline a request that is waiting and in the future	Doctor tries to decline a request that is waiting and in the future	Status 200	Status 200	Pass
ATC-M14-B39	M14-FR12	accept a request that is past its start time	Doctor tries to accept a request that is past its start time	Status 400	Status 400 ("The start time has passed, so this request expired. The patient can book another time.")	Pass
ATC-M14-B40	M14-FR12	decline a request that is past its start time	Doctor tries to decline a request that is past its start time	Status 400	Status 400 ("The start time has passed, so this request expired. The patient can book another time.")	Pass
ATC-M14-B41	M14-FR12	accept a request that is already declined	Doctor tries to accept a request that is already declined	Status 400	Status 400 ("This request is already declined.")	Pass
ATC-M14-B42	M14-FR12	accept a request that is already confirmed	Doctor tries to accept a request that is already confirmed	Status 400	Status 400 ("This request was already accepted.")	Pass
ATC-M14-B43	M14-FR12	accept a request that is sent to another specialist	Doctor tries to accept a request that is sent to another specialist	Status 404	Status 404 ("Appointment not found.")	Pass
ATC-M14-B44	M14-FR11	Application pending: an administrator chooses approve	Application is pending; an administrator sends "approve"	Status 200	Status 200	Pass
ATC-M14-B45	M14-FR11	Application pending: an administrator chooses reject	Application is pending; an administrator sends "reject"	Status 200	Status 200	Pass
ATC-M14-B46	M14-FR11	Application approved: an administrator chooses approve	Application is approved; an administrator sends "approve"	Status 409	Status 409 ("This application was already approved.")	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M14-B47	M14-FR11	Application approved: an administrator chooses reject	Application is approved; an administrator sends "reject"	Status 409	Status 409 ("This application was already approved.")	Pass
ATC-M14-B48	M14-FR11	Application holds another role: an administrator chooses approve	Application is holds another role; an administrator sends "approve"	Status 409	Status 409 ("This account already has the role "trainer".")	Pass
ATC-M14-B49	M14-FR11	Application pending: a doctor chooses approve	Application is pending; a doctor sends "approve"	Status 403	Status 403 ("This area is only for Baymax specialists.")	Pass
ATC-M14-B50	M14-FR11	Application pending: the applicant chooses approve	Application is pending; the applicant sends "approve"	Status 403	Status 403 ("This area is only for Baymax specialists.")	Pass
ATC-M14-B51	M14-FR14	Close account: confirmation Yes, sign-in recently	DELETE /api/me with Yes; sign-in recently	Status 200	Status 200; role kept false	Pass
ATC-M14-B52	M14-FR14	Close account: confirmation No, sign-in recently	DELETE /api/me with No; sign-in recently	Status 400	Status 400; role kept true	Pass
ATC-M14-B53	M14-FR14	Close account: confirmation Wrong word, sign-in recently	DELETE /api/me with Wrong word; sign-in recently	Status 400	Status 400; role kept true	Pass
ATC-M14-B54	M14-FR14	Close account: confirmation Yes, sign-in an hour ago	DELETE /api/me with Yes; sign-in an hour ago	Status 401	Status 401; role kept true	Pass
ATC-M16-B01	M16-FR3	Workout reminder: switch Yes, planned day Yes, workout done No, time passed Yes	Reminder switch Yes; planned today: Yes; workout done: No; 18:00 reached (now 19:00 or 17:00): Yes	Reminder raised	Reminder raised	Pass
ATC-M16-B02	M16-FR3	Workout reminder: switch No, planned day Yes, workout done No, time passed Yes	Reminder switch No; planned today: Yes; workout done: No; 18:00 reached (now 19:00 or 17:00): Yes	No reminder	No reminder	Pass
ATC-M16-B03	M16-FR3	Workout reminder: switch Yes, planned day No, workout done No, time passed Yes	Reminder switch Yes; planned today: No; workout done: No; 18:00 reached (now 19:00 or 17:00): Yes	No reminder	No reminder	Pass
ATC-M16-B04	M16-FR3	Workout reminder: switch Yes, planned day Yes, workout done Yes, time passed Yes	Reminder switch Yes; planned today: Yes; workout done: Yes; 18:00 reached (now 19:00 or 17:00): Yes	No reminder	No reminder	Pass
ATC-M16-B05	M16-FR3	Workout reminder: switch Yes, planned day Yes, workout done No, time passed No	Reminder switch Yes; planned today: Yes; workout done: No; 18:00 reached (now 19:00 or 17:00): No	No reminder	No reminder	Pass
ATC-M16-B06	M16-FR3	Meal reminder: switch Yes, food logged No, time passed Yes	Meal reminder switch Yes; food logged today: No; 20:00 reached (now 21:00 or 19:00): Yes	Reminder raised	Reminder raised	Pass
ATC-M16-B07	M16-FR3	Meal reminder: switch No, food logged No, time passed Yes	Meal reminder switch No; food logged today: No; 20:00 reached (now 21:00 or 19:00): Yes	No reminder	No reminder	Pass
ATC-M16-B08	M16-FR3	Meal reminder: switch Yes, food logged Yes, time passed Yes	Meal reminder switch Yes; food logged today: Yes; 20:00 reached (now 21:00 or 19:00): Yes	No reminder	No reminder	Pass
ATC-M16-B09	M16-FR3	Meal reminder: switch Yes, food logged No, time passed No	Meal reminder switch Yes; food logged today: No; 20:00 reached (now 21:00 or 19:00): No	No reminder	No reminder	Pass
ATC-M16-B10	M14-FR4	Shared data: patient ticked Yes, confirmed, 20 minutes before the start	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 20 minutes before the start	Shared data shown and the access logged	Shared data shown, access logged true	Pass
ATC-M16-B11	M14-FR4	Shared data: patient ticked No, confirmed, 20 minutes before the start	Patient ticked sharing: No; appointment confirmed; doctor opens it 20 minutes before the start	Shared data not shown	Shared data hidden	Pass
ATC-M16-B12	M14-FR4	Shared data: patient ticked Yes, confirmed, 40 minutes before the start	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 40 minutes before the start	Shared data not shown	Shared data hidden ("Shared data opens 30 minutes before the appointment.")	Pass
ATC-M16-B13	M14-FR4	Shared data: patient ticked Yes, confirmed, 10 minutes after the start	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 10 minutes after the start	Shared data shown and the access logged	Shared data shown, access logged true	Pass
ATC-M16-B14	M14-FR4	Shared data: patient ticked Yes, confirmed, 95 minutes after the start (slot ended + 65 min)	Patient ticked sharing: Yes; appointment confirmed; doctor opens it 95 minutes after the start (slot ended + 65 min)	Shared data not shown	Shared data hidden ("The viewing period for this appointment has ended.")	Pass
ATC-M16-B15	M14-FR4	Shared data: patient ticked Yes, requested, 20 minutes before the start	Patient ticked sharing: Yes; appointment requested; doctor opens it 20 minutes before the start	Shared data not shown	Shared data hidden ("Accept the request to see what the patient chose to share. It opens 30 minutes before the appointment.")	Pass
ATC-M16-B16	M14-FR4	Shared data: patient ticked Yes, cancelled, 20 minutes before the start	Patient ticked sharing: Yes; appointment cancelled; doctor opens it 20 minutes before the start	Shared data not shown	Shared data hidden ("This appointment was cancelled.")	Pass
ATC-INT-01	M2-FR10, M4-FR1, M9-FR6	A diabetic user with a dairy allergy gets a safe plan from profile to meals	Female, 165 cm, 72 kg, Diabetes, Dairy allergy, goal Lose Weight	Category Diabetic; meals have no dairy, no high-sugar or refined food; macros agree with the calorie target	Category Diabetic; 11 meal items; dairy items 0; high-sugar or refined items 0; macros give 1495 kcal against a target of 1494	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-INT-02	M2-FR10, M4-FR3, M9-FR1	A cardiac user gets a low-intensity plan and a calorie target above the safe floor	Male, 165 cm, 100 kg, Heart Disease, goal Lose Weight, asks for 6 days	Category Cardiac; intensity Low; at most 4 workout days; no HIIT; target not below 1500 kcal for a man	Category Cardiac; intensity Low; 4 workout days; 0 HIIT exercises; target 2107 kcal	Pass
ATC-INT-03	M8-FR8, M8-FR13	Typed meals become a daily summary with the correct totals and a fair verdict	Breakfast "2 rotis", Lunch "a bowl of daal"; burned 150 kcal; goal Lose Weight; target 1800; BMR 1400	Consumed 450, net 300, verdict not claiming a surplus	Consumed 450 kcal (Breakfast 220, Lunch 230); net 300; verdict: "That is a very low intake for you. If you skipped meals or did not log them, add them; if not, eat enough to fuel your body."	Pass
ATC-INT-04	M12-FR6, M12-FR8, M16-FR5	A week of activity earns a badge and raises no inactivity alert; a long gap raises one	7 consecutive active days; then no activity for 6 days with a plan for Mon/Wed/Fri	7-day streak badge earned and no "We miss you" notice; with the gap, a "We miss you" notice	Busy week: streak 7, badge daily_7 true, inactivity alert raised false; idle week: alert raised true	Pass
ATC-INT-05	M14-FR2	A booking is checked, paid through the test gateway and becomes a cancellable request without a video link	Valid slot on Monday 10:00, video, PKR 2,000 + PKR 150 platform fee, card 4242 4242 4242 4242	No booking problem; total PKR 2,150; payment succeeds with a reference; request is upcoming, cancellable, and cannot be joined yet	Booking problem: none; Pay enabled true; total PKR 2150; payment succeeded (reference sbx_c_3js4swwr2zra3f); upcoming true; cancellable true; joinable at the start false	Pass
ATC-INT-06	M14-FR5, M9-FR6	A specialist's diet plan changes the calorie target but never puts an allergen back into the meals	Baymax plan for a user allergic to Dairy and Nuts; nutritionist sets 1800 kcal	Target 1800 kcal; meals identical to Baymax's; still no allergen	Target 1800 kcal; meals unchanged true; allergen items 0 of 11	Pass
ATC-INT-07	M13-FR3	The chatbot answers calories left from the user's own plan and food log	Plan target for a Lose Weight user; 600 kcal logged; "how many calories left"	The answer contains target minus 600	Target 1494 kcal; expected "894" in the answer; answer starts: "You've eaten 600 of 1494 kcal today — 894 kcal left."	Pass
ATC-INT-10	M16-FR2	Nobody is an administrator until the one-time command is run on the server	Call the admin API with an ordinary account, run the one-time make-admin command on the server, then call again	Refused before the command, accepted after it	All checks in the scenario held (0.2 s)	Pass
ATC-INT-11	M14-FR11	A doctor registers; they get nothing until the administrator approves	A doctor registers through the API, tries the specialist screens, the administrator approves, the doctor tries again	No access while pending; access to the specialist API after approval	All checks in the scenario held (0.0 s)	Pass
ATC-INT-12	M14-FR15	The administrator adds a nutritionist directly (no password is ever set by the administrator)	The administrator adds a nutritionist by email	Account created without any password; role and public profile created	All checks in the scenario held (0.0 s)	Pass
ATC-INT-13	M14-FR2	The member (the phone) sees the specialists, holds a time, pays, and it becomes a request	A signed-in member (the phone) lists specialists, holds a time, pays	The booking becomes a request with the payment recorded and the time kept	All checks in the scenario held (0.0 s)	Pass
ATC-INT-14	M14-FR3, M14-FR12	The doctor sees the request, not the patient's data, and accepting creates the video room	The doctor opens the request, then accepts it	Request visible without the patient's data; after accepting a video room exists and the member is told	All checks in the scenario held (0.0 s)	Pass
ATC-INT-15	M14-FR13	My patients: the doctor now has a patient, and a record to work in	The doctor opens My patients and the patient's record	The patient is listed and the record opens	All checks in the scenario held (0.0 s)	Pass
ATC-INT-16	M14-FR5	The doctor sets a diet and a workout plan; the member can read them but never write them	The doctor sets a diet plan and a workout plan; the member then reads and tries to write them	Member can read both but the database refuses any write	All checks in the scenario held (0.0 s)	Pass
ATC-INT-17	M14-FR9	The nutritionist reviews a day of the food log (inside the sharing window); the member sees the verdict	The nutritionist reviews a day of the shared food log inside the sharing window	The verdict is stored and the member sees it	All checks in the scenario held (0.0 s)	Pass
ATC-INT-18	M14-FR14	A specialist closes their account: future bookings are cancelled and freed, the patient is told, the sign-in is deleted	A specialist closes their account	Future bookings cancelled and freed, patient told, sign-in deleted	All checks in the scenario held (0.0 s)	Pass
ATC-INT-19	M14-FR15	The administrator can delete another specialist, but not an administrator or themselves	The administrator deletes another specialist, then tries an administrator and themselves	The specialist is deleted; the other two are refused	All checks in the scenario held (0.0 s)	Pass
ATC-INT-20	M16-FR2	Nothing is readable without signing in, and members cannot read or write the protected nodes	Read the database without signing in, then as a member read and write protected nodes	Every read and write refused by the real database rules	All checks in the scenario held (0.3 s)	Pass
ATC-INT-21	M14-FR2	A member can hold a free time, and nobody else can take it while it is held	A member holds a free time; a second member tries the same time	First hold succeeds; the second is refused	All checks in the scenario held (0.1 s)	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-INT-22	M14-FR2	A hold must look right: own patient id, real fee, a hold of at most ten minutes, a real specialist, a future time	Holds with a wrong patient id, a wrong fee, a hold over ten minutes, a made-up specialist and a past time	Every malformed hold refused	All checks in the scenario held (0.0 s)	Pass
ATC-INT-25	M16-FR2	Only the two people involved can read an appointment, and a member cannot touch the specialist's notes or anyone else's booking	A third party and the two involved people read an appointment; a member edits notes and another member's booking	Only the two involved can read; edits refused	All checks in the scenario held (0.0 s)	Pass
ATC-INT-26	M14-FR2	Cancelling a held or requested booking frees the time; an expired hold can be taken over	Cancel a held and a requested booking; take over an expired hold	Times freed; the expired hold can be taken over	All checks in the scenario held (0.0 s)	Pass
ATC-INT-27	M14-FR3	Once the specialist accepts (portal server), the member keeps reading it but cannot cancel after the start or rewrite it	After the specialist accepts, the member reads, tries to cancel after the start and to rewrite the booking	Reading allowed; cancelling after the start and rewriting refused	All checks in the scenario held (0.0 s)	Pass
ATC-INT-30	M14-FR5, M14-FR9	Specialist plans and food log reviews: a member reads only their own and can never write them; the plan choice is theirs	A member reads and tries to write specialist plans and reviews; sets the plan choice	Reads only their own; writes refused; the plan choice is theirs	All checks in the scenario held (0.0 s)	Pass
ATC-INT-31	M16-FR2	The saved login signs in and is accepted by the admin API; the file is private and the password is not printed	Run create-admin; sign in with the saved login; call the admin API	The login works, the file is private and the password is never printed	All checks in the scenario held (0.2 s)	Pass
ATC-INT-32	M16-FR2	Running it again does not change the password or the file, unless a reset is requested	Run create-admin a second time for the same email, then with a reset	Password and file unchanged unless a reset is requested	All checks in the scenario held (0.3 s)	Pass
ATC-INT-33	M16-FR2	An ordinary member who is not an admin cannot use the admin API, and an existing member account is not hijacked	An ordinary member calls the admin API; create-admin on an existing member account	Refused; the existing member account is not hijacked	All checks in the scenario held (0.1 s)	Pass

Table 5.58 Automated test cases for Modules 4, 5 and 6 (headless and real device)

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M4-U07	M4-SFR1	The free-exercise-db dataset is imported with every usable record	exercise_catalog.json (pinned commit f00c92c)	876 records, 871 imported, 5 without instructions dropped	Records 876; imported 871; loaded 871; dropped 5	Pass
ATC-M4-U08	M4-SFR1	Exercise IDs are unique and a dataset duplicate of a curated exercise is not offered twice	unified catalogue (curated + dataset)	No duplicate ID; "Bodyweight_Squat" hidden behind curated "squat"	887 exercises; duplicates 0; Bodyweight_Squat listed: false	Pass
ATC-M4-U09	M4-SFR1	A missing optional field is tolerated, not invented	dataset record "Bicycling" (mechanic not stated)	Loaded with an empty mechanic	Loaded: true; mechanic ""	Pass
ATC-M4-U10	M4-SFR3	Equipment exercises are offered only for equipment the user owns	owned equipment {dumbbell}; then none	Every offered exercise needs a dumbbell; none offered without equipment	121 offered with a dumbbell, all dumbbell: true; without equipment: 0	Pass
ATC-M4-U11	M4-FR1	Expert-level dataset exercises are never scheduled (program rule)	healthy user; dataset exercise at level 3	Excluded by PRG-NO-MAXIMAL	Outcome excluded; rules [PRG-NO-MAXIMAL]	Pass
ATC-M4-U12	M4-FR2	Each health category maps to its training profile	categories Healthy, Cardiac, Obese, an unknown category	Healthy vigorous; Cardiac and Obese moderate; unknown falls back to At-Risk	Healthy vigorous; Cardiac moderate; Obese moderate; unknown → At-Risk	Pass
ATC-M4-U13	M4-FR1	Hypertension excludes high-intensity intervals and names the rule	condition "Hypertension"; HIIT exercise	Excluded; rule MED-HTN	Outcome excluded; rules [MED-HTN]	Pass
ATC-M4-U14	M4-FR1	Several conditions merge to the most restrictive limits	Hypertension + Diabetes; Hypertension alone	Both conditions apply; days capped at the smallest maximum	Rules applied: MED-HTN, MED-DIABETES; max days 5 (Hypertension alone 5)	Pass
ATC-M4-U15	M4-FR5	Owned equipment brings dataset exercises in, with the reason shown	Gain Muscle; owns dumbbells	At least one dataset exercise whose first reason starts "Uses equipment you have"	8 dataset exercises; first reason "Uses equipment you have (dumbbell)"	Pass
ATC-M4-U16	M4-FR5	The week has the chosen number of training days; the rest are rest days	healthy user, 4 days a week	7 days: 4 workouts, 3 rest days	7 days; 4 workouts; 3 rest days	Pass
ATC-M4-U17	M4-FR5	The same inputs always give the same plan (deterministic)	same profile and variant, generated twice	Identical exercise lists	Identical: true (36 entries)	Pass
ATC-M4-U18	M4-SFR2	MET values are matched to the 2024 Compendium with a recorded match type	all 871 dataset exercises	181 exact, 606 approximate, 84 unmatched	Exact 181; approximate 606; unmatched 84	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M4-U19	M4-SFR2	An exercise without a MET value is never offered for planning	every equipment type owned	All offered exercises have a MET value	482 offered; 0 without MET	Pass
ATC-M4-U20	M4-SFR2	Calories = MET × body weight × active hours	MET 3.8, 70 kg, 30 min active	133 kcal	kcal = 133.00	Pass
ATC-M4-U21	M4-SFR2	Intensity bands follow the WHO MET bands at the boundaries	MET 2.99, 3.0, 5.99, 6.0	light, moderate, moderate, vigorous	Bands: light, moderate, moderate, vigorous	Pass
ATC-M4-U22	M4-FR2	Diabetes: no more than two rest days in a row	condition "Diabetes", 2 days a week chosen	3 workout days; longest rest run 2 days	3 workout days; longest rest 2 days	Pass
ATC-M4-U23	M4-SFR3	Warning symptoms stop plan generation and give the reason	readiness: chest pain reported	No workout days; gate needsClearance (SCR-SYMPTOMS)	Workout days 0; gate needsClearance; rules [SCR-SYMPTOMS]	Pass
ATC-M4-U24	M4-SFR4	A planned exercise counts as completed only when its target is reached	manual session: 2 of 2 sets; then skipped	Completed true; skipped false	Completed: true; when skipped: false	Pass
ATC-M5-U01	M5-FR16.1	Joint angle at the middle landmark is computed correctly	points (1,0), (0,0), (0,1)	90°	Angle = 90.00°	Pass
ATC-M5-U02	M5-FR16.4	Landmarks below the visibility threshold are not used	squat frame with visibility 0.3 (threshold 0.5)	No measurement	Measurement: none	Pass
ATC-M5-U03	M5-FR16.2	Missing landmarks are named in the guidance	squat (dataset) with knees and ankles hidden	Feedback starts "Can't see your knees and ankles"	Feedback: "Can't see your knees and ankles — stand side-on to the camera, 2–3 m away, with your whole body in the frame"	Pass
ATC-M5-U04	M5-FR16.3	The movement phases follow top → descending → bottom → ascending → top	one squat to 88°	Phases seen in that order	Phases: unknown → top → descending → bottom → ascending → top	Pass
ATC-M5-U05	M5-FR16.3	Three full squats count three reps	3 squats to 88°	3 reps, 0 partial	Reps 3; partial 0	Pass
ATC-M5-U06	M5-FR16.3	A half repetition is not counted	1 squat that turns back at 125°	0 reps, 1 partial attempt	Reps 0; partial 1	Pass
ATC-M5-U07	M5-FR16.3	A movement faster than the minimum rep time is rejected as noise	squat down and up in 300 ms (minimum 600 ms)	0 reps	Reps 0	Pass
ATC-M5-U08	M5-FR16.3	The angle is smoothed (exponential average)	samples 170° then 130°	Smoothed angle 150°	Smoothed = 150.0	Pass
ATC-M5-U09	M5-SFR2	A plank hold is timed only while in position	5 s straight plank, then 3 s sagging	About 5 s held	Held 4 s	Pass
ATC-M5-U10	M5-FR16.2	The back-alignment rule passes upright and fails a deep forward lean	torso lean 30° and 70° (limit 60°)	Pass at 30°, fail at 70°	30°: pass; 70°: fail	Pass
ATC-M5-U11	M5-FR16.2	The hip rule gives the right correction for sagging and piking	hip offset -0.12 and +0.20	"Raise your hips…" and "Lower your hips…"	Sag: "Raise your hips slightly — don't let them sag"; pike: "Lower your hips into a straight line"	Pass
ATC-M5-U12	M5-FR18.1	Posture score = acceptable reps ÷ attempted reps	2 good squats and 1 shallow attempt	Score 67%	Score 67% (2 reps, 1 partial)	Pass
ATC-M5-U13	M5-FR16.2	A correction is shown only after the fault persists for 3 frames	forward lean of 75° held during the descent	"Keep your back straight" appears, never on the first faulty frame	First correction at frame 26 of 39	Pass
ATC-M5-U14	M5-SFR1	Exercises map to a movement pattern by movement, not muscle	Barbell_Squat, Leg_Extensions, Dumbbell_Bicep_Curl	squat, legExtension, curl	Patterns: squat, legExtension, curl	Pass
ATC-M5-U15	M5-FR15.1	An exercise the camera cannot follow is unsupported with a reason	Standing_Calf_Raises	Not supported; reason mentions a movement too small to measure	Level notSupported; reason "The movement is too small for the camera to measure reliably (for example wrists, calves, shoulder shrugs or a crunch)."	Pass
ATC-M5-U16	M5-FR15.1	Camera capability of every schedulable exercise (current code)	curated library + dataset exercises for any equipment, program rule applied	Counts reported from the registry; only the 4 tuned exercises are validated	532 schedulable: validated 4; experimental 300; basic 0; unsupported 228	Pass
ATC-M5-U17	M5-SFR1	Number of movement patterns with a tracker	MovementPattern enum	Every pattern except "none" builds a tracker	26 patterns, 26 with a tracker (2 hold, 24 repetition)	Pass
ATC-M5-U18	M5-FR16.3	An opening movement (overhead press) is counted on the mirrored angle	3 presses 110° → 150°; one-arm hanging arm ignored	3 reps	Reps 3	Pass
ATC-M6-U01	M6-SFR1	A rep question is recognised	"How many reps have I done?"	Intent repCount	Intent = repCount	Pass
ATC-M6-U02	M6-SFR1	A form question is recognised	"How's my form?"	Intent postureFeedback	Intent = postureFeedback	Pass
ATC-M6-U03	M6-SFR1	A pause command is recognised	"Pause workout"	Intent pauseSession	Intent = pauseSession	Pass
ATC-M6-U04	M6-SFR1	An unrelated question falls back to unknown	"what is the capital of France"	Intent unknown	Intent = unknown	Pass
ATC-M6-U05	M6-SFR1	Empty speech input is not treated as a command	" "	Intent unknown	Intent = unknown	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M6-U06	M6-SFR1	A pain report is recognised as a safety intent	"My knee hurts"	Intent painOrSymptom	Intent = painOrSymptom	Pass
ATC-M6-U07	M6-FR19.3	The reply uses the live rep counter, never an invented number	camera session with 2 counted squats; "how many reps"	Reply "You've completed 2 reps…"	Reply: "You've completed 2 reps of 36, 34 to go."	Pass
ATC-M6-U08	M6-FR19.3	With no session the assistant says so instead of inventing numbers	no active session; "How many reps?"	Reply mentions no active workout session	Reply: "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."	Pass
ATC-M6-U09	M6-FR19.5	Quick-action chips are contextual and at most three	"what is the capital of France"	1–3 chips	2 chips: What's my workout today?, How do I do a squat?	Pass
ATC-M6-U10	M6-FR19.2	Microphone denied: the assistant never listens	microphone permission denied	State unavailable; 0 listen calls	State unavailable; listen calls 0; message "Microphone access is needed for voice commands. You can still tap a suggestion below."	Pass
ATC-M6-U11	M6-FR19.2	Microphone permanently denied: the settings link is offered	microphone permanently denied	settingsNeeded true; no listening	Settings needed true; listen calls 0	Pass
ATC-M6-U12	M6-FR19.1	Recognition errors give a clear message and nothing is spoken	no-match error; then timeout error	"didn't catch that"; "didn't hear anything"; no speech output	No match: "I didn't catch that. Tap the mic and try again."; timeout: "I didn't hear anything. Tap the mic and try again."; spoken 0	Pass
ATC-M6-U13	M6-FR19.3	A speech-synthesis failure still shows the reply	text-to-speech fails; "hello"	Reply shown; speechOutputFailed true; state idle	Reply present true; output failed true; state idle	Pass
ATC-M6-U14	M6-FR19.4	Closing releases the microphone and ignores a pending command	listening; close; late "pause workout"	Recognizer cancelled once; session still active	Cancel calls 1; session active	Pass
ATC-M6-U15	M6-FR19.3	Replies never overlap; replay speaks the same reply once more	"hello", then replay	2 identical utterances; at most 1 at a time	Spoken 2; same true; max concurrent 1	Pass
ATC-M4-F03	M4-FR6	Exercise detail shows the prescribed dose and a start/end illustration	Planned "Jumping Jacks": 2 sets of 30 s, rest 40 s	Sets, per-set time and rest shown; illustration caption shown	Sets 1; "30s" 1; rest "40s" 1; illustration caption 1	Pass
ATC-M4-F04	M4-FR6	A dataset exercise shows its photos with the source caption	Planned "Dumbbell Bicep Curl": 3 × 10	Caption "Photos: free-exercise-db (public domain)…"; technique steps listed	Photo caption 1; Technique heading 1	Pass
ATC-M4-B01	M4-SFR3	Screening gate: symptoms No, heart condition No, clearance Not needed, profile data Yes	Conditions none; symptoms none; clearance null; weight 70 kg	Plan generated	Gate eligible; workout days 4	Pass
ATC-M4-B02	M4-SFR3	Screening gate: symptoms Yes, heart condition No, clearance Not needed, profile data Yes	Conditions none; symptoms chest_pain; clearance null; weight 70 kg	No plan; gate needsClearance (SCR-SYMPTOMS)	Gate needsClearance (SCR-SYMPTOMS); workout days 0	Pass
ATC-M4-B03	M4-SFR3	Screening gate: symptoms No, heart condition Yes, clearance No, profile data Yes	Conditions Heart Disease; symptoms none; clearance false; weight 70 kg	No plan; gate needsClearance (SCR-CARDIAC-CLEARANCE)	Gate needsClearance (SCR-CARDIAC-CLEARANCE); workout days 0	Pass
ATC-M4-B04	M4-SFR3	Screening gate: symptoms No, heart condition Yes, clearance Yes, profile data Yes	Conditions Heart Disease; symptoms none; clearance true; weight 70 kg	Plan generated	Gate eligible; workout days 4	Pass
ATC-M4-B05	M4-SFR3	Screening gate: symptoms No, heart condition No, clearance Not needed, profile data No	Conditions none; symptoms none; clearance null; weight 0 kg	No plan; gate needsInfo (SCR-MISSING-WEIGHT)	Gate needsInfo (SCR-MISSING-WEIGHT); workout days 0	Pass
ATC-M4-B06	M4-FR1	Exercise eligibility: Healthy, impact Low, tag None, injury No, MET Yes	Profile Healthy; exercise impact low, tags none, MET 4.0; injuries none	Outcome eligible	Outcome eligible	Pass
ATC-M4-B07	M4-FR1	Exercise eligibility: Hypertension, impact Low, tag HIIT, injury No, MET Yes	Profile At-Risk; exercise impact low, tags hiit, MET 4.0; injuries none	Outcome excluded	Outcome excluded; rules MED-HTN	Pass
ATC-M4-B08	M4-FR1	Exercise eligibility: Obese, impact High, tag None, injury No, MET Yes	Profile Obese; exercise impact high, tags none, MET 4.0; injuries none	Outcome excluded	Outcome excluded; rules BODY-OBESE	Pass
ATC-M4-B09	M4-FR1	Exercise eligibility: Healthy, impact High, tag None, injury Knee, MET Yes	Profile Healthy; exercise impact high, tags none, MET 4.0; injuries knee	Outcome excluded	Outcome excluded; rules INJ-KNEE	Pass
ATC-M4-B10	M4-FR1	Exercise eligibility: Healthy, impact Low, tag None, injury Knee, MET Yes	Profile Healthy; exercise impact low, tags none, MET 4.0; injuries knee	Outcome eligibleWithModifications	Outcome eligibleWithModifications; rules INJ-KNEE	Pass
ATC-M4-B11	M4-FR1	Exercise eligibility: Hypertension, impact Low, tag None, injury No, MET No	Profile At-Risk; exercise impact low, tags none, MET 0.0; injuries none	Outcome requiresInformation	Outcome requiresInformation; rules PRF-AT-RISK	Pass
ATC-M4-B12	M4-SFR3	Equipment rule: owns Dumbbell, exercise needs Dumbbell	Owned equipment dumbbell; exercise Dumbbell_Bicep_Curl	Offered for planning	Offered: true	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M4-B13	M4-SFR3	Equipment rule: owns None, exercise needs Dumbbell	Owned equipment none; exercise Dumbbell_Bicep_Curl	Not offered	Offered: false	Pass
ATC-M4-B14	M4-SFR3	Equipment rule: owns Dumbbell, exercise needs Barbell	Owned equipment dumbbell; exercise Barbell_Squat	Not offered	Offered: false	Pass
ATC-M5-B01	M5-FR15.1	Camera capability of squat: validatedForm	Exercise squat	Camera available; form checks yes; score labelled validated yes	Level validatedForm; camera true; form true; validated score true	Pass
ATC-M5-B02	M5-FR15.1	Camera capability of Barbell_Squat: experimentalForm	Exercise Barbell_Squat	Camera available; form checks yes; score labelled validated no	Level experimentalForm; camera true; form true; validated score false	Pass
ATC-M5-B03	M5-FR15.1	Camera capability of Standing_Calf_Raises: notSupported	Exercise Standing_Calf_Raises	Camera not available; form checks no; score labelled validated no	Level notSupported; camera false; form false; validated score false; reason "The movement is too small for the camera to measure reliably (for example wrists, calves, shoulder shrugs or a crunch)."	Pass
ATC-M5-B04	M5-FR16.3	Rep acceptance: depth reached Yes, duration ≥ minimum Yes	One squat (to 88°, 1.8 s)	1 rep, 0 partial	Reps 1; partial 0	Pass
ATC-M5-B05	M5-FR16.3	Rep acceptance: depth reached No, duration ≥ minimum Yes	One squat (turning at 125°, 1.8 s)	0 rep, 1 partial	Reps 0; partial 1	Pass
ATC-M5-B06	M5-FR16.3	Rep acceptance: depth reached Yes, duration ≥ minimum No	One squat (to 88°, 0.3 s)	0 rep, 0 partial	Reps 0; partial 0	Pass
ATC-M5-B07	M5-FR16.4	Landmark visibility 0.9 (threshold 0.5)	Squat frame at 120°, landmark visibility 0.9	Angle measured	Measured 120°	Pass
ATC-M5-B08	M5-FR16.4	Landmark visibility 0.3 (threshold 0.5)	Squat frame at 120°, landmark visibility 0.3	No measurement; nothing counted	Measurement: none	Pass
ATC-M5-B09	M5-FR16.2	Correction trigger: fault No, persists ≥ 3 frames n/a	Descent with 10° forward lean for 600 ms (limit 60°)	No correction	Frames with the correction: 0 of 45	Pass
ATC-M5-B10	M5-FR16.2	Correction trigger: fault Yes, persists ≥ 3 frames No	Descent with 75° forward lean for 66 ms (limit 60°)	No correction	Frames with the correction: 0 of 29	Pass
ATC-M5-B11	M5-FR16.2	Correction trigger: fault Yes, persists ≥ 3 frames Yes	Descent with 75° forward lean for 600 ms (limit 60°)	"Keep your back straight" shown	Frames with the correction: 16 of 45	Pass
ATC-M5-B12	M5-SFR3	Plan completion: all sets Yes, skipped No	Manual squat session, 2 sets planned; 2 set(s) recorded; Finish	completedTarget true	completedTarget true	Pass
ATC-M5-B13	M5-SFR3	Plan completion: all sets No, skipped No	Manual squat session, 2 sets planned; 1 set(s) recorded; Finish	completedTarget false	completedTarget false	Pass
ATC-M5-B14	M5-SFR3	Plan completion: all sets Yes, skipped Yes	Manual squat session, 2 sets planned; 2 set(s) recorded; Skip	completedTarget false	completedTarget false	Pass
ATC-M6-B01	M6-SFR3	Voice command "pause workout" with session active	Recognised text "pause workout"; session active	Session paused; reply contains "Paused"	Session paused; reply "Paused. Say resume when you are ready."	Pass
ATC-M6-B02	M6-SFR3	Voice command "resume workout" with session paused	Recognised text "resume workout"; session paused	Session active; reply contains "Resuming"	Session active; reply "Resuming. Let’s go."	Pass
ATC-M6-B03	M6-SFR3	Voice command "pause workout" with session none	Recognised text "pause workout"; session none	Session idle; reply contains "no active workout session"	Session idle; reply "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."	Pass
ATC-M6-B04	M6-SFR3	Voice command "my knee hurts" with session active	Recognised text "my knee hurts"; session active	Session paused; reply contains "Please stop the exercise now"	Session paused; reply "Please stop the exercise now and rest. Pain, dizziness or chest discomfort are signals to stop, not to push through. I can't diagnose injuries — if it is severe or doesn't settle, contact a doctor or emergency services. I've paused your session."	Pass
ATC-M6-B05	M6-FR19.2	Microphone permission granted	Assistant opened; microphone granted	State listening; settings link false	State listening; settings link false; listen calls 1	Pass
ATC-M6-B06	M6-FR19.2	Microphone permission denied	Assistant opened; microphone denied	State unavailable; settings link false	State unavailable; settings link false; listen calls 0	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M6-B07	M6-FR19.2	Microphone permission permanently denied	Assistant opened; microphone permanently denied	State unavailable; settings link true	State unavailable; settings link true; listen calls 0	Pass
ATC-M6-B08	M6-FR19.1	Recognition result: final text "how many reps"	Simulated recogniser returns final text "how many reps"	Reply shown and spoken	Reply "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."; spoken 1	Pass
ATC-M6-B09	M6-FR19.1	Recognition result: no match	Simulated recogniser returns no match	Message "didn't catch that"; nothing spoken	Message "I didn't catch that. Tap the mic and try again."; spoken 0	Pass
ATC-M6-B10	M6-FR19.1	Recognition result: timeout (silence)	Simulated recogniser returns timeout (silence)	Message "didn't hear anything"; nothing spoken	Message "I didn't hear anything. Tap the mic and try again."; spoken 0	Pass
ATC-M6-B11	M6-FR19.3	Text-to-speech works	"hello"; speech output available	Reply shown and spoken	Reply shown true; output failed false	Pass
ATC-M6-B12	M6-FR19.3	Text-to-speech fails	"hello"; speech output fails	Reply still shown; failure flagged	Reply shown true; output failed true	Pass
ATC-M4-F05	M4-SFR4	"Start exercise" opens mode selection linked to the planned exercise	Generated plan, Monday, 4th exercise	Mode selection opens with plan link (Monday, entry 3)	Opened Push-Ups; plan day 0; entry 3	Pass
ATC-M4-F06	M4-FR2	The workout tab shows the training profile of the health classification	Profile with "Heart Disease", doctor's clearance given	"Your training profile: Cardiac" with "Highest intensity scheduled: Moderate"	Profile title 1; intensity line 1	Pass
ATC-M5-F01	M5-SFR3	Posture Detection offers Today's workout, My plan and the Exercise library	Signed-in user with a 6-day plan	Three tabs	3 of 3 tabs present	Pass
ATC-M5-F02	M5-SFR3	Without a plan an empty state is shown, not invented exercises	No health profile	"No personalised plan yet" and "Create my health profile"	Empty state 1; button 1	Pass
ATC-INT-34	M4-FR2	The Module 2 health category sets the intensity ceiling of the generated plan	Profile with "Heart Disease" (classified Cardiac), clearance given	Every scheduled exercise with a MET value is light or moderate	Category Cardiac; 36 exercises checked; vigorous 0	Pass
ATC-INT-35	M4-FR1	A condition recorded in Module 2 removes restricted exercises from the Module 4 plan	Profile with "Hypertension"	No HIIT- or plyometric-tagged exercise in the week	44 planned entries; restricted entries: none	Pass
ATC-INT-36	M5-SFR3	A planned exercise starts a posture session with the planned dose and a camera configuration	Generated plan, Monday; every entry converted to a session	Sets, reps and seconds carried over; every entry resolves a camera capability	9 entries; dose mismatches 0; 4 with camera tracking, 5 Manual Guidance only	Pass
ATC-INT-37	M4-SFR4	Completing every main exercise of the day marks the plan day done once	Monday plan; each main entry finished in Manual Guidance with all sets	Day marked once in the Module 4 activity record	Marked 1× (2026-10-19 Full-Body Fitness); done today 5 entries	Pass
ATC-INT-38	M4-SFR4	A skipped or incomplete session does not complete the planned exercise	Monday entry 1: one set of several, then skipped	Entry not done; day not marked	Entry done false; day marks 0	Pass
ATC-INT-39	M6-SFR2	A posture correction from the camera engine is spoken once by the cue announcer	Camera squat session; about 3 s at the bottom with 75° forward lean (limit 60°)	"Keep your back straight" spoken exactly once (rate-limited)	Spoken 1: Keep your back straight — lift your chest	Pass
ATC-INT-40	M6-SFR2	Asking "how's my form" answers from the live posture analysis	Same faulty descent; then "How's my form?"	Reply "Here is what to fix: Keep your back straight…"	Reply "Here is what to fix: Keep your back straight — lift your chest."	Pass
ATC-INT-41	M6-FR19.3	The rep count spoken by the assistant equals the camera counter	3 squats counted by the camera engine; "how many reps"	Reply states 3 reps	Camera counter 3; reply "You've completed 3 reps of 36, 33 to go."	Pass
ATC-INT-42	M6-SFR3	A spoken "pause" stops rep counting until "resume"	Camera session; "pause workout"; 2 squats while paused; "resume workout"; 1 squat	Paused then active; frames while paused ignored	After pause: paused; reps shown while paused 0; after resume: active	Pass
ATC-INT-43	M6-SFR3	Ending a session by voice needs confirmation before the session is asked to finish	"stop workout", then "yes"	Confirmation asked first, session still running; after "yes" a finish is requested from the session screen	After "stop": still active true, asked "Do you want to end this Bodyweight Squat session?"; after "yes": finish requested true, active true	Pass
ATC-INT-44	M4-SFR4	Profile → plan → posture sessions with voice queries → plan day completed	Healthy profile; Monday plan; each main entry done in Manual Guidance; "how many reps" asked in each	Voice replies match the logged reps; Monday marked done once; every session saved	5 sessions; voice reply mismatches 0; saved 5; day marked 1×	Pass
ATC-M5-F03	M5-SFR3	My plan lists the real plan day with camera capability and starts an entry with its plan link	Tab "My plan", Monday, 3rd exercise	Entry and capability shown; mode selection receives exercise, sets and plan link	Capability label 1; opened Lunges (3 sets), plan entry 2	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-M5-F04	M5-SFR3	The generic exercise library is kept with camera and manual sections	Tab "Exercise library"	Note that it is not the personalised plan; "Camera posture detection" and "Manual guidance" sections	Generic-library note 1; camera section 1; manual section 1	Pass
ATC-M5-F05	M5-FR15.3	Camera permission is not requested before the consent toggle is on	Bodyweight Squat; Continue tapped with consent off	Both modes shown; 0 permission requests; consent prompt shown	Mode options 2; camera requests 0; camera screen opened 0	Pass
ATC-M5-F06	M5-FR15.4	"Skip for now" returns to the previous screen	Mode selection opened from a screen; tap "Skip for now"	Mode selection closed	Mode selection still open: 0	Pass
ATC-M5-F07	M5-FR15.1	Refused camera permission keeps the camera off and offers Manual Guidance	Consent on; camera permission denied	"Camera access needed" and "Use Manual Guidance"; camera screen not opened	Dialog 1; manual offered 1; camera screen 0	Pass
ATC-M5-F08	M5-FR16.3	Camera screen counts reps from pose frames and shows status and privacy notice	2 squats fed as pose frames	Reps 2; "Landmarks detected", "Angles calculated"; local-processing notice	Reps 2; status chips 1/1; notice 1	Pass
ATC-M5-F09	M5-FR16.5	"Stop Session" ends the session and shows the summary with score, stats and feedback	2 good squats, then Stop Session	Summary: "Session Complete!", posture score 100%, Total Reps, Accuracy, form feedback; session saved	Summary 1; score 100%; stats 1/1; feedback 1; saved 1	Pass
ATC-M5-F10	M5-FR17.2	Manual guidance: confirmed sets add reps; the form checklist and response buttons are shown	Bodyweight Squat 3 × 12; tap "Completed ✓" once	12 reps, 1 set; checklist step and Skip / Too Difficult buttons present	Reps 12; sets 1; checklist step 1; Too Difficult 1; Skip 1	Pass
ATC-M5-F11	M5-FR17.5	"Next Exercise" advances to the following exercise	Plank 1 × 20 s then Squat; hold finished	Next exercise is Bodyweight Squat; plank saved in history	Current exercise squat; last finished plank	Pass
ATC-M5-F12	M5-SFR2	A plank in camera mode shows a hold timer instead of a rep counter	Plank 2 × 20 s; 6 s in position	"HOLD" badge and "In position"; about 5–6 s held	HOLD badge 1; "In position" 1; held 5 s	Pass
ATC-M5-F13	M5-FR15.1	An exercise the camera cannot follow explains why and keeps Manual Guidance	Calf Raises	Camera not available with the reason; Continue opens Manual Guidance	Reason shown 1; session mode manual; camera requests 0	Pass
ATC-M6-F01	M6-FR19.1	Opening the assistant asks for the microphone and shows the listening state	Voice assistant opened	"Listening..." and "Cancel"; 1 microphone request	Listening 1; Cancel 1; microphone requests 1	Pass
ATC-M6-F02	M6-FR19.3	A spoken question shows the transcript, the reply card, at most three chips and is spoken aloud	Recognised speech "how do I do a squat" (simulated recogniser)	Transcript shown; reply about Bodyweight Squat spoken; ≤ 3 chips	Transcript 1; spoken "Here is how to do Bodyweight Squat"; chips 2	Pass
ATC-M6-F03	M6-FR19.2	A refused microphone is explained and a retry is offered	Microphone permission denied	"Voice input unavailable", the reason and "Try again"; not listening	Unavailable 1; reason 1; retry 1; listen calls 0	Pass
ATC-M6-F04	M6-FR19.4	Leaving the assistant releases the microphone without running a command	Assistant listening; screen closed; a late "pause workout"	Recognizer cancelled once; no command executed	Cancel calls 1; session active	Pass
ATC-DEV-S01	M3-FR7	The app shell starts on the phone and the bottom navigation opens the Workout tab and returns Home	Controlled test user "Device Tester" (in memory); tap "Workout", then "Home"	Home shows "TODAY'S WORKOUT"; Workout tab shows "Your workout plan"; Home shown again	Home shown (greeting "Device" 1); Workout tab "Your workout plan" 1; returned Home	Pass
ATC-DEV-M4-01	M4-FR5	The Workout tab shows the weekly plan generated for the profile	Test user: healthy, 6 training days a week; open the Workout tab	"Your workout plan", "<intensity> intensity · 6 days a week", 7 day buttons of which 1 is a rest day	Summary "Moderate intensity · 6 days a week" shown 1; day buttons 7; rest-day buttons 1	Pass
ATC-DEV-M4-02	M4-FR5	Selecting a workout day shows that day's session and its exercise count	Workout tab; tap the first training day that is not already selected	Header "<day name> · <focus>" and an "Exercises" count equal to the day's main exercises	Monday: header "Full-Body Fitness · Total body" shown 1; 9 planned entries (5 main); exercise cards on screen 9	Pass
ATC-DEV-M4-03	M4-FR6	An exercise opens its detail screen with the planned sets, reps or time and rest, and Back returns to the plan	Workout tab; tap the first exercise card; then Back	Exercise Detail shows the exercise name and the planned dose; the plan is shown again after Back	"Dynamic Warm-Up" 1 sets × 3m, rest 0 s: name 1, sets 1, reps/time 2, rest 1; back on the plan	Pass
ATC-DEV-M4-04	M4-FR5	Regenerate plan produces a different exercise selection (cloud save unavailable in the test session)	Workout tab; "Regenerate plan" → confirm "Regenerate"	The weekly plan changes; the app reports that the new plan could not be saved to the server (no account in the test)	Plan changed: true; message "New plan generated, but it could not be saved to the server."	Pass
ATC-DEV-M4-05	M4-FR2	The training profile card names the profile of the health classification	Test user with "Hypertension"; Workout tab	"Your training profile: <category>" for the classifier's category	Category At-Risk; card "Your training profile: At-Risk" shown 1	Pass
ATC-DEV-M4-06	M4-SFR3	A reported warning symptom blocks plan generation and the reason is shown (negative case)	Test user; readiness answer: chest pain; Workout tab	"A doctor's clearance comes first" and rule SCR-SYMPTOMS; no exercise cards	Gate shown; rule SCR-SYMPTOMS shown 1; exercise cards 0	Pass
ATC-DEV-M4-07	M4-FR1	Without a health profile (no classification) no plan is invented (negative case)	Test session with no profile; Workout tab	"Complete your health profile to get a personalised workout plan."	Empty-state message shown 1; plan null	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-DEV-M5-01	M5-SFR3	Posture check opens Posture Detection with Today's workout, My plan and the Exercise library	Home → menu → "Posture check"	Tabs "Today's workout", "My plan", "Exercise library"	Tabs shown: Today's workout, My plan, Exercise library	Pass
ATC-DEV-M5-02	M5-SFR3	Today's workout lists today's plan entries (or the rest day) from the personalised plan	Posture Detection → "Today's workout"	One entry per exercise of today's plan day, or the rest-day message when today is a rest day	Today (weekday 6) is a rest day in the plan; rest-day message shown 1	Pass
ATC-DEV-M5-03	M5-SFR3	My plan shows every day of the complete plan with each entry's camera capability	Posture Detection → "My plan" → each training day	For every training day: its entries with a capability label	6 training days checked. Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1; Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1	Pass
ATC-DEV-M5-04	M5-SFR3	The generic Exercise library keeps its camera and manual sections	Posture Detection → "Exercise library"	"Camera posture detection" and "Manual guidance" sections	Camera section shown; manual section shown 1	Pass
ATC-DEV-M5-05	M5-FR15.3	A planned camera exercise opens mode selection; Continue without consent does not start the camera	My plan → a camera-tracked planned exercise → Continue with the consent toggle off	Both modes and the consent switch; the consent message; no camera screen	Entry 0-1: modes 2, consent switch 1, consent message shown, camera screen 0	Pass
ATC-DEV-M5-06	M5-FR16.5	With consent the real camera session starts on the phone, and Stop Session ends it with a summary	Planned camera exercise → consent on → Continue → about 8 s of live camera → "Stop Session"	Live camera overlay running, exercise name or setup hint, privacy notice; after Stop: "Session Summary" whose reps equal the session counter; saved in memory only	Blocked: camera permission for the installed test build is "denied" on the phone (a reinstall resets it); the tests do not grant permissions. Allow Camera for Baymax in Android Settings and rerun	Blocked
ATC-DEV-M5-07	M5-FR17.2	Manual Guidance shows the set counter, checklist and controls, and Pause stops the clock	Planned exercise → Manual Guidance → Continue → Pause → Resume	"Manual Guidance", "Proper Form", "Completed ✓", "Skip", "Too Difficult"; Pause switches to "Resume" and back	Shown: Proper Form, Completed ✓, Skip, Too Difficult; Resume after Pause 1; Pause after Resume 1	Pass
ATC-DEV-M5-08	M5-FR15.1	A refused camera permission keeps the camera off and offers Manual Guidance (denial simulated in the app, no system dialog)	Camera permission answer "denied" (AppPermissions override); consent on → Continue → "Use Manual Guidance" → Continue	"Camera access needed" dialog; no camera screen; the mode switches to Manual Guidance and Continue opens it	Dialog shown; camera screen 0; mode switched to Manual Guidance; Continue opened Manual Guidance	Pass
ATC-DEV-M5-09	M5-FR15.1	An exercise the camera cannot follow says why and still offers Manual Guidance	Exercise library → first exercise in "Manual guidance" section	Mode selection explains that camera detection is not available, with the reason; Manual Guidance available	Ankle Circles: reason "Stretches and mobility moves are held in many different positions, so there is no single movement the camera could follow reliably" shown true; Manual Guidance option 1	Pass
ATC-DEV-M6-01	M6-FR19.1	The chatbot's mic opens the Module 6 voice assistant, which shows the state matching the phone's microphone permission	Home → AI chatbot (Module 13) → "Voice assistant"; microphone permission as set on the phone	Voice assistant screen (not the chatbot); granted → "Listening..."; not granted → "Voice input unavailable" with the reason and "Try again"	Phone microphone permission: denied; assistant state: unavailable; reason shown 1; "Try again" 1; chatbot still on top: 0	Pass
ATC-DEV-M6-02	M6-FR19.3	A quick-action chip is answered from the real session state and spoken with the phone's text-to-speech	No active workout; tap the chip "How many reps?"	Reply card says there is no active workout session (no invented numbers); text-to-speech reports no failure; at most 3 chips	Reply: "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."; engine On-device rules (Dialogflow unavailable); speech output failed false; chips 2	Pass
ATC-DEV-M6-03	M6-FR19.3	"What's my workout today?" is answered from the personalised plan	Test user's plan; tap the chip "What's my workout today?"	Reply names today's plan day, or says today is a rest day, matching the plan	Plan today: rest day; reply "Today is a rest day in your plan. Light walking or stretching is fine if you feel like moving."	Pass
ATC-DEV-M6-04	M6-FR19.4	The close button leaves the voice assistant and the app stays usable	Home → chatbot → voice assistant → "Close voice assistant" → Back	Voice assistant closed; chatbot shown; Back returns to Home	Voice assistant closed; chatbot shown 1; Home shown after Back	Pass

Test Case ID	Related Requirement	Test Objective	Test Input / Condition	Expected Result	Actual Result	Status
ATC-DEV-INT-01	M4-SFR4	Home → plan → exercise → Manual Guidance session → summary → the planned exercise is recorded as completed (Modules 4 → 5 → 4)	Test user (6 days a week); Workout tab → training day (today, or the first training day when today is a rest day) → its quickest repetition exercise → Start exercise → Manual Guidance → "Completed ✓" for every set (rest timers run) → Finish → Back to Workout Plan	Summary "Confirmed N reps" with N = sets × reps; target reached; session saved with its plan link; plan progress records the entry as done	Monday (today is a rest day): Bodyweight Squat 3×12 (entry 1, plan link 0/1); summary opened by itself after the last set, label "Confirmed 36 reps" 1; target reached true; saved with plan link true (in memory); plan progress done true	Pass
ATC-DEV-INT-02	M6-SFR2	During a live camera session the voice assistant answers from that session and the session continues (Modules 5 ↔ 6)	Planned camera exercise → consent → Continue → menu "Ask Baymax (voice)" → chip "How many reps?" → close	Reply states the camera counter's reps for that exercise; after closing, the camera session is still active	Blocked: camera permission for the installed test build is "denied" on the phone (a reinstall resets it); the tests do not grant permissions. Allow Camera for Baymax in Android Settings and rerun	Blocked
ATC-DEV-INT-03	M6-SFR3	The voice chip "Pause workout" pauses the running posture session (Module 6 → 5)	Planned exercise → Manual Guidance → "Ask Baymax" → chip "How many reps?" → chip "Pause workout" → close	Session paused; the guidance screen shows "Resume"	Session paused; "Resume" shown 1	Pass

Table 5.59 Test source mapping for Modules 4, 5 and 6

Test Case ID	Testing category	Test file	Test name	Framework	-	Automated
ATC-M4-U07	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U07 The free-exercise-db dataset is imported with every usable record	Flutter test (headless)	-	Yes
ATC-M4-U08	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U08 Exercise IDs are unique and a dataset duplicate of a curated exercise is not offered twice	Flutter test (headless)	-	Yes
ATC-M4-U09	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U09 A missing optional field is tolerated, not invented	Flutter test (headless)	-	Yes
ATC-M4-U10	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U10 Equipment exercises are offered only for equipment the user owns	Flutter test (headless)	-	Yes
ATC-M4-U11	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U11 Expert-level dataset exercises are never scheduled (program rule)	Flutter test (headless)	-	Yes
ATC-M4-U12	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U12 Each health category maps to its training profile	Flutter test (headless)	-	Yes
ATC-M4-U13	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U13 Hypertension excludes high-intensity intervals and names the rule	Flutter test (headless)	-	Yes
ATC-M4-U14	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U14 Several conditions merge to the most restrictive limits	Flutter test (headless)	-	Yes
ATC-M4-U15	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U15 Owned equipment brings dataset exercises in, with the reason shown	Flutter test (headless)	-	Yes
ATC-M4-U16	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U16 The week has the chosen number of training days; the rest are rest days	Flutter test (headless)	-	Yes
ATC-M4-U17	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U17 The same inputs always give the same plan (deterministic)	Flutter test (headless)	-	Yes
ATC-M4-U18	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U18 MET values are matched to the 2024 Compendium with a recorded match type	Flutter test (headless)	-	Yes
ATC-M4-U19	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U19 An exercise without a MET value is never offered for planning	Flutter test (headless)	-	Yes
ATC-M4-U20	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U20 Calories = MET × body weight × active hours	Flutter test (headless)	-	Yes
ATC-M4-U21	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U21 Intensity bands follow the WHO MET bands at the boundaries	Flutter test (headless)	-	Yes
ATC-M4-U22	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U22 Diabetes: no more than two rest days in a row	Flutter test (headless)	-	Yes

Test Case ID	Testing category	Test file	Test name	Framework	-	Automated
ATC-M4-U23	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U23 Warning symptoms stop plan generation and give the reason	Flutter test (headless)	-	Yes
ATC-M4-U24	Unit	test/atc/atc_m456_unit_test.dart	ATC-M4-U24 A planned exercise counts as completed only when its target is reached	Flutter test (headless)	-	Yes
ATC-M5-U01	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U01 Joint angle at the middle landmark is computed correctly	Flutter test (headless)	-	Yes
ATC-M5-U02	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U02 Landmarks below the visibility threshold are not used	Flutter test (headless)	-	Yes
ATC-M5-U03	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U03 Missing landmarks are named in the guidance	Flutter test (headless)	-	Yes
ATC-M5-U04	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U04 The movement phases follow top → descending → bottom → ascending → top	Flutter test (headless)	-	Yes
ATC-M5-U05	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U05 Three full squats count three reps	Flutter test (headless)	-	Yes
ATC-M5-U06	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U06 A half repetition is not counted	Flutter test (headless)	-	Yes
ATC-M5-U07	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U07 A movement faster than the minimum rep time is rejected as noise	Flutter test (headless)	-	Yes
ATC-M5-U08	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U08 The angle is smoothed (exponential average)	Flutter test (headless)	-	Yes
ATC-M5-U09	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U09 A plank hold is timed only while in position	Flutter test (headless)	-	Yes
ATC-M5-U10	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U10 The back-alignment rule passes upright and fails a deep forward lean	Flutter test (headless)	-	Yes
ATC-M5-U11	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U11 The hip rule gives the right correction for sagging and piking	Flutter test (headless)	-	Yes
ATC-M5-U12	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U12 Posture score = acceptable reps ÷ attempted reps	Flutter test (headless)	-	Yes
ATC-M5-U13	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U13 A correction is shown only after the fault persists for 3 frames	Flutter test (headless)	-	Yes
ATC-M5-U14	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U14 Exercises map to a movement pattern by movement, not muscle	Flutter test (headless)	-	Yes
ATC-M5-U15	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U15 An exercise the camera cannot follow is unsupported with a reason	Flutter test (headless)	-	Yes
ATC-M5-U16	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U16 Camera capability of every schedulable exercise (current code)	Flutter test (headless)	-	Yes
ATC-M5-U17	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U17 Number of movement patterns with a tracker	Flutter test (headless)	-	Yes
ATC-M5-U18	Unit	test/atc/atc_m456_unit_test.dart	ATC-M5-U18 An opening movement (overhead press) is counted on the mirrored angle	Flutter test (headless)	-	Yes
ATC-M6-U01	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U01 A rep question is recognised	Flutter test (headless)	-	Yes
ATC-M6-U02	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U02 A form question is recognised	Flutter test (headless)	-	Yes
ATC-M6-U03	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U03 A pause command is recognised	Flutter test (headless)	-	Yes
ATC-M6-U04	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U04 An unrelated question falls back to unknown	Flutter test (headless)	-	Yes
ATC-M6-U05	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U05 Empty speech input is not treated as a command	Flutter test (headless)	-	Yes
ATC-M6-U06	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U06 A pain report is recognised as a safety intent	Flutter test (headless)	-	Yes
ATC-M6-U07	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U07 The reply uses the live rep counter, never an invented number	Flutter test (headless)	-	Yes
ATC-M6-U08	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U08 With no session the assistant says so instead of inventing numbers	Flutter test (headless)	-	Yes
ATC-M6-U09	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U09 Quick-action chips are contextual and at most three	Flutter test (headless)	-	Yes
ATC-M6-U10	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U10 Microphone denied: the assistant never listens	Flutter test (headless)	-	Yes

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ATC-M6-U11	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U11 Microphone permanently denied: the settings link is offered	Flutter test (headless)	-	Yes
ATC-M6-U12	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U12 Recognition errors give a clear message and nothing is spoken	Flutter test (headless)	-	Yes
ATC-M6-U13	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U13 A speech-synthesis failure still shows the reply	Flutter test (headless)	-	Yes
ATC-M6-U14	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U14 Closing releases the microphone and ignores a pending command	Flutter test (headless)	-	Yes
ATC-M6-U15	Unit	test/atc/atc_m456_unit_test.dart	ATC-M6-U15 Replies never overlap; replay speaks the same reply once more	Flutter test (headless)	-	Yes
ATC-M4-F03	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M4-F03 Exercise detail shows the prescribed dose and a start/end illustration	Flutter test (headless)	-	Yes
ATC-M4-F04	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M4-F04 A dataset exercise shows its photos with the source caption	Flutter test (headless)	-	Yes
ATC-M4-B01	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B01 Screening gate: symptoms No, heart condition No, clearance Not needed, profile data Yes	Flutter test (headless)	-	Yes
ATC-M4-B02	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B02 Screening gate: symptoms Yes, heart condition No, clearance Not needed, profile data Yes	Flutter test (headless)	-	Yes
ATC-M4-B03	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B03 Screening gate: symptoms No, heart condition Yes, clearance No, profile data Yes	Flutter test (headless)	-	Yes
ATC-M4-B04	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B04 Screening gate: symptoms No, heart condition Yes, clearance Yes, profile data Yes	Flutter test (headless)	-	Yes
ATC-M4-B05	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B05 Screening gate: symptoms No, heart condition No, clearance Not needed, profile data No	Flutter test (headless)	-	Yes
ATC-M4-B06	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B06 Exercise eligibility: Healthy, impact Low, tag None, injury No, MET Yes	Flutter test (headless)	-	Yes
ATC-M4-B07	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B07 Exercise eligibility: Hypertension, impact Low, tag HIIT, injury No, MET Yes	Flutter test (headless)	-	Yes
ATC-M4-B08	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B08 Exercise eligibility: Obese, impact High, tag None, injury No, MET Yes	Flutter test (headless)	-	Yes
ATC-M4-B09	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B09 Exercise eligibility: Healthy, impact High, tag None, injury Knee, MET Yes	Flutter test (headless)	-	Yes
ATC-M4-B10	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B10 Exercise eligibility: Healthy, impact Low, tag None, injury Knee, MET Yes	Flutter test (headless)	-	Yes
ATC-M4-B11	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B11 Exercise eligibility: Hypertension, impact Low, tag None, injury No, MET No	Flutter test (headless)	-	Yes
ATC-M4-B12	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B12 Equipment rule: owns Dumbbell, exercise needs Dumbbell	Flutter test (headless)	-	Yes
ATC-M4-B13	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B13 Equipment rule: owns None, exercise needs Dumbbell	Flutter test (headless)	-	Yes
ATC-M4-B14	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M4-B14 Equipment rule: owns Dumbbell, exercise needs Barbell	Flutter test (headless)	-	Yes
ATC-M5-B01	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B01 Camera capability of squat: validatedForm	Flutter test (headless)	-	Yes
ATC-M5-B02	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B02 Camera capability of Barbell_Squat: experimentalForm	Flutter test (headless)	-	Yes
ATC-M5-B03	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B03 Camera capability of Standing_Calf_Raises: notSupported	Flutter test (headless)	-	Yes
ATC-M5-B04	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B04 Rep acceptance: depth reached Yes, duration ≥ minimum Yes	Flutter test (headless)	-	Yes
ATC-M5-B05	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B05 Rep acceptance: depth reached No, duration ≥ minimum Yes	Flutter test (headless)	-	Yes

Test Case ID	Testing category	Test file	Test name	Framework	-	Automated
ATC-M5-B06	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B06 Rep acceptance: depth reached Yes, duration ≥ minimum No	Flutter test (headless)	-	Yes
ATC-M5-B07	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B07 Landmark visibility 0.9 (threshold 0.5)	Flutter test (headless)	-	Yes
ATC-M5-B08	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B08 Landmark visibility 0.3 (threshold 0.5)	Flutter test (headless)	-	Yes
ATC-M5-B09	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B09 Correction trigger: fault No, persists ≥ 3 frames n/a	Flutter test (headless)	-	Yes
ATC-M5-B10	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B10 Correction trigger: fault Yes, persists ≥ 3 frames No	Flutter test (headless)	-	Yes
ATC-M5-B11	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B11 Correction trigger: fault Yes, persists ≥ 3 frames Yes	Flutter test (headless)	-	Yes
ATC-M5-B12	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B12 Plan completion: all sets Yes, skipped No	Flutter test (headless)	-	Yes
ATC-M5-B13	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B13 Plan completion: all sets No, skipped No	Flutter test (headless)	-	Yes
ATC-M5-B14	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M5-B14 Plan completion: all sets Yes, skipped Yes	Flutter test (headless)	-	Yes
ATC-M6-B01	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B01 Voice command "pause workout" with session active	Flutter test (headless)	-	Yes
ATC-M6-B02	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B02 Voice command "resume workout" with session paused	Flutter test (headless)	-	Yes
ATC-M6-B03	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B03 Voice command "pause workout" with session none	Flutter test (headless)	-	Yes
ATC-M6-B04	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B04 Voice command "my knee hurts" with session active	Flutter test (headless)	-	Yes
ATC-M6-B05	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B05 Microphone permission granted	Flutter test (headless)	-	Yes
ATC-M6-B06	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B06 Microphone permission denied	Flutter test (headless)	-	Yes
ATC-M6-B07	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B07 Microphone permission permanently denied	Flutter test (headless)	-	Yes
ATC-M6-B08	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B08 Recognition result: final text "how many reps"	Flutter test (headless)	-	Yes
ATC-M6-B09	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B09 Recognition result: no match	Flutter test (headless)	-	Yes
ATC-M6-B10	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B10 Recognition result: timeout (silence)	Flutter test (headless)	-	Yes
ATC-M6-B11	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B11 Text-to-speech works	Flutter test (headless)	-	Yes
ATC-M6-B12	Business rule	test/atc/atc_m456_rules_test.dart	ATC-M6-B12 Text-to-speech fails	Flutter test (headless)	-	Yes
ATC-M4-F05	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M4-F05 "Start exercise" opens mode selection linked to the planned exercise	Flutter test (headless)	-	Yes
ATC-M4-F06	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M4-F06 The workout tab shows the training profile of the health classification	Flutter test (headless)	-	Yes
ATC-M5-F01	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F01 Posture Detection offers Today's workout, My plan and the Exercise library	Flutter test (headless)	-	Yes
ATC-M5-F02	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F02 Without a plan an empty state is shown, not invented exercises	Flutter test (headless)	-	Yes
ATC-INT-34	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-34 The Module 2 health category sets the intensity ceiling of the generated plan	Flutter test (headless)	-	Yes
ATC-INT-35	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-35 A condition recorded in Module 2 removes restricted exercises from the Module 4 plan	Flutter test (headless)	-	Yes
ATC-INT-36	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-36 A planned exercise starts a posture session with the planned dose and a camera configuration	Flutter test (headless)	-	Yes
ATC-INT-37	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-37 Completing every main exercise of the day marks the plan day done once	Flutter test (headless)	-	Yes
ATC-INT-38	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-38 A skipped or incomplete session does not complete the planned exercise	Flutter test (headless)	-	Yes

Test Case ID	Testing category	Test file	Test name	Framework	-	Automated
ATC-INT-39	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-39 A posture correction from the camera engine is spoken once by the cue announcer	Flutter test (headless)	-	Yes
ATC-INT-40	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-40 Asking "how's my form" answers from the live posture analysis	Flutter test (headless)	-	Yes
ATC-INT-41	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-41 The rep count spoken by the assistant equals the camera counter	Flutter test (headless)	-	Yes
ATC-INT-42	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-42 A spoken "pause" stops rep counting until "resume"	Flutter test (headless)	-	Yes
ATC-INT-43	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-43 Ending a session by voice needs confirmation before the session is asked to finish	Flutter test (headless)	-	Yes
ATC-INT-44	Integration	test/atc/atc_m456_integration_test.dart	ATC-INT-44 Profile → plan → posture sessions with voice queries → plan day completed	Flutter test (headless)	-	Yes
ATC-M5-F03	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F03 My plan lists the real plan day with camera capability and starts an entry with its plan link	Flutter test (headless)	-	Yes
ATC-M5-F04	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F04 The generic exercise library is kept with camera and manual sections	Flutter test (headless)	-	Yes
ATC-M5-F05	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F05 Camera permission is not requested before the consent toggle is on	Flutter test (headless)	-	Yes
ATC-M5-F06	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F06 "Skip for now" returns to the previous screen	Flutter test (headless)	-	Yes
ATC-M5-F07	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F07 Refused camera permission keeps the camera off and offers Manual Guidance	Flutter test (headless)	-	Yes
ATC-M5-F08	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F08 Camera screen counts reps from pose frames and shows status and privacy notice	Flutter test (headless)	-	Yes
ATC-M5-F09	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F09 "Stop Session" ends the session and shows the summary with score, stats and feedback	Flutter test (headless)	-	Yes
ATC-M5-F10	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F10 Manual guidance: confirmed sets add reps; the form checklist and response buttons are shown	Flutter test (headless)	-	Yes
ATC-M5-F11	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F11 "Next Exercise" advances to the following exercise	Flutter test (headless)	-	Yes
ATC-M5-F12	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F12 A plank in camera mode shows a hold timer instead of a rep counter	Flutter test (headless)	-	Yes
ATC-M5-F13	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M5-F13 An exercise the camera cannot follow explains why and keeps Manual Guidance	Flutter test (headless)	-	Yes
ATC-M6-F01	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M6-F01 Opening the assistant asks for the microphone and shows the listening state	Flutter test (headless)	-	Yes
ATC-M6-F02	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M6-F02 A spoken question shows the transcript, the reply card, at most three chips and is spoken aloud	Flutter test (headless)	-	Yes
ATC-M6-F03	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M6-F03 A refused microphone is explained and a retry is offered	Flutter test (headless)	-	Yes
ATC-M6-F04	Functional (widget)	test/atc/atc_m456_functional_test.dart	ATC-M6-F04 Leaving the assistant releases the microphone without running a command	Flutter test (headless)	-	Yes
ATC-DEV-S01	Functional (real device)	integration_test/smoke_test.dart	ATC-DEV-S01 The app shell starts on the phone and the bottom navigation opens the Workout tab and returns Home	Flutter integration_test (real device)	-	Yes
ATC-DEV-M4-01	Functional (real device)	integration_test/module4_workout_test.dart	ATC-DEV-M4-01 The Workout tab shows the weekly plan generated for the profile	Flutter integration_test (real device)	-	Yes
ATC-DEV-M4-02	Functional (real device)	integration_test/module4_workout_test.dart	ATC-DEV-M4-02 Selecting a workout day shows that day's session and its exercise count	Flutter integration_test (real device)	-	Yes
ATC-DEV-M4-03	Functional (real device)	integration_test/module4_workout_test.dart	ATC-DEV-M4-03 An exercise opens its detail screen with the planned sets, reps or time and rest, and Back returns to the plan	Flutter integration_test (real device)	-	Yes

Test Case ID	Testing category	Test file	Test name	Framework	-	Automated
ATC-DEV-M4-04	Functional (real device)	integration_test/module4_workout_test.dart	ATC-DEV-M4-04 Regenerate plan produces a different exercise selection (cloud save unavailable in the test session)	Flutter integration_test (real device)	-	Yes
ATC-DEV-M4-05	Functional (real device)	integration_test/module4_workout_test.dart	ATC-DEV-M4-05 The training profile card names the profile of the health classification	Flutter integration_test (real device)	-	Yes
ATC-DEV-M4-06	Functional (real device)	integration_test/module4_workout_test.dart	ATC-DEV-M4-06 A reported warning symptom blocks plan generation and the reason is shown (negative case)	Flutter integration_test (real device)	-	Yes
ATC-DEV-M4-07	Functional (real device)	integration_test/module4_workout_test.dart	ATC-DEV-M4-07 Without a health profile (no classification) no plan is invented (negative case)	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-01	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-01 Posture check opens Posture Detection with Today's workout, My plan and the Exercise library	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-02	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-02 Today's workout lists today's plan entries (or the rest day) from the personalised plan	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-03	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-03 My plan shows every day of the complete plan with each entry's camera capability	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-04	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-04 The generic Exercise library keeps its camera and manual sections	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-05	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-05 A planned camera exercise opens mode selection; Continue without consent does not start the camera	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-06	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-06 With consent the real camera session starts on the phone, and Stop Session ends it with a summary	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-07	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-07 Manual Guidance shows the set counter, checklist and controls, and Pause stops the clock	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-08	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-08 A refused camera permission keeps the camera off and offers Manual Guidance (denial simulated in the app, no system dialog)	Flutter integration_test (real device)	-	Yes
ATC-DEV-M5-09	Functional (real device)	integration_test/module5_posture_test.dart	ATC-DEV-M5-09 An exercise the camera cannot follow says why and still offers Manual Guidance	Flutter integration_test (real device)	-	Yes
ATC-DEV-M6-01	Functional (real device)	integration_test/module6_voice_test.dart	ATC-DEV-M6-01 The chatbot's mic opens the Module 6 voice assistant, which shows the state matching the phone's microphone permission	Flutter integration_test (real device)	-	Yes
ATC-DEV-M6-02	Functional (real device)	integration_test/module6_voice_test.dart	ATC-DEV-M6-02 A quick-action chip is answered from the real session state and spoken with the phone's text-to-speech	Flutter integration_test (real device)	-	Yes
ATC-DEV-M6-03	Functional (real device)	integration_test/module6_voice_test.dart	ATC-DEV-M6-03 "What's my workout today?" is answered from the personalised plan	Flutter integration_test (real device)	-	Yes
ATC-DEV-M6-04	Functional (real device)	integration_test/module6_voice_test.dart	ATC-DEV-M6-04 The close button leaves the voice assistant and the app stays usable	Flutter integration_test (real device)	-	Yes
ATC-DEV-INT-01	Integration (real device)	integration_test/cross_module_test.dart	ATC-DEV-INT-01 Home → plan → exercise → Manual Guidance session → summary → the planned exercise is recorded as completed (Modules 4 → 5 → 4)	Flutter integration_test (real device)	-	Yes
ATC-DEV-INT-02	Integration (real device)	integration_test/cross_module_test.dart	ATC-DEV-INT-02 During a live camera session the voice assistant answers from that session and the session continues (Modules 5 ↔ 6)	Flutter integration_test (real device)	-	Yes
ATC-DEV-INT-03	Integration (real device)	integration_test/cross_module_test.dart	ATC-DEV-INT-03 The voice chip "Pause workout" pauses the running posture session (Module 6 → 5)	Flutter integration_test (real device)	-	Yes

5.5.4 Automated Test Execution and Results

```
tools/run_all.sh        (add --full to also run the complete existing test suites)
```

Table 5.30 How each suite is run

Suite	Command	Type
Flutter unit cases	cd app && flutter test test/atc/atc_unit_test.dart	Unit
Flutter functional cases	cd app && flutter test test/atc/atc_functional_test.dart	Functional
Flutter business-rule cases	cd app && flutter test test/atc/atc_rules_test.dart	Business rules
Flutter integration cases	cd app && flutter test test/atc/atc_integration_test.dart	Integration
API functional and business-rule cases	cd baymax-api && node --test test/atc/	Functional / Business rules
End-to-end cases (emulators)	cd baymax-api/e2e && node --test --test-force-exit rules.e2e.mjs (then flow.e2e.mjs, then admin-account.e2e.mjs, one at a time)	Integration

Table 5.60 How the Modules 4, 5 and 6 suites are run

Suite	Command	Type
Modules 4-6 unit, functional, business-rule and integration cases (headless)	cd app && flutter test test/atc/atc_m456_unit_test.dart test/atc/atc_m456_functional_test.dart test/atc/atc_m456_rules_test.dart test/atc/atc_m456_integration_test.dart	Unit / Functional / Business rules / Integration
Real-device smoke test	cd app && flutter drive --driver=test_driver/integration_test.dart --target=integration_test/smoke_test.dart -d f3fd6781 --keep-app-running	Functional (real device)
All real-device cases (one install)	cd app && bash integration_test/run_on_device.sh integration_test/all_device_test.dart <run-name>	Functional / Integration (real device)
Complete headless regression	cd app && flutter test --concurrency=2	All app tests

Table 5.31 Execution results (from the latest run)

Framework	Automated tests executed	Passed	Failed	Not run
Flutter test (Dart)	326	326	0	0
Node test (API)	66	66	0	0
Node test + Firebase emulator	20	20	0	0
Total	412	412	0	0

Table 5.31 counts the headless Flutter cases; the 123 Modules 4-6 cases were added to the Flutter row. The real-device integration_test run is reported separately below and is not double-counted here.

Figure 5.1: Automated Test Execution Result

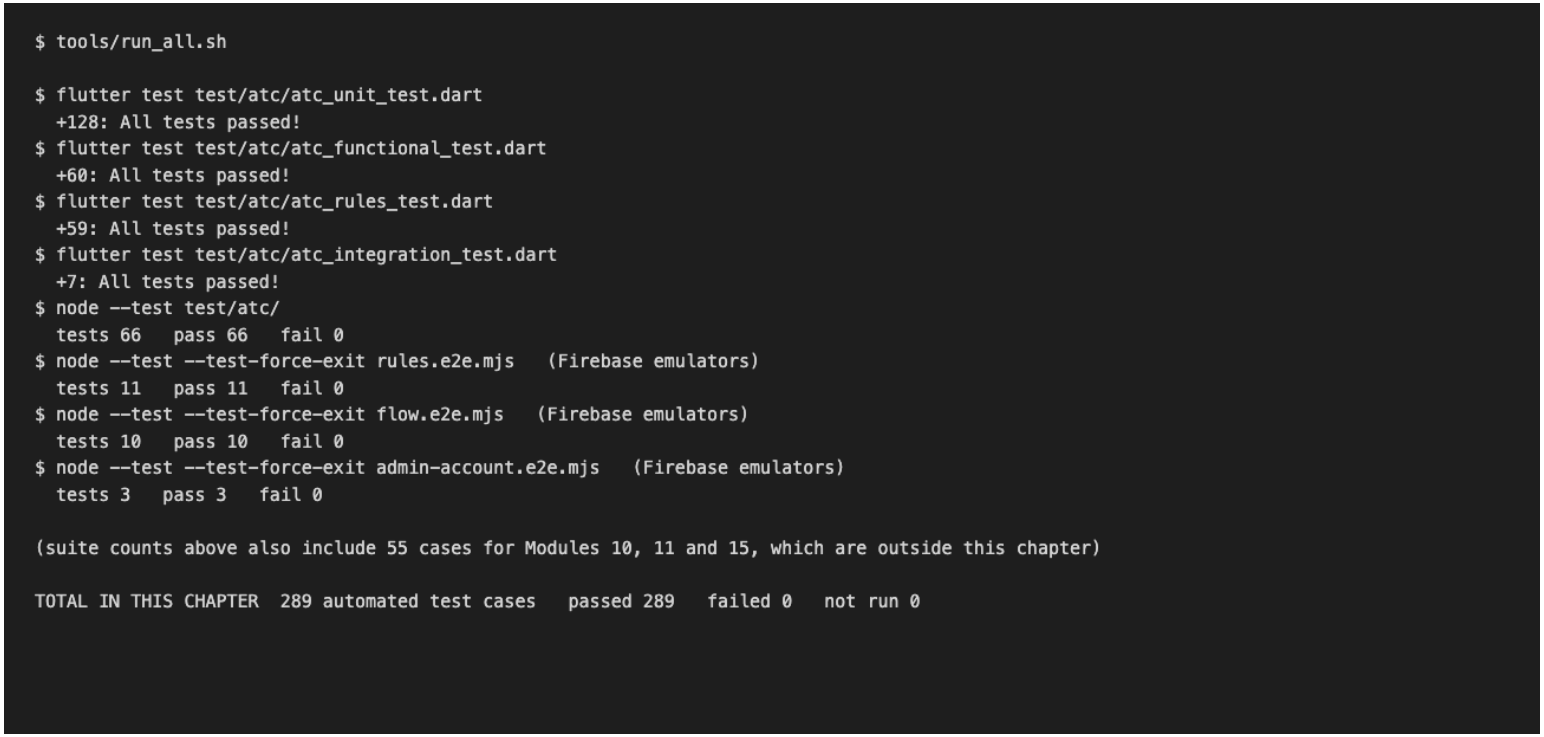


Table 5.61 Execution results for Modules 4, 5 and 6 (verified on 2026-10-18)

Framework	Automated tests executed	Passed	Failed	Not run
Flutter test (Dart), headless: Modules 4-6 ATC cases	123	123	0	0
Flutter integration_test on the Redmi Note 11 (2201117TG): Modules 4-6 cases	24	22	0	2
Total newly verified for Modules 4, 5 and 6	147	145	0	2
Complete headless Flutter regression, all modules (context, not added to the totals)	993	993	0	13 (skipped in source)

Real-device runs on Redmi Note 11 (2201117TG), Android 13 (API 33), MIUI V816, each started with integration_test/run_on_device.sh (flutter drive --keep-app-running): run3_smoke (integration_test/smoke_test.dart, started 2026-10-18T17:28:20+05:00): device runner "+2: All tests passed!" after 00:24; run6_module4_workout (integration_test/module4_workout_test.dart, started 2026-10-18T17:49:46+05:00): device runner "+8: All tests passed!" after 00:35; run7_module5_posture (integration_test/module5_posture_test.dart, started 2026-10-18T18:04:37+05:00): device runner "+9 -1: Some tests failed." after 01:25; run6_module6_voice (integration_test/module6_voice_test.dart, started 2026-10-18T17:56:11+05:00): device runner "+5: All tests passed!" after 00:37; run7_cross_module (integration_test/cross_module_test.dart, started 2026-10-18T18:07:18+05:00): device runner

"+3 -1: Some tests failed." after 02:19. The runner counts include the framework's own tear-down step. Per-case results, durations and screenshots are in docs/chapter5/device_evidence/<run>/. Earlier runs (run1_smoke_timeout, run2_smoke, run4_all, run5_all, run6_cross_module, run6_module5_posture) are kept as evidence and are not counted: they failed because of defects in the test code (a screenshot call without a frame, a finder that looked for a semantic label instead of a tooltip, a time formatted differently from the screen, a flow that assumed today is a training day), which were corrected, and one combined run lost its debug connection to the phone midway. In the counted runs only the first case of each run saved a screenshot: the helper converted the Android surface once per run although the framework resets it after every test (corrected in the helper; results are unaffected, screenshots are evidence only); the application was not changed to make tests pass. The headless figures above come from the ATC_LOG run of the same day; the historical team results in Table 5.31 are not repeated here.

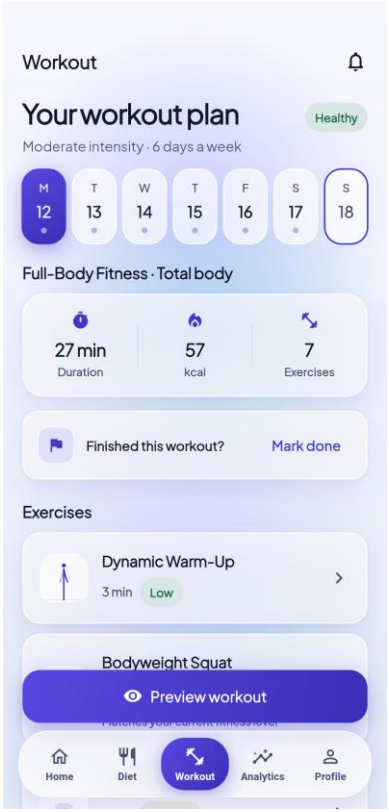


Figure 5.2: ATC-DEV-M4-01 on the Redmi Note 11 (2201117TG) (screenshot taken by the test at the end of the case)

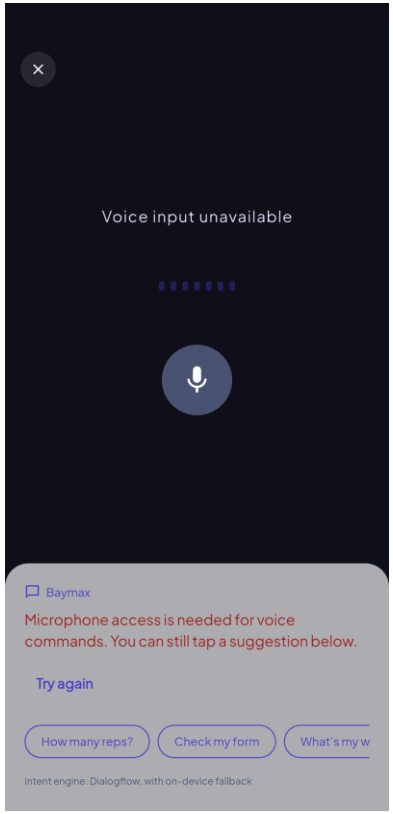


Figure 5.3: ATC-DEV-M6-01 on the Redmi Note 11 (2201117TG) (screenshot taken by the test at the end of the case)

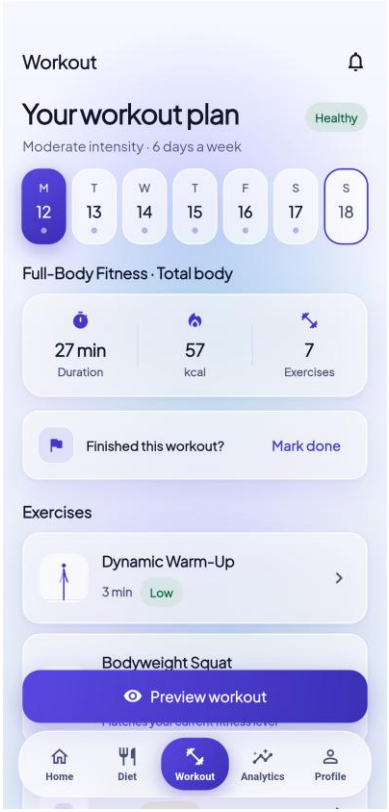


Figure 5.4: ATC-DEV-INT-01 on the Redmi Note 11 (2201117TG) (screenshot taken by the test at the end of the case)

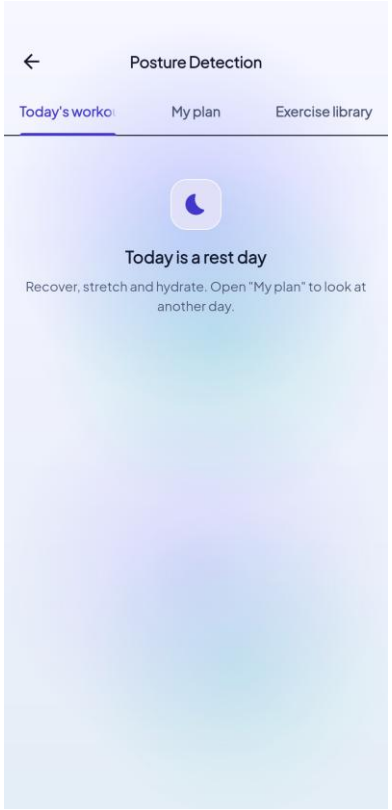


Figure 5.5: ATC-DEV-M5-01 on the Redmi Note 11 (2201117TG) (screenshot taken by the test at the end of the case)

5.5.5 Requirement Traceability for Automated Testing

Table 5.32 Requirement Traceability Matrix for Automated Testing

Requirement ID	Requirement Description	Use Case / Feature	Automated Test Case ID	Automation Tool / Framework	Automated (Y/N)	Status
M1-FR2	Login with email/password credentials	User Authentication & Onboarding	ATC-M1-U15, ATC-M1-U16, ATC-M1-U17	Flutter test	Yes	Pass (3/3)
M1-FR4	Allow new account registration	User Authentication & Onboarding	ATC-M1-F01, ATC-M1-F03	Flutter test	Yes	Pass (2/2)
M1-FR5	Validate input fields during registration	User Authentication & Onboarding	ATC-M1-F02, ATC-M1-F04, ATC-M1-U01, ATC-M1-U02, ATC-M1-U03, ATC-M1-U04, ATC-M1-U05, ATC-M1-U06, ATC-M1-U07, ATC-M1-U08, ATC-M1-U09, ATC-M1-U10, ATC-M1-U11, ATC-M1-U12, ATC-M1-U13, ATC-M1-U14	Flutter test	Yes	Pass (16/16)
M2-FR1	Collect personal information and physical measurements	AI Health Classification Engine	ATC-M2-U06	Flutter test	Yes	Pass (1/1)
M2-FR2	Automatically calculate BMI from height and weight	AI Health Classification Engine	ATC-M2-F01, ATC-M2-U01, ATC-M2-U02, ATC-M2-U05	Flutter test	Yes	Pass (4/4)
M2-FR5	Calculate BMR (Mifflin-St Jeor) and daily calorie requirement	AI Health Classification Engine	ATC-M2-U03, ATC-M2-U04	Flutter test	Yes	Pass (2/2)

Requirement ID	Requirement Description	Use Case / Feature	Automated Test Case ID	Automation Tool / Framework	Automated (Y/N)	Status
M2-FR9	Run Logistic Regression model for classification	AI Health Classification Engine	ATC-M2-U16, ATC-M2-U17, ATC-M2-U18	Flutter test	Yes	Pass (3/3)
M2-FR10	Classify user as Healthy, Overweight, Obese, At-Risk, Cardiac or Diabetic	AI Health Classification Engine	ATC-INT-01, ATC-INT-02, ATC-M2-B01, ATC-M2-B02, ATC-M2-B03, ATC-M2-B04, ATC-M2-B05, ATC-M2-B06, ATC-M2-B07, ATC-M2-B08, ATC-M2-B09, ATC-M2-F02, ATC-M2-U07, ATC-M2-U08, ATC-M2-U09, ATC-M2-U10, ATC-M2-U11, ATC-M2-U12, ATC-M2-U13, ATC-M2-U14	Flutter test	Yes	Pass (20/20)
M2-FR11	Display AI health insight, risk factors and recommendations	AI Health Classification Engine	ATC-M2-U15	Flutter test	Yes	Pass (1/1)
M3-FR1	Personalized greeting with user name	Home Dashboard	ATC-M3-F01, ATC-M3-F02, ATC-M3-F08	Flutter test	Yes	Pass (3/3)
M3-FR2	Today's activity stats: Calories Burned, Active Minutes, Step Count	Home Dashboard (partly implemented)	ATC-M3-F03	Flutter test	Yes	Pass (1/1)
M3-FR3	Upcoming workout card	Home Dashboard	ATC-M3-F04, ATC-M3-U03	Flutter test	Yes	Pass (2/2)
M3-FR4	Quick mood selector	Home Dashboard	ATC-M3-F05, ATC-M3-U01, ATC-M3-U02	Flutter test	Yes	Pass (3/3)
M3-FR5	Quick Access grid	Home Dashboard (partly implemented)	ATC-M3-F06, ATC-M3-F07	Flutter test	Yes	Pass (2/2)
M3-FR7	Bottom navigation bar	Home Dashboard	ATC-M3-F10	Flutter test	Yes	Pass (1/1)
M3-FR8	Notification bell icon	Home Dashboard	ATC-M3-F09	Flutter test	Yes	Pass (1/1)
M4-FR1	Apply medical restriction rules based on health classification before generating a plan	Workout & Exercise	ATC-INT-01, ATC-M4-U01, ATC-M4-U02	Flutter test	Yes	Pass (3/3)
M4-FR2	Select workout template appropriate to the health category	Workout & Exercise	ATC-M4-U05	Flutter test	Yes	Pass (1/1)
M4-FR3	Adjust exercise intensity based on health category	Workout & Exercise	ATC-INT-02, ATC-M4-U03	Flutter test	Yes	Pass (2/2)
M4-FR5	Generate a weekly workout schedule with exercises, sets, reps and intensity badges	Workout & Exercise	ATC-M4-F01, ATC-M4-U04, ATC-M4-U06	Flutter test	Yes	Pass (3/3)
M8-FR2	Macronutrient breakdown (Protein / Carbs / Fat)	Smart Diet & Calorie Tracking	ATC-M8-U06	Flutter test	Yes	Pass (1/1)
M8-FR4	Organise logged meals by slot: Breakfast, Lunch, Dinner, Snacks	Smart Diet & Calorie Tracking	ATC-M8-U07	Flutter test	Yes	Pass (1/1)
M8-FR7	Text entry: search food items, set quantity, assign to a meal slot	Smart Diet & Calorie Tracking	ATC-M8-F03, ATC-M8-F05, ATC-M8-F06, ATC-M8-U05	Flutter test	Yes	Pass (4/4)
M8-FR8	Extract food entities from text (NLP) and fetch calorie data	Smart Diet & Calorie Tracking	ATC-INT-03, ATC-M8-F04, ATC-M8-U01, ATC-M8-U02, ATC-M8-U03, ATC-M8-U04	Flutter test	Yes	Pass (6/6)
M8-FR13	Calorie Summary screen with 7-day chart, macro donut and daily verdict	Smart Diet & Calorie Tracking	ATC-INT-03, ATC-M8-F01, ATC-M8-F02, ATC-M8-U08, ATC-M8-U09, ATC-M8-U10	Flutter test	Yes	Pass (6/6)
M9-FR1	Generate a personalized Nutrition Plan from classification, BMR, calorie target and goal	Nutrition Plan	ATC-INT-02, ATC-M9-U05, ATC-M9-U06, ATC-M9-U07	Flutter test	Yes	Pass (4/4)
M9-FR2	Display daily calorie target and macronutrient targets	Nutrition Plan	ATC-M9-U01	Flutter test	Yes	Pass (1/1)
M9-FR5	Total calories per meal slot	Nutrition Plan	ATC-M9-U04	Flutter test	Yes	Pass (1/1)
M9-FR6	'Foods to Avoid' section tailored to allergies and classification	Nutrition Plan	ATC-INT-01, ATC-INT-06, ATC-M9-F01, ATC-M9-U02, ATC-M9-U03	Flutter test	Yes	Pass (5/5)
M9-FR7	Personalized dietary Pro Tips	Nutrition Plan	ATC-M9-U08	Flutter test	Yes	Pass (1/1)
M12-FR1	Calculate fitness score using a weighted formula	Community Ranking & Gamification	ATC-M12-U01, ATC-M12-U02, ATC-M12-U03	Flutter test	Yes	Pass (3/3)
M12-FR4	Cluster users using K-Means into Beginner / Intermediate / Advanced	Community Ranking & Gamification	ATC-M12-U04, ATC-M12-U05	Flutter test	Yes	Pass (2/2)
M12-FR6	Award badges for completed milestones	Community Ranking & Gamification (partly implemented)	ATC-INT-04, ATC-M12-U07	Flutter test	Yes	Pass (2/2)
M12-FR7	Distinguish earned and locked badges	Community Ranking & Gamification	ATC-M12-F02, ATC-M12-U08	Flutter test	Yes	Pass (2/2)
M12-FR8	Streak counter and bonus badge at streak milestones	Community Ranking & Gamification	ATC-INT-04, ATC-M12-U06	Flutter test	Yes	Pass (2/2)
M12-FR9	Show locked badges with their unlock requirements	Community Ranking & Gamification	ATC-M12-F01	Flutter test	Yes	Pass (1/1)
M12-FR10	XP and level (added: lifetime XP, level L starts at 100 x L x (L-1))	Community Ranking & Gamification (added requirement)	ATC-M12-U09	Flutter test	Yes	Pass (1/1)
M13-FR1	Detect user intent from typed or voice input	AI Chatbot (partly implemented)	ATC-M13-U01	Flutter test	Yes	Pass (1/1)
M13-FR2	Answer FAQs	AI Chatbot	ATC-M13-U07	Flutter test	Yes	Pass (1/1)
M13-FR3	Provide calorie/workout information	AI Chatbot	ATC-INT-07, ATC-M13-U02, ATC-M13-U03, ATC-M13-U06	Flutter test	Yes	Pass (4/4)
M13-FR4	Recommend specialist consultation when a health concern is detected	AI Chatbot	ATC-M13-B01, ATC-M13-B02, ATC-M13-B03, ATC-M13-B04, ATC-M13-B05, ATC-M13-U04, ATC-M13-U05	Flutter test	Yes	Pass (7/7)
M14-FR1	Specialist List screen filterable by Doctor, Nutritionist and Trainer	Online Consultation	ATC-M14-F01, ATC-M14-F02, ATC-M14-F03, ATC-M14-F04, ATC-M14-F05, ATC-M14-U08, ATC-M14-U09, ATC-M14-U10	Flutter test	Yes	Pass (8/8)

Requirement ID	Requirement Description	Use Case / Feature	Automated Test Case ID	Automation Tool / Framework	Automated (Y/N)	Status
M14-FR2	User can book an appointment	Online Consultation	ATC-INT-05, ATC-INT-13, ATC-INT-21, ATC-INT-22, ATC-INT-26, ATC-M14-B01, ATC-M14-B02, ATC-M14-B03, ATC-M14-B04, ATC-M14-B05, ATC-M14-B06, ATC-M14-B07, ATC-M14-F06, ATC-M14-F07, ATC-M14-F08, ATC-M14-F09, ATC-M14-F10, ATC-M14-F16, ATC-M14-U01, ATC-M14-U02, ATC-M14-U03, ATC-M14-U04, ATC-M14-U05, ATC-M14-U06, ATC-M14-U07, ATC-M14-U13, ATC-M14-U14	Flutter test / Node test + emulator	Yes	Pass (27/27)
M14-FR3	System generates a video meeting	Online Consultation	ATC-INT-14, ATC-INT-27, ATC-M14-B08, ATC-M14-B09, ATC-M14-B10, ATC-M14-B11, ATC-M14-B12, ATC-M14-B13, ATC-M14-B14, ATC-M14-B15, ATC-M14-F11, ATC-M14-F12, ATC-M14-F13, ATC-M14-F14, ATC-M14-F15, ATC-M14-F17, ATC-M14-F22, ATC-M14-U11, ATC-M14-U12	Flutter test / Node test (API) / Node test + emulator	Yes	Pass (19/19)
M14-FR4	Doctor can access the user's analytics dashboard (only what the user chose to share)	Online Consultation	ATC-M16-B10, ATC-M16-B11, ATC-M16-B12, ATC-M16-B13, ATC-M16-B14, ATC-M16-B15, ATC-M16-B16	Node test (API)	Yes	Pass (7/7)
M14-FR5	Doctor, trainer and nutritionist can modify the user's workout / diet plan	Online Consultation	ATC-INT-06, ATC-INT-16, ATC-INT-30, ATC-M14-B16, ATC-M14-B17, ATC-M14-B18, ATC-M14-B19, ATC-M14-B20, ATC-M14-B21, ATC-M14-B22, ATC-M14-B23, ATC-M14-B24, ATC-M14-B25, ATC-M14-B26, ATC-M14-B27, ATC-M14-B28, ATC-M14-B29, ATC-M14-B30, ATC-M14-F30, ATC-M14-F31, ATC-M14-F33, ATC-M14-U15, ATC-M14-U16, ATC-M14-U17, ATC-M14-U18	Flutter test / Node test (API) / Node test + emulator	Yes	Pass (25/25)
M14-FR6	Doctor can upload prescriptions	Online Consultation	ATC-M14-F27	Node test (API)	Yes	Pass (1/1)
M14-FR7	Specialist can provide consultation	Online Consultation	ATC-M14-B31, ATC-M14-B32, ATC-M14-B33, ATC-M14-B34, ATC-M14-B35, ATC-M14-B36, ATC-M14-F28	Node test (API)	Yes	Pass (7/7)
M14-FR8	Specialist uploads consultation notes to the user's record	Online Consultation	ATC-M14-F26	Node test (API)	Yes	Pass (1/1)
M14-FR9	Nutritionist validates diet logs and plans, approving or flagging day logs	Online Consultation	ATC-INT-17, ATC-INT-30, ATC-M14-F34	Node test (API) / Node test + emulator	Yes	Pass (3/3)
M14-FR10	Trainer validates workout plans and gives exercise guidance	Online Consultation	ATC-M14-F32	Node test (API)	Yes	Pass (1/1)
M14-FR11	ADDED: specialists register on the web portal; an administrator approves or rejects (licence number checked)	Online Consultation (added requirement)	ATC-INT-11, ATC-M14-B44, ATC-M14-B45, ATC-M14-B46, ATC-M14-B47, ATC-M14-B48, ATC-M14-B49, ATC-M14-B50, ATC-M14-F35, ATC-M14-F36, ATC-M14-F37, ATC-M14-F38	Node test (API) / Node test + emulator	Yes	Pass (12/12)
M14-FR12	ADDED: request workflow: paid request, specialist accepts (video room made) or declines, unanswered requests expire	Online Consultation (added requirement)	ATC-INT-14, ATC-M14-B37, ATC-M14-B38, ATC-M14-B39, ATC-M14-B40, ATC-M14-B41, ATC-M14-B42, ATC-M14-B43, ATC-M14-F21, ATC-M14-F22, ATC-M14-F23, ATC-M14-F24, ATC-M14-F25	Node test (API) / Node test + emulator	Yes	Pass (13/13)
M14-FR13	ADDED: specialist patient list and patient record in the web portal	Online Consultation (added requirement)	ATC-INT-15, ATC-M14-F29	Node test (API) / Node test + emulator	Yes	Pass (2/2)
M14-FR14	ADDED: specialist can close their own account (future bookings cancelled and freed)	Online Consultation (added requirement)	ATC-INT-18, ATC-M14-B51, ATC-M14-B52, ATC-M14-B53, ATC-M14-B54, ATC-M14-F40, ATC-M14-F41	Node test (API) / Node test + emulator	Yes	Pass (7/7)
M14-FR15	ADDED: administrator panel: add, list and delete specialists	Online Consultation (added requirement)	ATC-INT-12, ATC-INT-19, ATC-M14-F39	Node test (API) / Node test + emulator	Yes	Pass (3/3)
M16-FR2	Enforce role-based access control	Secure Data Management & Smart Alerts	ATC-INT-10, ATC-INT-20, ATC-INT-25, ATC-INT-31, ATC-INT-32, ATC-INT-33, ATC-M16-F04, ATC-M16-F05, ATC-M16-F06, ATC-M16-F07, ATC-M16-U01, ATC-M16-U02, ATC-M16-U03, ATC-M16-U04	Flutter test / Node test (API) / Node test + emulator	Yes	Pass (14/14)
M16-FR3	Workout reminders	Secure Data Management & Smart Alerts	ATC-M16-B01, ATC-M16-B02, ATC-M16-B03, ATC-M16-B04, ATC-M16-B05, ATC-M16-B06, ATC-M16-B07, ATC-M16-B08, ATC-M16-B09, ATC-M16-U07	Flutter test	Yes	Pass (10/10)
M16-FR5	Detect inactivity and send reminders	Secure Data Management & Smart Alerts	ATC-INT-04, ATC-M16-U05, ATC-M16-U06	Flutter test	Yes	Pass (3/3)
M16-FR8	Categorized Notifications screen with read/unread states	Secure Data Management & Smart Alerts	ATC-M16-F01, ATC-M16-F03	Flutter test	Yes	Pass (2/2)
M16-FR9	Mark all notifications as read	Secure Data Management & Smart Alerts	ATC-M16-F02	Flutter test	Yes	Pass (1/1)
M17-FR2	Calorie intake vs burn chart	Analytics & Progress Dashboard	ATC-M17-F01, ATC-M17-F02	Flutter test	Yes	Pass (2/2)
M17-FR5	Moving average smoothing	Analytics & Progress Dashboard	ATC-M17-U01, ATC-M17-U02	Flutter test	Yes	Pass (2/2)

Table 5.62 Requirement Traceability Matrix for Modules 4, 5 and 6

Requirement ID	Requirement Description	Use Case / Feature	Automated Test Case ID	Automation Tool / Framework	Automated (Y/N)	Status
M3-FR7	Bottom navigation bar	Home Dashboard (navigation); Functional (real device)	ATC-DEV-S01	Flutter integration_test (real device)	Yes	Pass (1/1)
M4-FR1	Apply medical restriction rules based on health classification before generating a plan	Workout & Exercise (plan generation); Business rule, Functional (real device), Integration, Unit	ATC-M4-U11, ATC-M4-U13, ATC-M4-U14, ATC-M4-B06, ATC-M4-B07, ATC-M4-B08, ATC-M4-B09, ATC-M4-B10, ATC-M4-B11, ATC-INT-35, ATC-DEV-M4-07	Flutter integration_test (real device); Flutter test	Yes	Pass (11/11)
M4-FR2	Select workout template appropriate to the health category	Workout & Exercise (plan generation); Functional (real device), Functional (widget), Integration, Unit	ATC-M4-U12, ATC-M4-U22, ATC-M4-F06, ATC-INT-34, ATC-DEV-M4-05	Flutter integration_test (real device); Flutter test	Yes	Pass (5/5)
M4-FR5	Generate a weekly workout schedule with exercises, sets, reps and intensity badges	Workout & Exercise (plan generation); Functional (real device), Unit	ATC-M4-U15, ATC-M4-U16, ATC-M4-U17, ATC-DEV-M4-01, ATC-DEV-M4-02, ATC-DEV-M4-04	Flutter integration_test (real device); Flutter test	Yes	Pass (6/6)
M4-FR6	Exercise Detail screen with muscles, metrics and technique steps	Workout & Exercise (plan generation); Functional (real device), Functional (widget)	ATC-M4-F03, ATC-M4-F04, ATC-DEV-M4-03	Flutter integration_test (real device); Flutter test	Yes	Pass (3/3)
M4-SFR1	Import the free-exercise-db exercise dataset with stable, unique IDs (added during implementation)	Workout & Exercise (plan generation); Unit	ATC-M4-U07, ATC-M4-U08, ATC-M4-U09	Flutter test	Yes	Pass (3/3)
M4-SFR2	Match each exercise to a Compendium MET value; estimate energy as MET x weight x time (added)	Workout & Exercise (plan generation); Unit	ATC-M4-U18, ATC-M4-U19, ATC-M4-U20, ATC-M4-U21	Flutter test	Yes	Pass (4/4)
M4-SFR3	Screen readiness (warning symptoms, clearance, missing data) and owned equipment before planning (added)	Workout & Exercise (plan generation); Business rule, Functional (real device), Unit	ATC-M4-U10, ATC-M4-U23, ATC-M4-B01, ATC-M4-B02, ATC-M4-B03, ATC-M4-B04, ATC-M4-B05, ATC-M4-B12, ATC-M4-B13, ATC-M4-B14, ATC-DEV-M4-06	Flutter integration_test (real device); Flutter test	Yes	Pass (11/11)
M4-SFR4	Record completion of planned exercises from sessions and mark the plan day done (added)	Workout & Exercise (plan generation); Functional (widget), Integration, Integration (real device), Unit	ATC-M4-U24, ATC-M4-F05, ATC-INT-37, ATC-INT-38, ATC-INT-44, ATC-DEV-INT-01	Flutter integration_test (real device); Flutter test	Yes	Pass (6/6)
M5-FR15.1	The system shall provide a 'Camera Detection' mode option that uses the device camera for real-time posture tracking and angle analysis.	Posture Detection; Business rule, Functional (real device), Functional (widget), Unit	ATC-M5-U15, ATC-M5-U16, ATC-M5-B01, ATC-M5-B02, ATC-M5-B03, ATC-M5-F05, ATC-M5-F07, ATC-M5-F13, ATC-DEV-M5-08, ATC-DEV-M5-09	Flutter integration_test (real device); Flutter test	Yes	Pass (10/10)
M5-FR15.2	The system shall provide a 'Manual Guidance' mode option that delivers guided exercise cues without active camera tracking.	Posture Detection; Functional (real device), Functional (widget)	ATC-M5-F05, ATC-M5-F07, ATC-DEV-M5-05	Flutter integration_test (real device); Flutter test	Yes	Pass (3/3)
M5-FR15.3	The system shall display a camera access consent toggle with a privacy notice before enabling camera-based posture detection.	Posture Detection; Functional (real device), Functional (widget)	ATC-M5-F05, ATC-DEV-M5-05	Flutter integration_test (real device); Flutter test	Yes	Pass (2/2)
M5-FR15.4	The system shall provide 'Continue' and 'Skip for now' buttons on the mode selection screen.	Posture Detection; Functional (widget)	ATC-M5-F06	Flutter test	Yes	Pass (1/1)
M5-FR16.1	The system shall display a live camera feed with a skeletal landmark overlay showing detected body joints and angle measurements.	Posture Detection; Unit	ATC-M5-U01	Flutter test	Yes	Pass (1/1)
M5-FR16.2	The system shall display a real-time posture correction feedback banner (e.g., 'Keep your back straight') during camera detection sessions.	Posture Detection; Business rule, Unit	ATC-M5-U03, ATC-M5-U10, ATC-M5-U11, ATC-M5-U13, ATC-M5-B09, ATC-M5-B10, ATC-M5-B11	Flutter test	Yes	Pass (7/7)
M5-FR16.3	The system shall display a live rep counter on the camera detection screen that increments as the system detects completed repetitions.	Posture Detection; Business rule, Functional (widget), Unit	ATC-M5-U04, ATC-M5-U05, ATC-M5-U06, ATC-M5-U07, ATC-M5-U08, ATC-M5-U18, ATC-M5-B04, ATC-M5-B05, ATC-M5-B06, ATC-M5-F08	Flutter test	Yes	Pass (10/10)
M5-FR16.4	The system shall display status indicators confirming 'Landmarks detected' and 'Angles calculated' during an active posture session.	Posture Detection; Business rule, Functional (widget), Unit	ATC-M5-U02, ATC-M5-B07, ATC-M5-B08, ATC-M5-F08	Flutter test	Yes	Pass (4/4)
M5-FR16.5	The system shall provide a 'Stop Session' button that ends the camera detection session and navigates to the Session Summary screen.	Posture Detection; Functional (real device), Functional (widget)	ATC-M5-F09, ATC-DEV-M5-06	Flutter integration_test (real device); Flutter test	Yes	Pass (1/1); Blocked (1)
M5-FR16.6	The system shall display a notice stating 'Processing is done locally. No video is stored.' on the camera detection screen.	Posture Detection; Functional (real device), Functional (widget)	ATC-M5-F08, ATC-DEV-M5-06	Flutter integration_test (real device); Flutter test	Yes	Pass (1/1); Blocked (1)
M5-FR17.1	The system shall display a live elapsed timer on the manual guidance screen showing time spent on the current exercise.	Posture Detection	-	-	No	Not covered by an automated case

Requirement ID	Requirement Description	Use Case / Feature	Automated Test Case ID	Automation Tool / Framework	Automated (Y/N)	Status
M5-FR17.2	The system shall display a rep counter in format 'current/target' (e.g., 8/12) on the manual guidance screen.	Posture Detection; Functional (real device), Functional (widget)	ATC-M5-F10, ATC-DEV-M5-07	Flutter integration_test (real device); Flutter test	Yes	Pass (2/2)
M5-FR17.3	The system shall display a stepwise proper form checklist with visual status indicators for each form cue during manual guidance.	Posture Detection; Functional (real device), Functional (widget)	ATC-M5-F10, ATC-DEV-M5-07	Flutter integration_test (real device); Flutter test	Yes	Pass (2/2)
M5-FR17.4	The system shall provide 'Completed', 'Skip', and 'Too Difficult' response buttons on the manual guidance screen.	Posture Detection; Functional (real device), Functional (widget)	ATC-M5-F10, ATC-DEV-M5-07	Flutter integration_test (real device); Flutter test	Yes	Pass (2/2)
M5-FR17.5	The system shall provide a 'Next Exercise' button that advances to the following exercise in the session sequence.	Posture Detection; Functional (widget)	ATC-M5-F11	Flutter test	Yes	Pass (1/1)
M5-FR18.1	The system shall display a 'Session Complete!' confirmation screen with a circular posture score indicator at the end of a posture check session.	Posture Detection; Functional (widget), Unit	ATC-M5-U12, ATC-M5-F09	Flutter test	Yes	Pass (2/2)
M5-FR18.2	The system shall display session statistics including Total Reps, Duration, and Accuracy percentage on the session summary screen.	Posture Detection; Functional (widget)	ATC-M5-F09	Flutter test	Yes	Pass (1/1)
M5-FR18.3	The system shall display a list of personalized form feedback items (passed and failed) with tick/cross indicators on the session summary screen.	Posture Detection; Functional (widget)	ATC-M5-F09	Flutter test	Yes	Pass (1/1)
M5-FR18.4	The system shall display an injury risk assessment notice (e.g., 'Injury risk remains Low — good form consistency!') on the session summary screen.	Posture Detection	-	-	No	Not implemented in this release (no test)
M5-SFR1	Map every exercise to a movement pattern with a camera tracker or a stated reason (added)	Posture Detection; Unit	ATC-M5-U14, ATC-M5-U17	Flutter test	Yes	Pass (2/2)
M5-SFR2	Time static holds (plank, wall sit) in camera mode (added)	Posture Detection; Functional (widget), Unit	ATC-M5-U09, ATC-M5-F12	Flutter test	Yes	Pass (2/2)
M5-SFR3	Start posture sessions from the personalised plan (Today's workout / My plan) as well as the library (added)	Posture Detection; Business rule, Functional (real device), Functional (widget), Integration	ATC-M5-B12, ATC-M5-B13, ATC-M5-B14, ATC-M5-F01, ATC-M5-F02, ATC-INT-36, ATC-M5-F03, ATC-M5-F04, ATC-DEV-M5-01, ATC-DEV-M5-02, ATC-DEV-M5-03, ATC-DEV-M5-04	Flutter integration_test (real device); Flutter test	Yes	Pass (12/12)
M6-FR19.1	The system shall display a full-screen voice command interface with an animated audio waveform and a 'Listening...' status label when activated.	Voice Assistant; Business rule, Functional (real device), Functional (widget), Unit	ATC-M6-U12, ATC-M6-B08, ATC-M6-B09, ATC-M6-B10, ATC-M6-F01, ATC-DEV-M6-01	Flutter integration_test (real device); Flutter test	Yes	Pass (6/6)
M6-FR19.2	The system shall display a central microphone button that toggles the voice listening state on or off.	Voice Assistant; Business rule, Functional (real device), Functional (widget), Unit	ATC-M6-U10, ATC-M6-U11, ATC-M6-B05, ATC-M6-B06, ATC-M6-B07, ATC-M6-F01, ATC-M6-F03, ATC-DEV-M6-01	Flutter integration_test (real device); Flutter test	Yes	Pass (8/8)
M6-FR19.3	The system shall display Baymax's voice response in a bottom card panel with the response text and contextual quick-action buttons (e.g., 'Start workout', 'Lower intensity', 'Show exercises').	Voice Assistant; Business rule, Functional (real device), Functional (widget), Integration, Unit	ATC-M6-U07, ATC-M6-U08, ATC-M6-U13, ATC-M6-U15, ATC-M6-B11, ATC-M6-B12, ATC-INT-41, ATC-M6-F02, ATC-DEV-M6-02, ATC-DEV-M6-03	Flutter integration_test (real device); Flutter test	Yes	Pass (10/10)
M6-FR19.4	The system shall provide a close (×) button to exit the voice command interface without executing any command.	Voice Assistant; Functional (real device), Functional (widget), Unit	ATC-M6-U14, ATC-M6-F04, ATC-DEV-M6-04	Flutter integration_test (real device); Flutter test	Yes	Pass (3/3)
M6-FR19.5	The system shall present contextual quick-action chips in the AI response card based on the nature of the user's voice query.	Voice Assistant; Functional (real device), Functional (widget), Unit	ATC-M6-U09, ATC-M6-F02, ATC-DEV-M6-02	Flutter integration_test (real device); Flutter test	Yes	Pass (3/3)
M6-SFR1	Recognise workout intents from the spoken (transcribed) text (added)	Voice Assistant; Unit	ATC-M6-U01, ATC-M6-U02, ATC-M6-U03, ATC-M6-U04, ATC-M6-U05, ATC-M6-U06	Flutter test	Yes	Pass (6/6)
M6-SFR2	Speak posture corrections and answer form questions from the live session (added)	Voice Assistant; Integration, Integration (real device)	ATC-INT-39, ATC-INT-40, ATC-DEV-INT-02	Flutter integration_test (real device); Flutter test	Yes	Pass (2/2); Blocked (1)
M6-SFR3	Control the posture session by voice (pause, resume, stop with confirmation) (added)	Voice Assistant; Business rule, Integration, Integration (real device)	ATC-M6-B01, ATC-M6-B02, ATC-M6-B03, ATC-M6-B04, ATC-INT-42, ATC-INT-43, ATC-DEV-INT-03	Flutter integration_test (real device); Flutter test	Yes	Pass (7/7)

5.5.6 Scope and limitations of the Modules 4, 5 and 6 evidence

Posture detection: the engine cases use synthetic landmark frames and the real-photo replay uses landmarks from public-domain exercise photos (held-out half: 66 % of repetitions counted, 13 % false alarms); they test the counting, timing and feedback logic. The real-device cases show that the camera and on-device MediaPipe start, run and stop on the phone and that results reach the plan; they were run without a person exercising in front of the camera, so their rep counts are not accuracy evidence. Real-person validation is tracked separately (docs/chapter5/REAL_PERSON_VALIDATION.md): only the squat has a live check (2026-10-15); the other representative exercises have not been performed and are not claimed.

Voice assistant: headless cases use a simulated recogniser and speech output. On the phone, the real speech and text-to-speech engines were used, and quick-action chips supplied fixed phrases; nothing was spoken to the phone, so this is not live speech-recognition validation. The microphone permission of the test build was not granted on the phone during the run; the case asserted the app's "Voice input unavailable" path, and the listening path is covered only by the headless cases. Permission dialogs were never answered by the tests; the camera-denied case simulates the answer inside the app.

Workout plan: rule cases show the screening and safety rules are applied as written (rule set pending professional review); they are not evidence of clinical safety. Saving to the cloud (plan regeneration save, "Mark done", session upload) was not exercised on the device because the tests never sign in; the app's offline message was asserted instead. The day-level completion mark is only set on the plan's own weekday, which was a rest day on the test date. M5-FR18.4 (injury-risk notice) is not implemented. Requirement IDs marked "added" (SFR) were added during implementation and are not yet in Chapter 2.